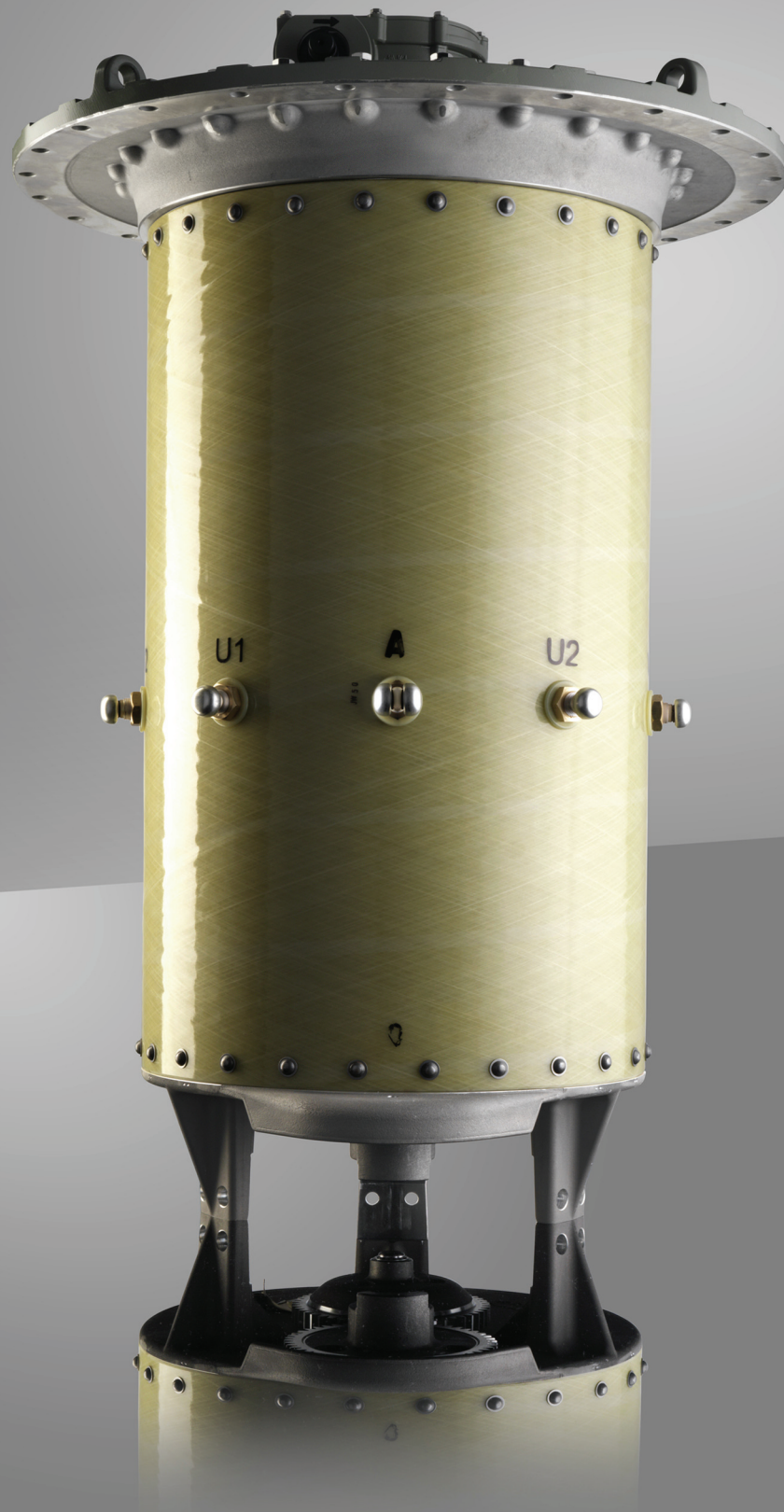


On-load tap-changer VACUTAP® VM®

Technical Data 2332907/02



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The original operating instructions were drawn up in German.



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1 General / Technical Data

1.1 On-load tap-changer designations

Example: **VM** **III** **650** **Y** - **72.5** / **C** - **10** **19** **1W** **R**

Basic connection diagram

Type	VACUTAP® VM®	VM®	
Number of phases	1 phase	I	
	2 phases	II	
	3 phases	III	
I _{um} (in A)	VM I, VM II, VM III	300	
	VM I, VM II, VM III	350	
	VM I, VM II, VM III	500	
	VM I, VM II, VM III	650	
	VM I	802	
	VM I	1002	
	VM I	1203	
	VM I	1503	
	1 (3-phase Y)	0	
	1 (1-phase)	1	
Number of parallel main switching contacts	2 (1-phase)	2	
	3 (1-phase)	3	
Applications	For use with neutral point only	Y	
U _m (in kV)	VM®	72.5	
	VM®	123	
	VM®	170	
	VM®	245	
	VM®	300	
Tap selector size		B	
		C	
		D	
	not with multiple coarse change-over selector	DE	
Number of maximum operating positions without change-over selector	10	10	
	12	12	
	14	14	
	16	16	
	18	18	
	22	22	
Number of maximum operating positions with change-over selector (reversing change-over selector or coarse tap selector) ¹⁾	10	19	
	12	23	
	14	27	
	16	31	
	18	35	
Number of maximum operating positions with multiple coarse change-over selector	10	59	
	12	71	
	14	83	
	16	95	
	18	107	
Mid-positions	0 mid-positions (without change-over selector)	0	
	1 mid-position	1	
	3 mid-positions	3	
Change-over selector	Reversing change-over selector	W	
	Coarse tap connection	G	
Tie-in measures	Tie-in resistor	R	
	Potential switch	S	
	Potential switch with tie-in resistor	P	

¹⁾ 300 amp variants with a maximum of 27 operating positions available



1.2 Technical data for VACUTAP® VM®

1.2.1 Technical data for VACUTAP® VM I

On-load tap-changer VM I	301	351	501	651	802	1002	1203	1503
Number of phases and application	1	1	1	1	1	1	1	1
Maximum rated through current I_{um} (in A)	300	350	500	650	800	1000	1200	1500
Rated short-time current (in kA)	4	4.2	5	6.5	8	10	12	15
Rated duration of short-circuits (in s)	3	3	3	3	3	3	3	3
Rated peak withstand current (in kA)	10	10.5	12.5	16.25	20	25	30	37.5
Maximum rated step voltage U_{im} (in V) ¹⁾	3300	3300	3300	3300	3300	3300	3300	3300
Step capacity (PStN, in kVA)	990	1155	1625	1625	2600	2600	3500	3500
Rated frequency (in Hz)	50...60							
Operating positions	Without change-over selector: maximum 18 With change-over selector: maximum 35 ²⁾ With multiple coarse change-over selector: maximum 107 ²⁾							
Oil compartment	Pressure-tight up to 0.3 bar permanent differential pressure (test pressure 0.6 bar), head and cover of the diverter switch oil compartment are vacuum-proof.							
Temperature range	The on-load tap-changer VACUTAP® VM® can be operated in the rated load range at surrounding oil temperatures of between -25 ° and +105 °C and with overload up to +115 °C in accordance with IEC 60214-1. For details of operation under Arctic conditions, please refer to the General Technical Data TD 61.							
Dimensions	Weight, displacement volume and oil content of the diverter switch oil compartment are shown in the relevant dimension drawings.							

Table 1 Technical data for VACUTAP® VM I

- ¹⁾ The maximum rated step voltage may be exceeded by 10 % due to overexcitation of the transformer if the step capacity is limited to its rated value.
- ²⁾ 300 amp variants with a maximum of 27 operating positions available



1.2.2 Technical data for VACUTAP® VM II

On-load tap-changer VM II	302	352	502	652
Number of phases and application	2	2	2	2
Maximum rated through current I_{um} (in A)	300	350	500	650
Rated short-time current (in kA)	4	4.2	5	6.5
Rated duration of short-circuits (in s)	3	3	3	3
Rated peak withstand current (in kA)	10	10.5	12.5	16.25
Maximum rated step voltage U_{im} (in V) ¹⁾	3300	3300	3300	3300
Step capacity (PStN, in kVA)	990	1155	1625	1625
Rated frequency (in Hz)	50...60			
Operating positions	Without change-over selector: maximum 18 With change-over selector: maximum 35 ²⁾ With multiple coarse change-over selector: maximum 107 ²⁾			
Oil compartment	Pressure-tight up to 0.3 bar permanent differential pressure (test pressure 0.6 bar), head and cover of the diverter switch oil compartment are vacuum-proof.			
Temperature range	The on-load tap-changer VACUTAP® VM® can be operated in the rated load range at surrounding oil temperatures of between -25 ° and +105 °C and with overload up to +115 °C in accordance with IEC 60214-1. For details of operation under Arctic conditions, please refer to the General Technical Data for TD 61.			
Dimensions	Weight, displacement volume and oil content of the diverter switch oil compartment are shown in the relevant dimension drawings.			

Table 2 Technical data for VACUTAP® VM II

- 1) The maximum rated step voltage may be exceeded by 10 % due to overexcitation of the transformer if the step capacity is limited to its rated value.
- 2) 300 amp variants with a maximum of 27 operating positions available



1.2.3 Technical data for VACUTAP® VM III

On-load tap-changer VM III	300 Y	350 Y	500 Y	650 Y
Number of phases and application	3	3	3	3
Maximum rated through current I_{um} (in A)	300	350	500	650
Rated short-time current (in kA)	4	4.2	5	6.5
Rated duration of short-circuits (in s)	3	3	3	3
Rated peak withstand current (in kA)	10	10.5	12.5	16.25
Maximum rated step voltage U_{im} (in V) ¹⁾	3300	3300	3300	3300
Step capacity (PStN, in kVA)	990	1155	1625	1625
Rated frequency (in Hz)	50...60			
Operating positions	Without change-over selector: maximum 18 With change-over selector: maximum 35 ²⁾ With multiple coarse change-over selector: maximum 107 ²⁾			
Oil compartment	Pressure-tight up to 0.3 bar permanent differential pressure (test pressure 0.6 bar), head and cover of the diverter switch oil compartment are vacuum-proof.			
Temperature range	The on-load tap-changer VACUTAP® VM® can be operated in the rated load range at surrounding oil temperatures of between -25 ° and +105 °C and with overload up to +115 °C in accordance with IEC 60214-1. For details of operation under Arctic conditions, please refer to the General Technical Data for TD 61.			
Dimensions	Weight, displacement volume and oil content of the diverter switch oil compartment are shown in the relevant dimension drawings.			

Table 3 Technical data for VACUTAP® VM III

- ¹⁾ The maximum rated step voltage may be exceeded by 10 % due to overexcitation of the transformer if the step capacity is limited to its rated value.
- ²⁾ 300 amp variants with a maximum of 27 operating positions available



1.2.4 Step capacity diagram VM®

1.2.4.1 Step capacity diagram for network application

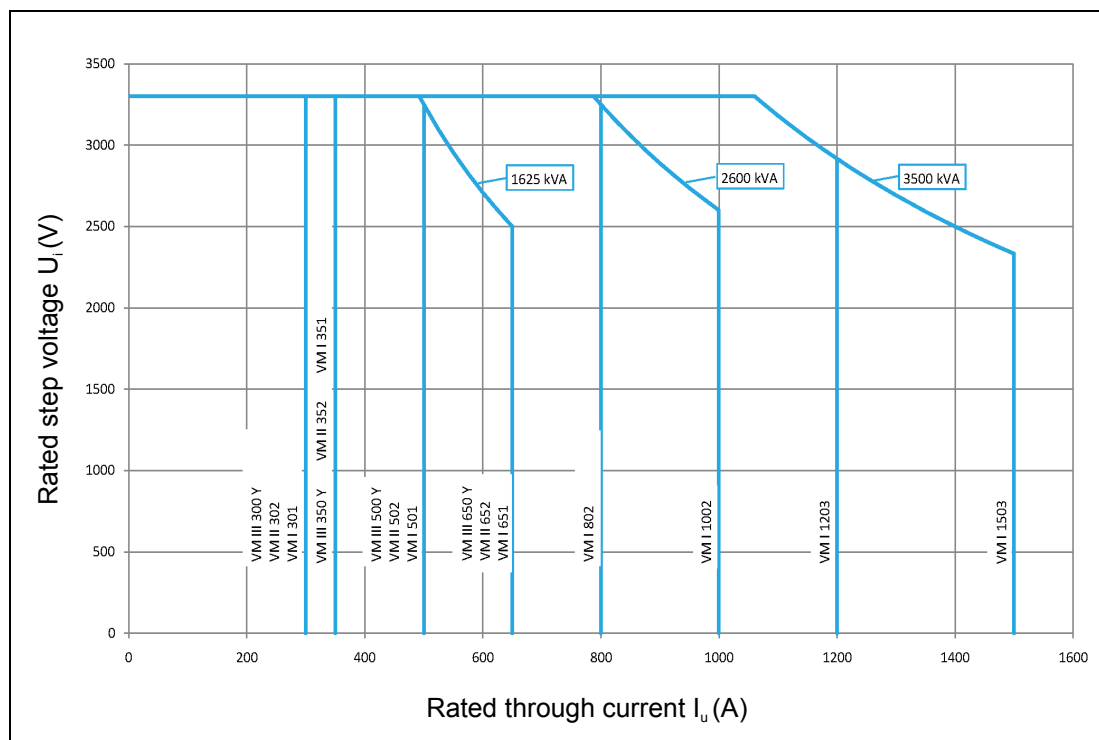


Figure 1 Step capacities (rated step voltage U_i , rated through-current I_u)

1.2.4.2 Step capacity diagram for electrical arc furnace application

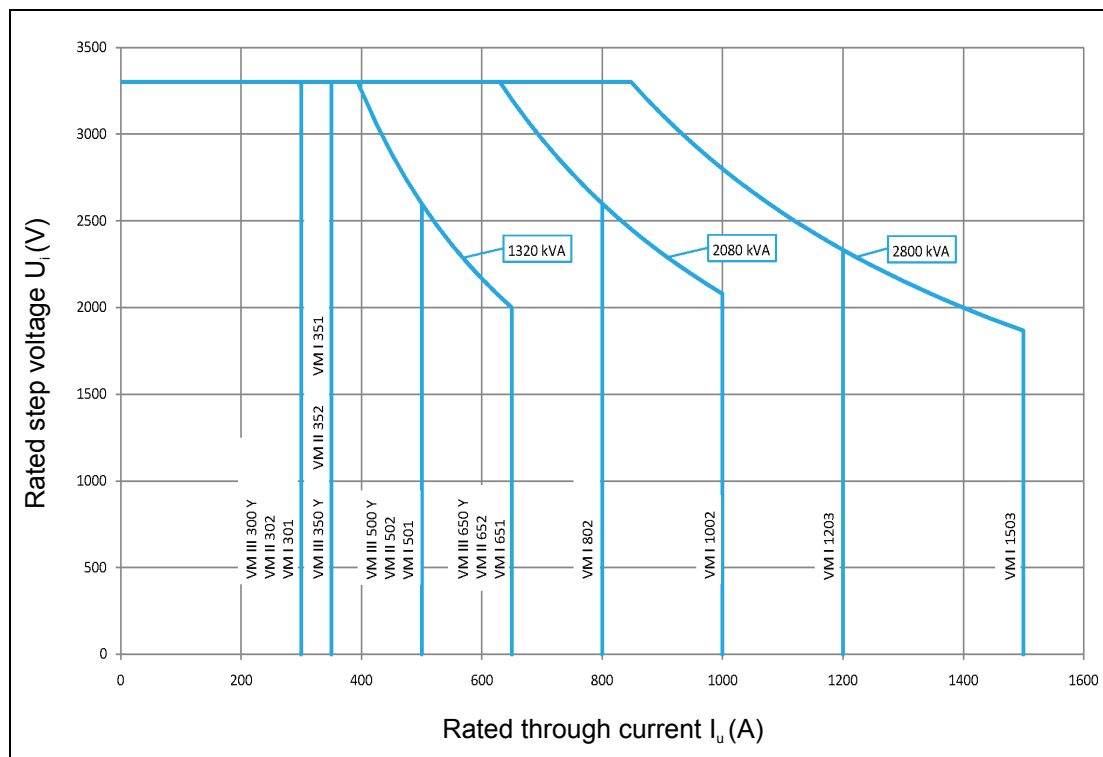


Figure 2 Step capacities (rated step voltage U_i , rated through-current I_u)



1.3 Rated insulation level

Rated insulation level	For all on-load tap-changer variants				
Highest voltage for equipment U_m (in kV) ¹	72.5	123	170	245	300 ²⁾
Rated lightning impulse withstand voltage (in kV, 1.2/50 μ s)	350	550	750	1050	1050
Rated switching impulse voltage (in kV)				850	850
Rated short-duration power frequency withstand voltage (in kV, 50 Hz, 1 min.)	140	230	325	460	460

Table 4 Rated insulation level

- 1) In accordance with IEC 60214-1, chapter 3.57: highest effective value for phase-to-phase voltage in a three-phase system for which an on-load tap-changer is designed with respect to its insulation.
- 2) Star-point on-load tap-changer and 300 A variants with a maximum of 245 kV available.

1.4 Rated withstand voltages

Figure 3 shows diagrams of the voltage stress present on the tap winding of the three primary basic connections of three-pole on-load tap-changers and single-pole on-load tap-changers.

When selecting the on-load tap-changer, you must check that the highest stress on the tap selector does not exceed the related rated withstand voltages.

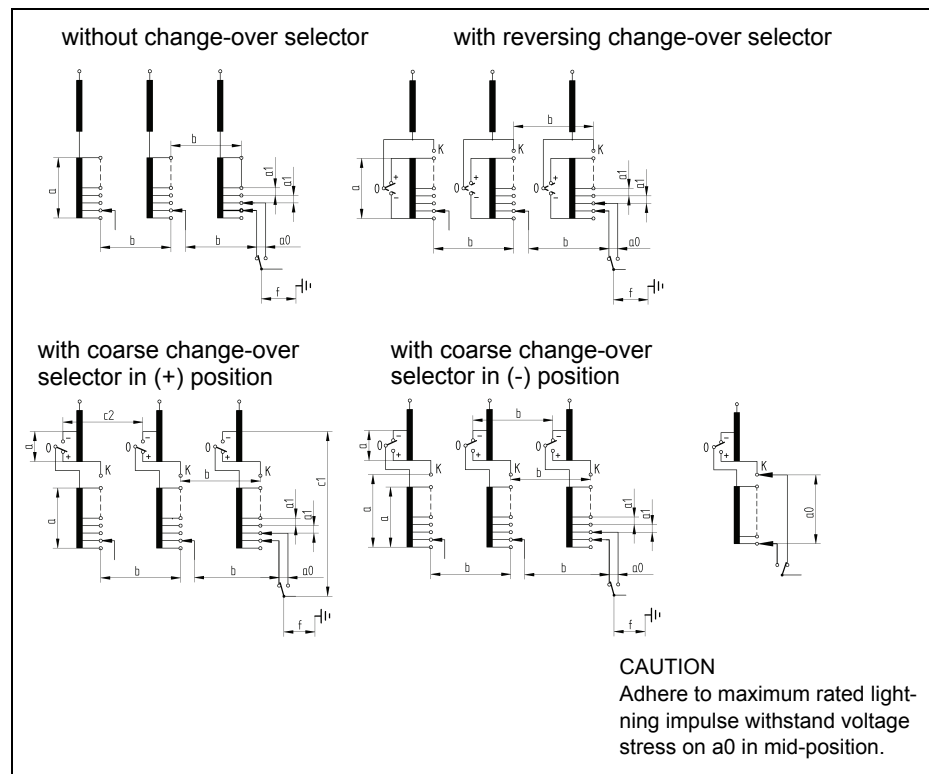


Figure 3 Rated withstand voltages

- a0** = Between selected and preselected tap on the diverter switch and tap selector.
 - a1** = Between tap selector contacts of the winding of one tap position (connected or not connected)
 - a** = Between beginning and end of a tapped winding and, with respect to a coarse winding, between beginning and end of a coarse winding.
Note for coarse tap selector connection (-) position of the change-over selector:
When stressed with impulse voltage, the permissible withstand voltage "a" must be adhered to between the end of a coarse tap winding connected with the K fine tap selector contact and the fine tap selector contact at the end of the tapped winding of the same phase.
 - b** = Between the fine tap selector contacts of different phases and between change-over selector contacts of different phases, which are connected with the beginning/end of a tapped winding or with a fine tap selector contact.
 - f** = Between diverter switch output terminal and ground
- Additionally for coarse tapping arrangement in (+) position of the change-over selector:**
- c1** = From one (-) change-over selector contact to take-off lead of the same phase.
 - c2** = Between (-) change-over selector contacts of different phases.



1.4.1 Rated withstand voltages of the internal on-load tap-changer insulation (with the exception of VACUTAP® VM 300)

The tap selector size (ID letters B, C, D, DE, E) characterizes the internal insulation of the tap selector, whose rated withstand voltages must be matched to the requirements of the transformer winding.

Insulation distances	Tap selector size B		Tap selector size C		Tap selector size D		Tap selector size DE	
	kV 1.2/ 50 µs	kV 50 Hz 1 min	kV 1.2/50 µs	kV 50 Hz 1 min	kV 1.2/50 µs	kV 50 Hz 1 min	kV 1.2/50 µs	kV 50 Hz 1 min
a0	Power-frequency voltage test (50 Hz/1 min): 20 kV 50 Hz 1 min; Response voltage with 1 mA leakage current: 35 kV 1.2/50 µs; 3 kA residual voltage (= lightning impulse protection level): 56 kV 1.2/50 µs with ZnO							
a1	150	30	150	30	150	30	150	30
a	265	50	350	82	490	105	550	120
b ¹⁾	265	50	350	82	490	146	550	160
c1	485	143	545	178	590	208	660	230
c2 ¹⁾	495	150	550	182	590	225	660	250

Table 5 Rated withstand voltages of the internal on-load tap-changer insulation (with the exception of VACUTAP® VM 300)

1) Insulation distance omitted for single-pole on-load tap-changers

The admissible maximum operating voltage on the individual tap selector distances corresponds to half the value of the above mentioned rated short-duration power frequency withstand voltages.



1.4.2 Rated withstand voltages of the internal on-load tap-changer insulation for VACUTAP® VM 300

Insulation distances	VM III 300 Y		VM I 301		VM II 302	
	kV 1.2/ 50 μ s	kV, 50 Hz, 1 min	kV 1.2/ 50 μ s	kV, 50 Hz, 1 min	kV 1,2/50 μ s	kV, 50 Hz, 1 min
a0	Power-frequency voltage test (50 Hz/1 min): 20 kV 50 Hz 1 min; Response voltage with 1 mA leakage current: 35 kV 1.2/50 μ s; 3 kA residual voltage (= lightning impulse protection level): 56 kV 1.2/50 μ s with ZnO					
a	300	70	300	70	300	70
b	300	70	-	-	300	70
c1	400	120	400	120	400	120
c2	400	120	-	-	400	120

Table 6 Rated withstand voltages of the internal on-load tap-changer insulation for VACUTAP® VM 300

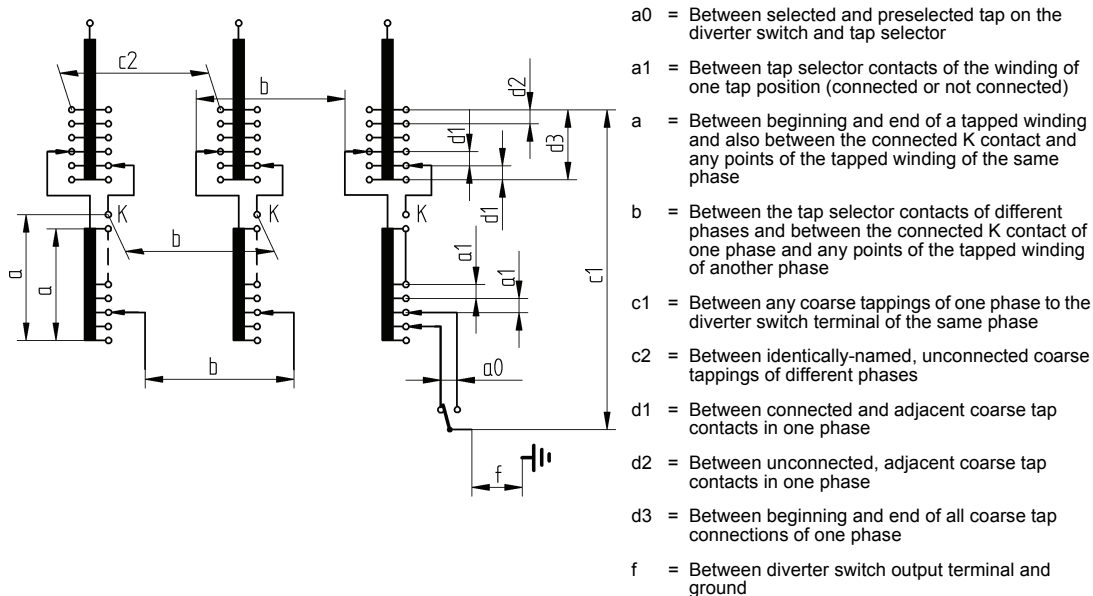
The admissible maximum operating voltage on the individual tap selector distances corresponds to half the value of the above mentioned rated short-duration power frequency withstand voltages.



without change-over selector		with reversing change-over selector		with coarse change-over selector	
Connection	Tap selector size	Connection	Tap selector size	Connection	Tap selector size
10050	B/C/D/DE	10071W	B/C/D/DE	10071G	B/C/D/DE
10060	B/C/D/DE	10081W	B/C/D/DE	10081G	B/C/D/DE
10070	B/C/D/DE	10091W	B/C/D/DE	10091G	B/C/D/DE
10080	B/C/D/DE	12101W	B/C/D/DE	12101G	B/C/D/DE
10090	B/C/D/DE	12111W	B/C	12111G	B/C
10100	B/C/D/DE	14111W	D/DE	14111G	D/DE
12110	B/C/D/DE	14121W	B/C	14121G	B/C
12120	B/C/D/DE	14131W	B/C	14131G	B/C
14130	B/C/D/DE	16121W	D/DE	16121G	D/DE
14140	B/C/D/DE	16131W	D/DE	16131G	D/DE
16150	B/C/D/DE	16141W	B/C/D/DE	16141G	B/C/D/DE
16160	B/C/D/DE	16151W	B/C	16151G	B/C
18170	B/C/D/DE	18151W	D/DE	18151G	D/DE
18180	B/C/D/DE	18161W	B/C	18161G	B/C
22190	B/C/D/DE	18171W	B/C	18171G	B/C
22200	B/C/D/DE	10191W	B/C/D/DE	10191G	B/C/D/DE
22210	B/C	12231W	B/C/D/DE	12231G	B/C/D/DE
22220	B/C	14271W	B/C/D/DE	14271G	B/C/D/DE
		16311W	B/C/D/DE	16311G	B/C/D/DE
		18351W	B/C/D/DE	18351G	B/C/D/DE

Table 7 Available connections (also available as 3 W, 3 G)

1.4.3 VM[®] rated withstand voltages – multiple coarse change-over selector



Insulation distances	Tap selector size B		Tap selector size C		Tap selector size D	
	kV 1.2/ 50 μ s	kV 50 Hz, 1 min	kV 1.2/50 μ s	kV 50 Hz, 1 min	kV 1.2/50 μ s	kV 50 Hz, 1 min
a0	Power-frequency voltage test (50 Hz/60 s): 20 kV 50 Hz 1 min; Response voltage with 1 mA leakage current: 35 kV 1.2/50 μ s; 3 kA residual voltage (= lightning impulse protection level): 56 kV 1.2/50 μ s with ZnO					
a1	150	30	150	30	150	30
a	265	50	350	82	450	105
b ¹⁾	265	50	350	82	450	146
c1	455	127	525	165	590	210
c2 ¹⁾	455	127	525	165	590	215
d1	265	50	350	82	450	105
d2	350	82	450	105	450	105
d3	350	82	450	105	490	120

Table 8 On-load tap-changer VACUTAP[®] VM[®] with multiple coarse change-over selector, rated withstand voltages of the internal on-load tap-changer insulation

1) Insulation distance omitted for single-pole on-load tap-changers

The admissible maximum operating voltage on the individual tap selector distances corresponds to half the value of the above mentioned rated short-duration power frequency withstand voltages.

2 Special designs

2.1 Parallel bridges for parallel connections

Current division on the connection contacts of 2 tap selector planes only for on-load tap-changers VACUTAP® VM I 802 (803)/1002 and of 3 tap selector planes only for on-load tap-changers VACUTAP® VM I 1203/1503, see **Chapter 3.1.37**

Parallel bridges on the tap selector terminals are mandatory if the tap winding has been wound in two or more branches and each of these branch taps is connected to the terminals of the tap selector.

This measure reliably prevents the following:

- a) Introduction of circulating currents into the current paths of tap selector and diverter switch
- b) Commutating arc on movable tap selector contact bridges
- c) Overvoltage between adjacent tap selector connection contacts connected in parallel

On-load tap-changer combination for delta connection

The on-load tap-changers can also be used with the single-phase on-load tap-changer VM I 351 as a two column on-load tap-changer combination VM I 351/ VM II 352 for adjusting the voltage of transformer windings in a delta connection (analogous to VM I 301/VM II 302, VM I 501/VM II 502 and VM I 651/VM II 652). This combination is labeled as VM III K ("K", to represent the German word for combination), see **Figure 4**.

The tap windings should correspond to those in **Figure 4**.

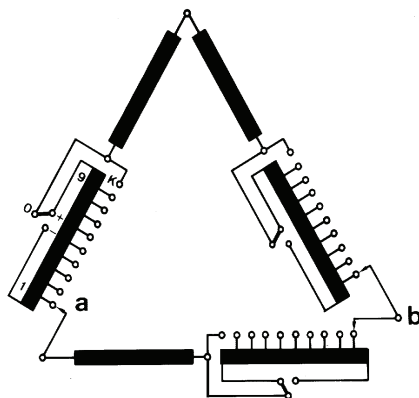


Figure 4

On-load tap-changer combination VM III K for delta connection:
VM I 351/VM II 352 (a – VM I 351, b – VM II 352)



2.2 On-load tap-changer VACUTAP® VM 650 Y...VM I 1503 with multiple coarse change-over selector (up to maximum of 5 coarse tap connections)

Extremely fine voltage adjustment requires a great number of operating positions which sometimes can only be achieved with a multiple coarse tapping arrangement.

For instance, 107 operating positions can be obtained by using a 5-tap coarse winding and a tapped winding with 18 taps.

The multiple coarse change-over selector is attached to both sides of the tap selector.

The on-load tap-changers are available (on request) for $U_m = 72.5$ up to max. 300 kV and for 2 to 5 coarse taps (tap selector sizes B and C) or 2 and 3 coarse tap connections (tap selector size D).

2.3 Two-pole on-load tap-changer VACUTAP® VM II 302/352/502/652

On-load tap-changer VM II 302/352/502/652 can be supplied as a two-pole on-load tap-changer for single-phase center point tap-change operations with the same technical data as on-load tap-changers VM III 350 Y, VM III 500 Y and/or VM III 650 Y (see dimensional drawings, **Chapter 3**).

2.4 On-load tap-changer VACUTAP® VM III 350 Y...VM I 1503 for linear voltage adjustment, up to a maximum of 34 operating positions

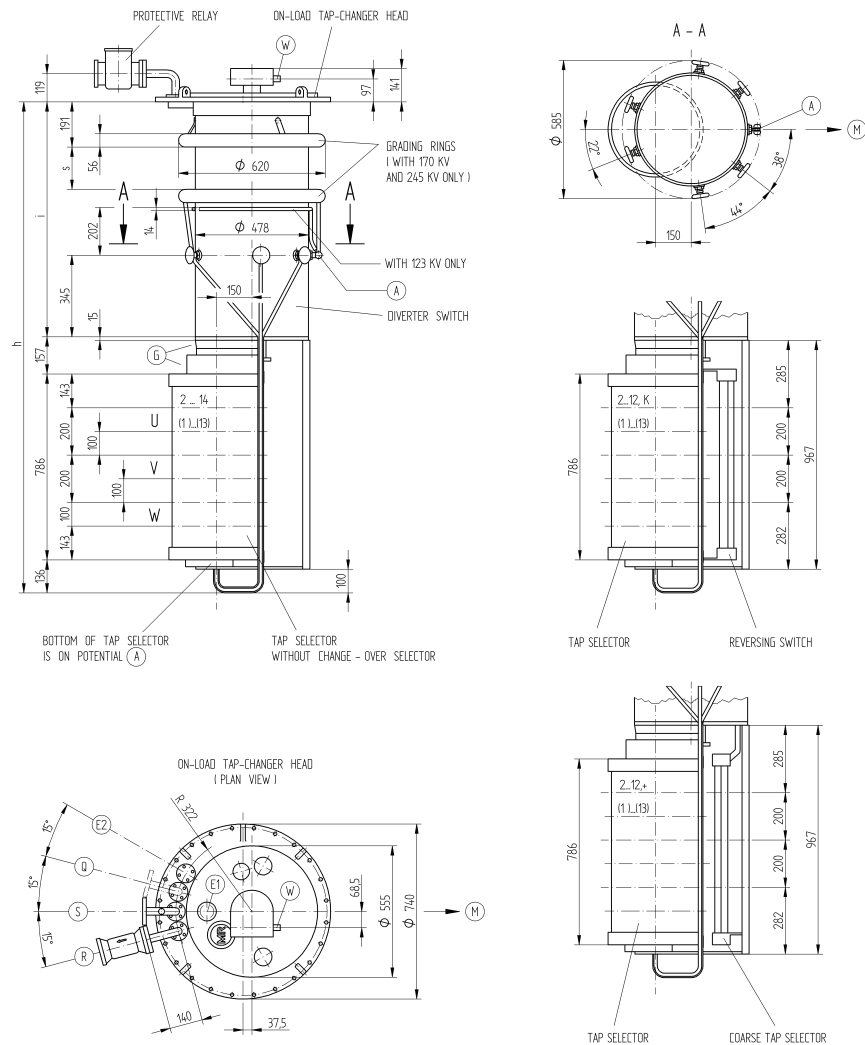
The high number of operating positions without change-over selector is achieved by doubling the tap selector contact planes compared with the standard design (requires a larger installation length).

The on-load tap-changers are available for $U_m = 72.5...300$ kV and several tap selector sizes (rated withstand voltages and dimensional drawings on request).

3 Appendix

3.1 Dimensional drawings/connection diagrams

3.1.1 VACUTAP® VM III 300 Y-0/W/G (768698)



- (A) ON-LOAD TAP-CHANGER CURRENT TAKE-OFF TERMINAL (NEUTRAL)
- (E1) BLEEDING OF ON-LOAD TAP-CHANGER HEAD
- (E2) BLEEDING OF TRANSFORMER TANK
- (G) TAP SELECTOR GEAR (ON POTENTIAL (A))
- (Q) CONNECTING FLANGE FOR OIL RETURN
(WITH FILTER UNIT ONLY)
- (R) CONNECTING FLANGE FOR PROTECTIVE RELAY
- (S) CONNECTING FLANGE FOR SUCTION PIPE
- (W) DRIVE SHAFT
- (M) DRIVE SIDE OF TAP SELECTOR

FOR INHERENT DRAWINGS
REFER TO 898026

FOR BINDING DESIGNATIONS OF TERMINALS AND PHASES
REFER TO THE CONNECTION DIAGRAM OF THE TAP CHANGER

TAP SELECTOR SIZE		B			
U _M [KV]		72.5	123	170	245
DIMENSIONS [MM]	h	1942	2072	2202	2302
	i	863	993	1123	1223
	s	—		267	367
OIL VOLUME [DM ³]		130	150	170	190
DISPLACEMENT [DM ³]		190	220	240	260
MAX. WEIGHT [KG]		280	285	290	295

PROTECTIVE RELAY ON-LOAD TAP-CHANGER HEAD

GRADING RINGS (WITH 170 KV & 245 KV ONLY)

WITH 123 KV ONLY

DIVERTER SWITCH

TAP SELECTOR WITHOUT CHANGE - OVER SELECTOR

2 - 14 (1 L113)

2.12 K (1 L113)

2.12 K (1 L113)

COARSE TAP SELECTOR

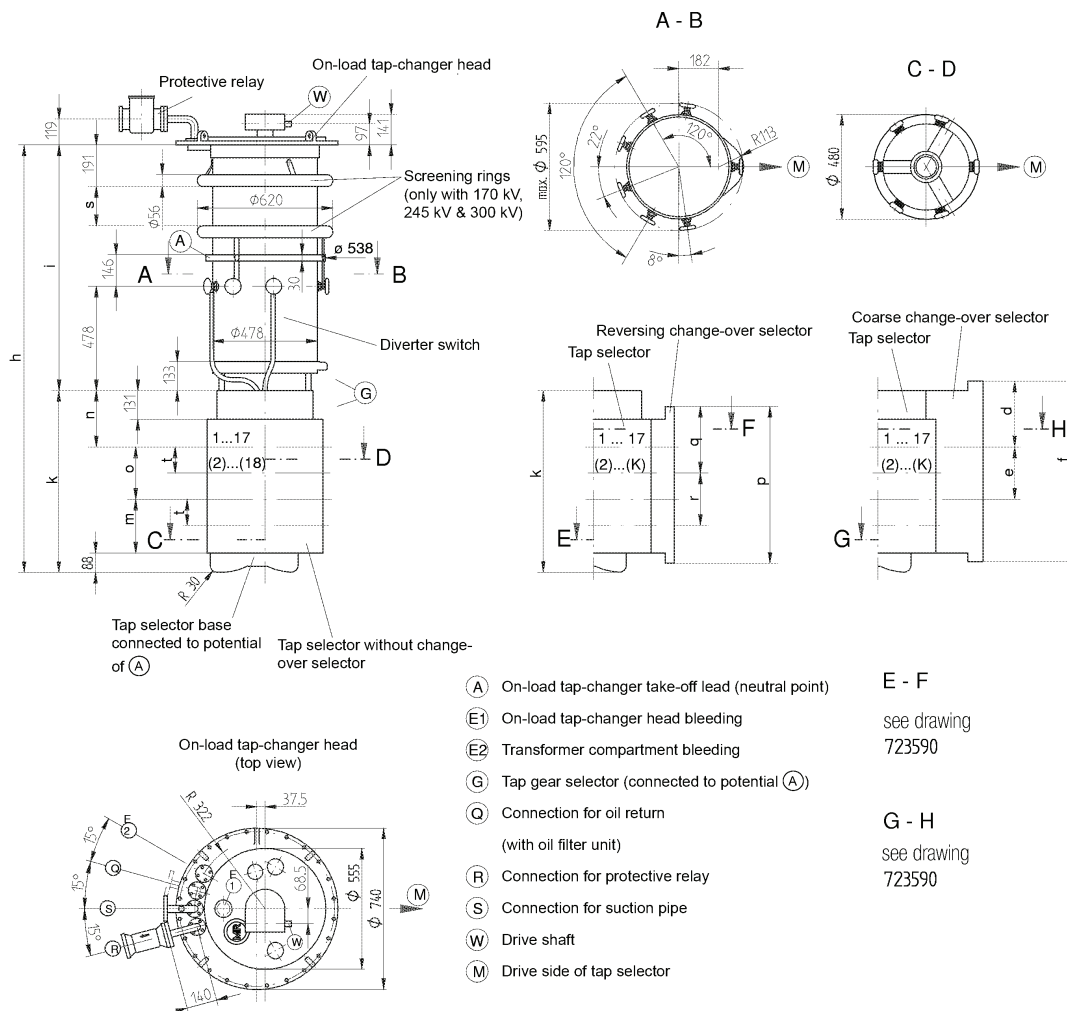
ON-LOAD TAP-CHANGER HEAD (PLAN VIEW)

FOR INHERENT DRAWINGS REFER TO 898026

FOR BINDING DESIGNATIONS OF TERMINALS AND PHASES REFER TO THE CONNECTION DIAGRAM OF THE TAP CHANGER

TAP SELECTOR SIZE		B			
U_M [KV]		72.5	123	170	245
DIMENSIONS [MM]	h	1542	1672	1802	1902
	i	863	993	1123	1223
	s	—		267	367
OIL VOLUME [DM ³]		130	150	170	190
DISPLACEMENT [DM ³]		160	190	210	230
MAX. WEIGHT [KG]		240	245	250	255

3.1.5 VACUTAP® VM II 352/502/652-0/W/G (746220)



For associated drawings,
see: 898012

The connection diagram is binding for the designation of connection terminals and phases

Tap selector size		B					C					D/DE				
U _m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1704	1834	1964	2064	2216	1829	1959	2089	2189	2341	2154	2284	2414	2514	2666
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	-		267	367	520	-		267	367	520	-		267	367	520
	k	708					833					1158				
	n	233					258					323				
	o	190					240					370				
	m	197					247					377				
	t	95					120					185				
	r	190					240					370				
	q	255					305					435				
	p	593					718					1043				
	e	190					240					370				
	d	276.5					301.5					366.5				
	f	702					827					1152				
Oil content [dm³]		130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement [dm³]		196	221	241	261	281	196	221	241	261	281	199	224	244	264	284
Max. weight [kg]		310	315	320	325	330	320	325	330	335	340	330	335	340	345	350

A - B

C - D

E - F

G - H

Protective relay
On-load tap-changer head
Screening rings (only with 170 kV, 245 kV & 300 kV)
only with 123 kV
Diverter switch
Tap selector base connected to potential of (A)
Tap selector without change-over selector
Reversing change-over selector Tap selector
Coarse change-over selector Tap selector

(A) On-load tap-changer take-off lead (neutral point)
(E1) On-load tap-changer head bleeding
(E2) Transformer compartment bleeding
(G) Tap gear selector (connected to potential (A))
(Q) Connection for oil return (with oil filter unit)
(R) Connection for protective relay
(S) Connection for suction pipe
(W) Drive shaft
(M) Drive side of tap selector
(P) connected to potential of (A)

For associated drawings, see: 898012

The connection diagram is binding for the designation of connection terminals and phases

A - B

C - D

E - F

G - H

On-load tap-changer head (top view)

Labels: Protective relay, On-load tap-changer head, Screening rings (only with 170 kV, 245 kV & 300 kV), Divertor switch, Tap selector without change-over selector, Tap selector base connected to potential of (A).

Dimensions: h, i, s, A, B, C, D, E, F, G, H, K, L, M, N, O, P, Q, R, S, W, X, Y, Z.

Legend:

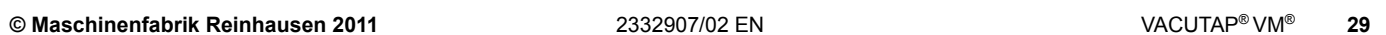
- (A) On-load tap-changer take-off lead (neutral point)
- (E1) On-load tap-changer head bleeding
- (E2) Transformer compartment bleeding
- (G) Tap gear selector (connected to potential (A))
- (Q) Connection for oil return (with oil filter unit)
- (R) Connection for protective relay
- (S) Connection for suction pipe
- (W) Drive shaft
- (M) Drive side of tap selector

For associated drawings,
see: 898012

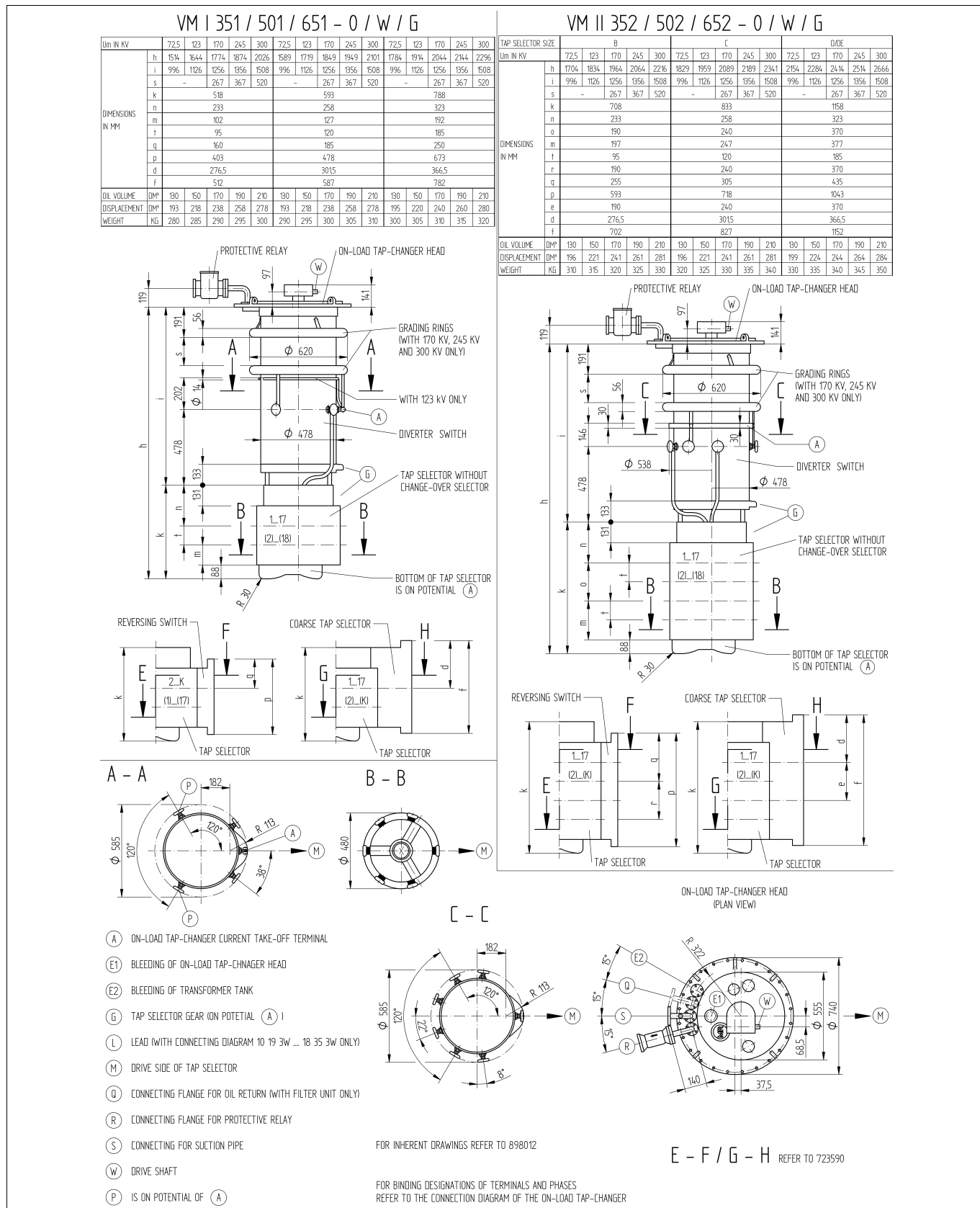
The connection diagram is binding for the designation of
connection terminals and phases

Tap selector size		B					C					D/DE				
U_m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	-	-	267	367	520	-	-	267	367	520	-	-	267	367	520
	k	-	-	728	-	-	-	-	803	-	-	-	-	998	-	-
	n	-	-	233	-	-	-	-	258	-	-	-	-	323	-	-
	o	-	-	95	-	-	-	-	120	-	-	-	-	185	-	-
	m	-	-	102	-	-	-	-	127	-	-	-	-	192	-	-
	t	-	-	105	-	-	-	-	105	-	-	-	-	105	-	-
	r	-	-	105	-	-	-	-	105	-	-	-	-	105	-	-
	q	-	-	160	-	-	-	-	185	-	-	-	-	250	-	-
	p	-	-	613	-	-	-	-	688	-	-	-	-	883	-	-
	e	-	-	105	-	-	-	-	105	-	-	-	-	105	-	-
	d	-	-	276.5	-	-	-	-	301.5	-	-	-	-	366.5	-	-
	f	-	-	722	-	-	-	-	797	-	-	-	-	992	-	-
		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Oil content [dm ³]		130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement [dm ³]		196	221	241	261	281	196	221	241	261	281	199	224	244	264	284
Max. weight [kg]		310	315	320	325	330	320	325	330	335	340	330	335	340	345	350

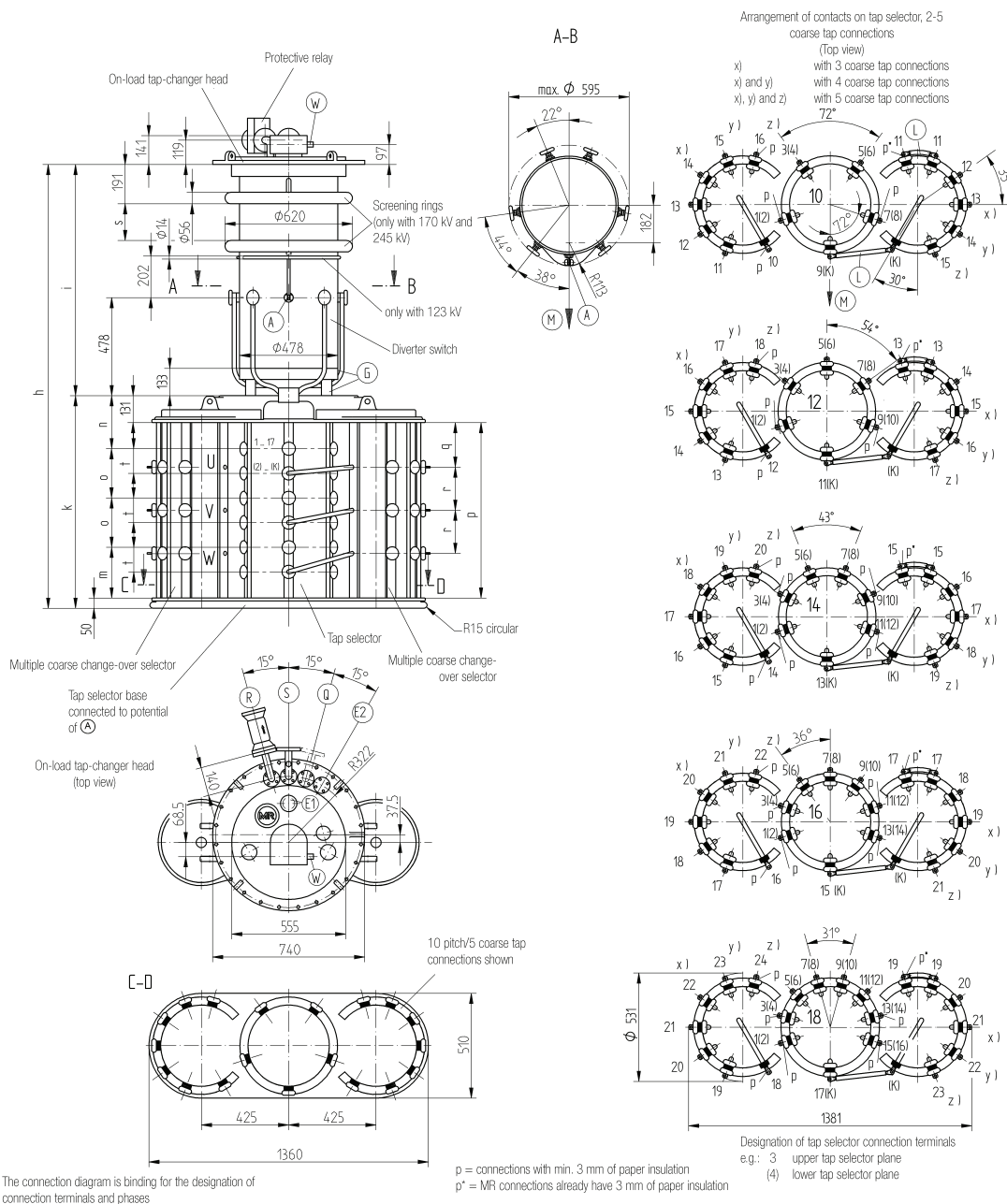
3.1.9 VACUTAP® VM III 300 K-0/W/G (768851)



3.1.10

VACUTAP® VM III K-0/W/G (746224)


3.1.11 VACUTAP® VM III 650 Y with multiple coarse change-over selector (746226)

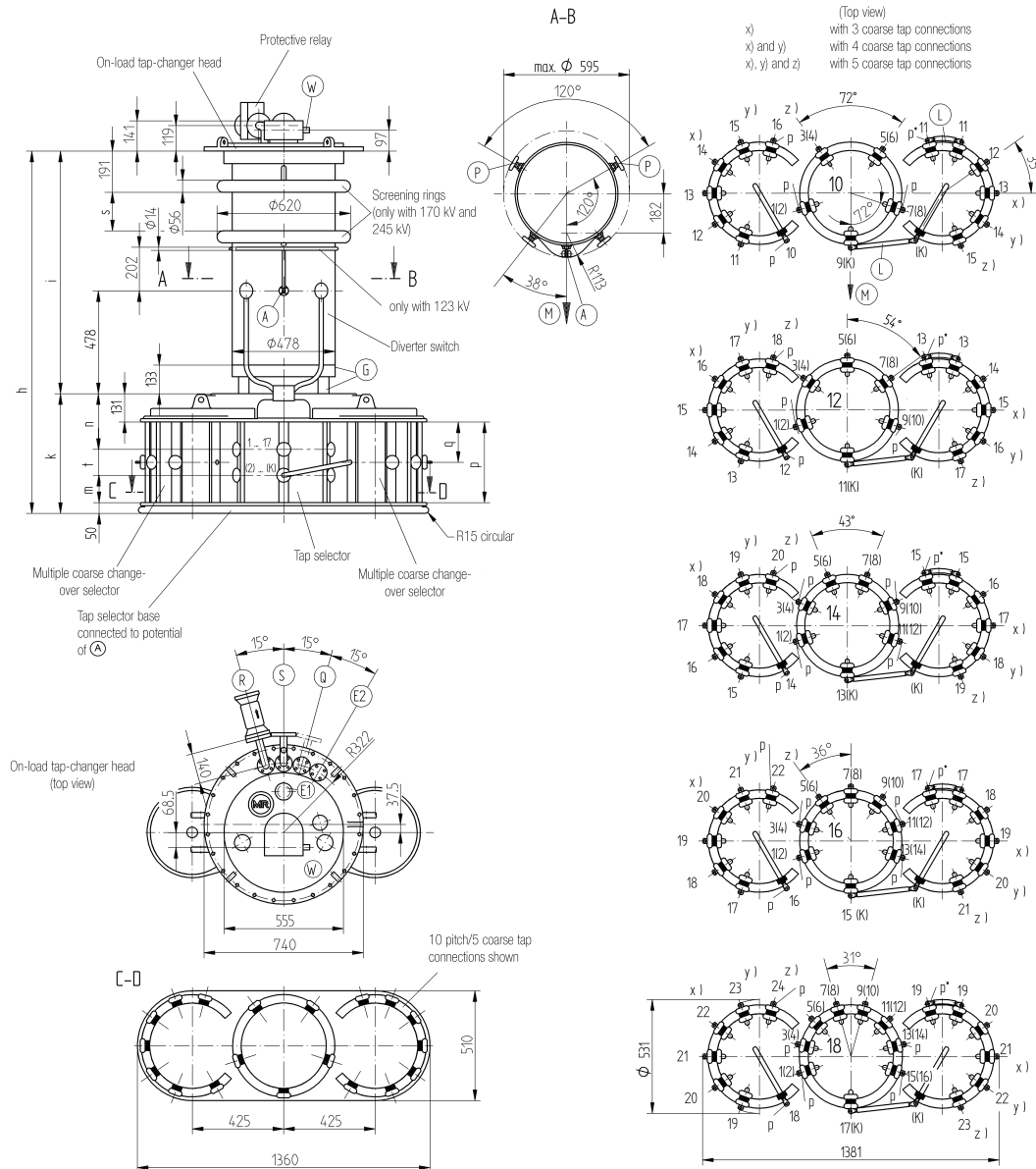


The connection diagram is binding for the designation of connection terminals and phases

- (A) On-load tap-changer take-off lead
- (E1) On-load tap-changer head bleeding
- (E2) Transformer compartment bleeding
- (G) Tap selector gear (connected to potential (A))
- (Q) Connection for oil return (with oil filter unit)
- (R) Connection for protective relay
- (S) Connection for suction pipe
- (W) Drive shaft
- (L) Connecting lead
- (M) Drive side of the tap selector

Tap selector size		B				C				D			
U_m [kV]		72.5	123	170	245	72.5	123	170	245	72.5	123	170	245
Dimensions [mm]	h	1856	1986	2116	2216	2031	2161	2291	2391	2486	2616	2746	2846
	i	996	1126	1256	1356	996	1126	1256	1356	996	1126	1256	1356
	s	—	—	267	367	—	—	267	367	—	—	267	367
	k	—	860	—	—	—	1035	—	—	—	1490	—	—
	n	—	233	—	—	—	258	—	—	—	323	—	—
	o	—	190	—	—	—	240	—	—	—	370	—	—
	m	—	197	—	—	—	247	—	—	—	377	—	—
	t	—	95	—	—	—	120	—	—	—	185	—	—
	r	—	166,5	—	—	—	210	—	—	—	327	—	—
	q	—	173	—	—	—	217	—	—	—	327,5	—	—
	p	—	679	—	—	—	854	—	—	—	1309	—	—
Oil content [dm ³]		130	150	170	190	130	150	170	190	130	150	170	190
Displacement [dm ³]		321	346	366	386	321	346	366	386	333	353	375	395
Max. weight [kg]		460	465	470	475	470	475	480	485	485	490	495	500

3.1.12 VACUTAP® VM I 651 with multiple coarse change-over selector (746227)



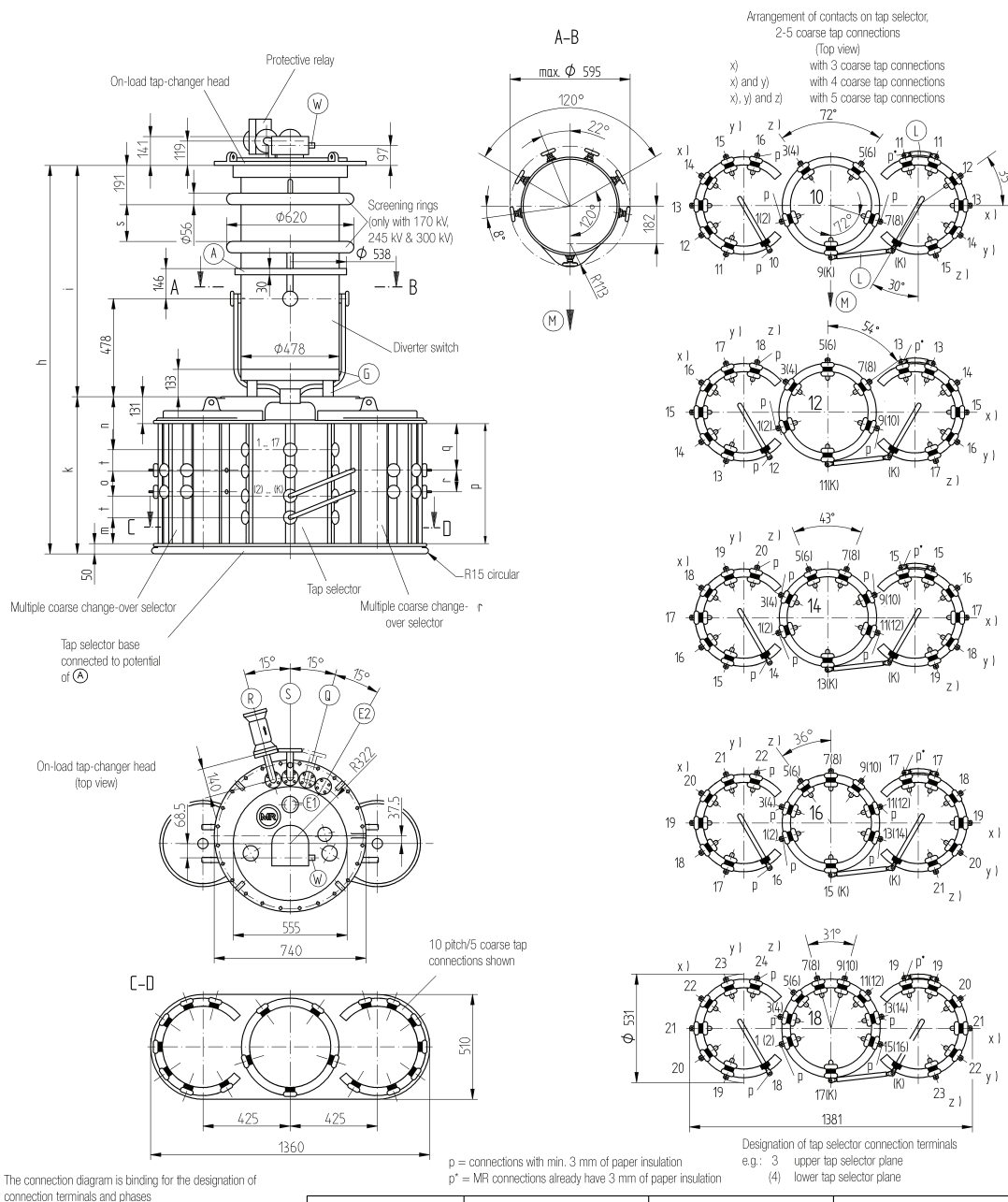
The connection diagram is binding for the designation of connection terminals and phases

- (A) On-load tap-changer take-off lead
- (E1) On-load tap-changer head bleeding
- (E2) Transformer compartment bleeding
- (G) Tap selector gear (connected to potential (A))
- (Q) Connection for oil return (with oil filter unit)
- (R) Connection for protective relay
- (S) Connection for suction pipe
- (W) Drive shaft
- (L) Connecting lead
- (M) Drive side of the tap selector
- (P) connected to potential (A)

p = connections with min. 3 mm of paper insulation
p* = MR connections already have 3 mm of paper insulation

Tap selector size		B					C					D				
U _m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1476	1606	1736	1836	1988	1551	1681	1811	1911	2063	1746	1876	2006	2106	2258
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	—	267	367	520	—	—	267	367	520	—	—	267	367	520	—
	k	—	—	480	—	—	—	—	555	—	—	—	—	750	—	—
	n	—	—	233	—	—	—	—	258	—	—	—	—	323	—	—
	m	—	—	102	—	—	—	—	127	—	—	—	—	192	—	—
	t	—	—	95	—	—	—	—	120	—	—	—	—	185	—	—
	q	—	—	149,5	—	—	—	—	187	—	—	—	—	284,5	—	—
	p	—	—	299	—	—	—	—	374	—	—	—	—	569	—	—
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Oil content [dm ³]		130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
Displacement [dm ³]		198	223	243	263	283	198	223	243	263	283	198	223	243	263	283
Max. weight [kg]		370	375	380	385	390	380	385	390	395	400	390	395	400	405	410

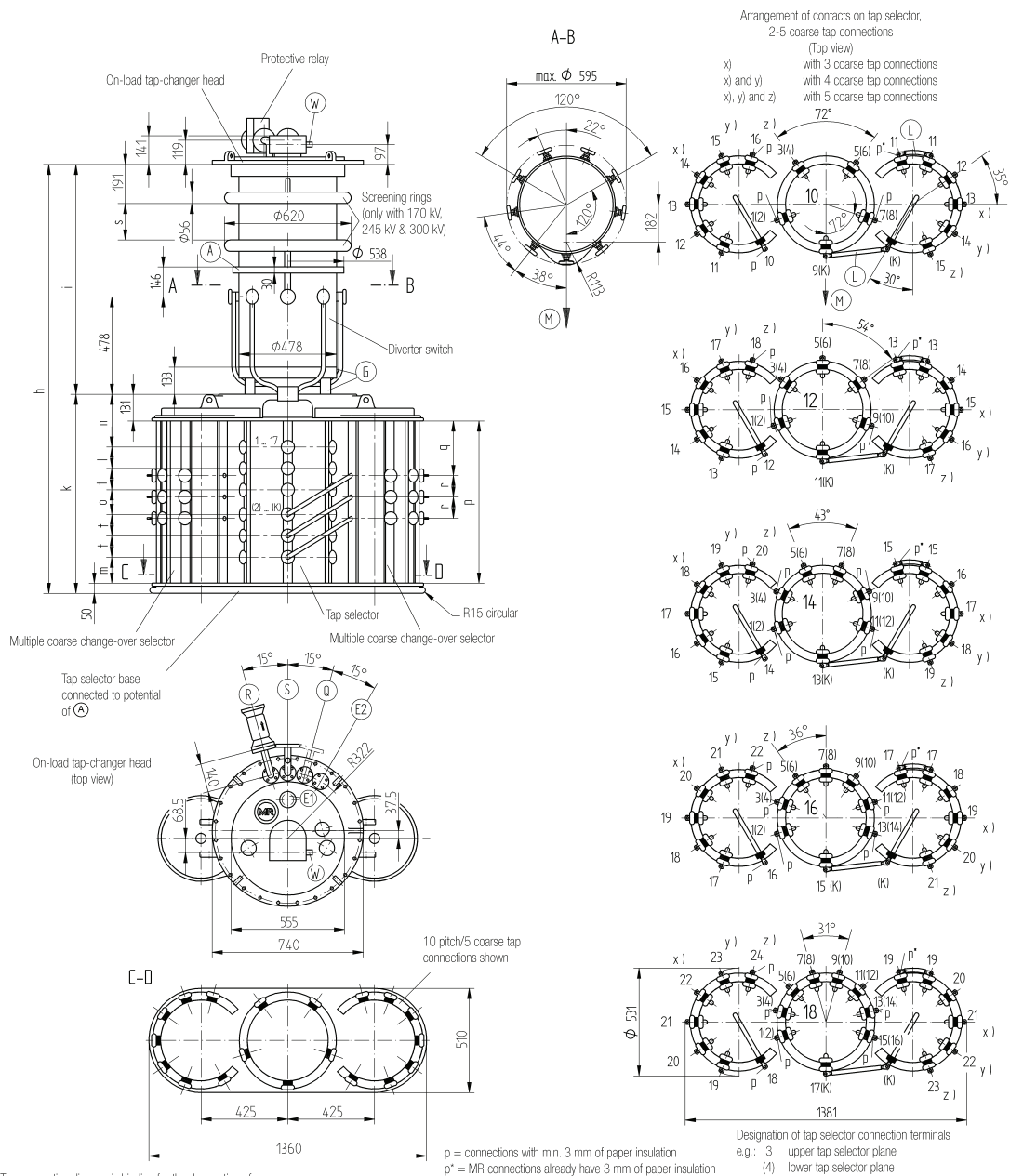
3.1.13 VACUTAP® VM I 802/1002 with multiple coarse change-over selector (746228)



- (A) On-load tap-changer take-off lead
- (E1) On-load tap-changer head bleeding
- (E2) Transformer compartment bleeding
- (B) Tap selector gear (connected to potential (A))
- (U) Connection for oil return (with oil filter unit)
- (R) Connection for protective relay
- (S) Connection for suction pipe
- (W) Drive shaft
- (L) Connecting lead
- (M) Drive side of the tap selector

Tap selector size		B					C					D				
U _m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1686	1816	1946	2046	2198	1761	1891	2021	2121	2273	1956	2086	2216	2316	2468
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	—	267	367	520	—	—	267	367	520	—	—	267	367	520	—
	k	—	—	690	—	—	—	—	765	—	—	—	—	960	—	—
	n	—	—	233	—	—	—	—	258	—	—	—	—	323	—	—
	o	—	—	95	—	—	—	—	120	—	—	—	—	185	—	—
	m	—	—	102	—	—	—	—	127	—	—	—	—	192	—	—
	t	—	—	105	—	—	—	—	105	—	—	—	—	105	—	—
	r	—	—	105	—	—	—	—	105	—	—	—	—	105	—	—
	q	—	—	189,5	—	—	—	—	227	—	—	—	—	324,5	—	—
	p	—	—	509	—	—	—	—	584	—	—	—	—	779	—	—
	Oil content [dm ³]	130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
	Displacement [dm ³]	200	225	245	265	285	200	225	245	265	285	208	233	253	273	293
	Max. weight [kg]	410	415	420	425	430	420	425	430	435	440	430	435	440	445	450

3.1.14 VACUTAP® VM I 1203/1503 with multiple coarse change-over selector (746229)

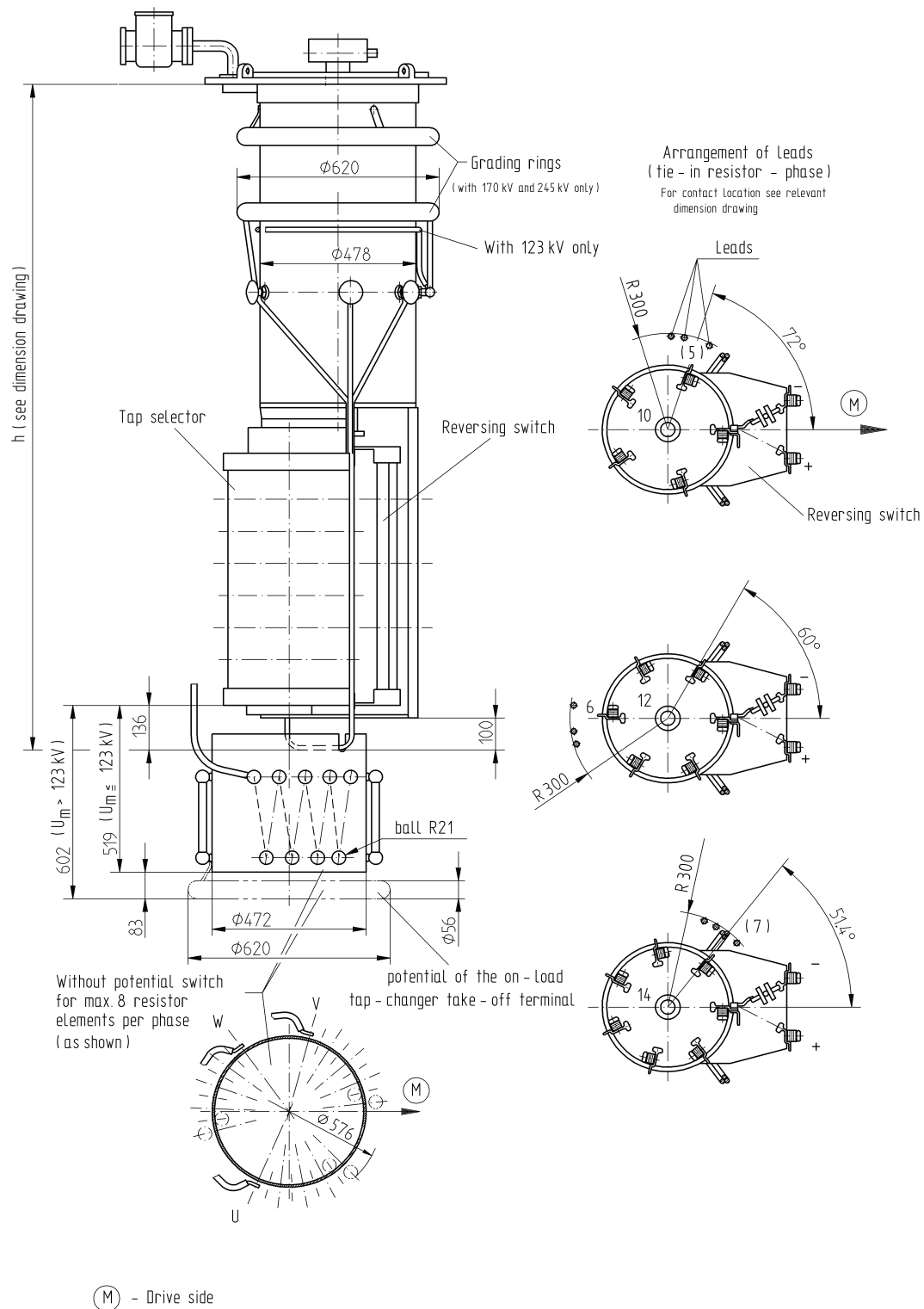


The connection diagram is binding for the designation of connection terminals and phases

- (A) On-load tap-changer take-off lead
- (E1) On-load tap-changer head bleeding
- (E2) Transformer compartment bleeding
- (G) Tap selector gear (connected to potential (A))
- (Q) Connection for oil return (with oil filter unit)
- (R) Connection for protective relay
- (S) Connection for suction pipe
- (W) Drive shaft
- (L) Connecting lead
- (M) Drive side of the tap selector

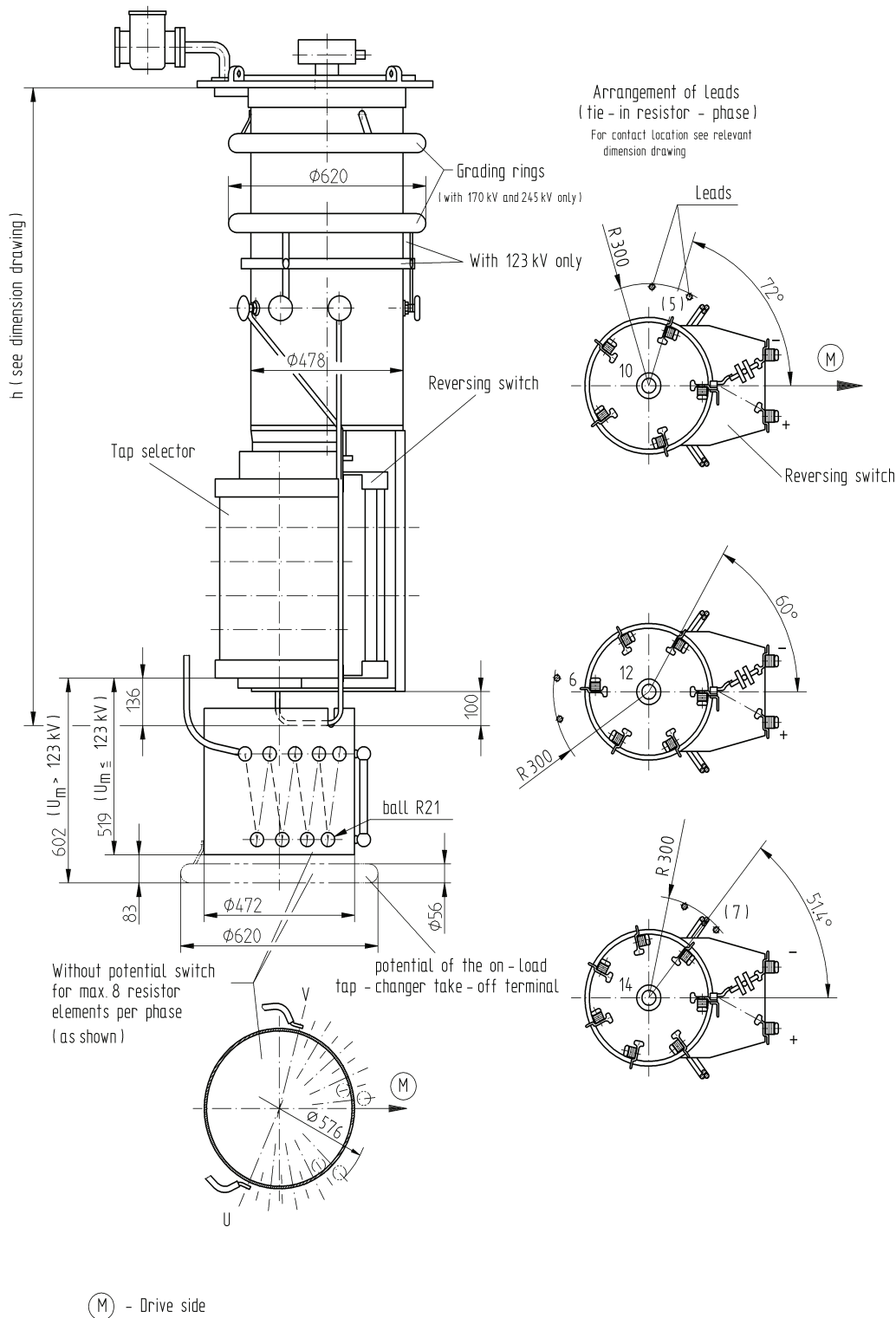
Tap selector size		B					C					D				
U _m [kV]		72,5	123	170	245	300	72,5	123	170	245	300	72,5	123	170	245	300
Dimensions [mm]	h	1896	2026	2156	2256	2408	1971	2101	2231	2331	2483	2166	2296	2426	2526	2678
	i	996	1126	1256	1356	1508	996	1126	1256	1356	1508	996	1126	1256	1356	1508
	s	—	—	267	367	520	—	—	267	367	520	—	—	267	367	520
	k	—	—	900	—	—	—	—	975	—	—	—	—	1170	—	—
	n	—	—	233	—	—	—	—	258	—	—	—	—	323	—	—
	o	—	—	95	—	—	—	—	120	—	—	—	—	185	—	—
	m	—	—	102	—	—	—	—	127	—	—	—	—	192	—	—
	t	—	—	105	—	—	—	—	105	—	—	—	—	105	—	—
	r	—	—	105	—	—	—	—	105	—	—	—	—	105	—	—
	q	—	—	229,5	—	—	—	—	267	—	—	—	—	364,5	—	—
	p	—	—	719	—	—	—	—	794	—	—	—	—	989	—	—
	Oil content [dm ³]	130	150	170	190	210	130	150	170	190	210	130	150	170	190	210
	Displacement [dm ³]	320	345	365	385	405	320	345	365	385	405	320	345	365	385	405
	Max. weight [kg]	460	465	470	475	480	470	475	480	485	490	485	490	495	500	505

3.1.15 VACUTAP® VM III 300 - tie-in resistors without potential switch (898695)



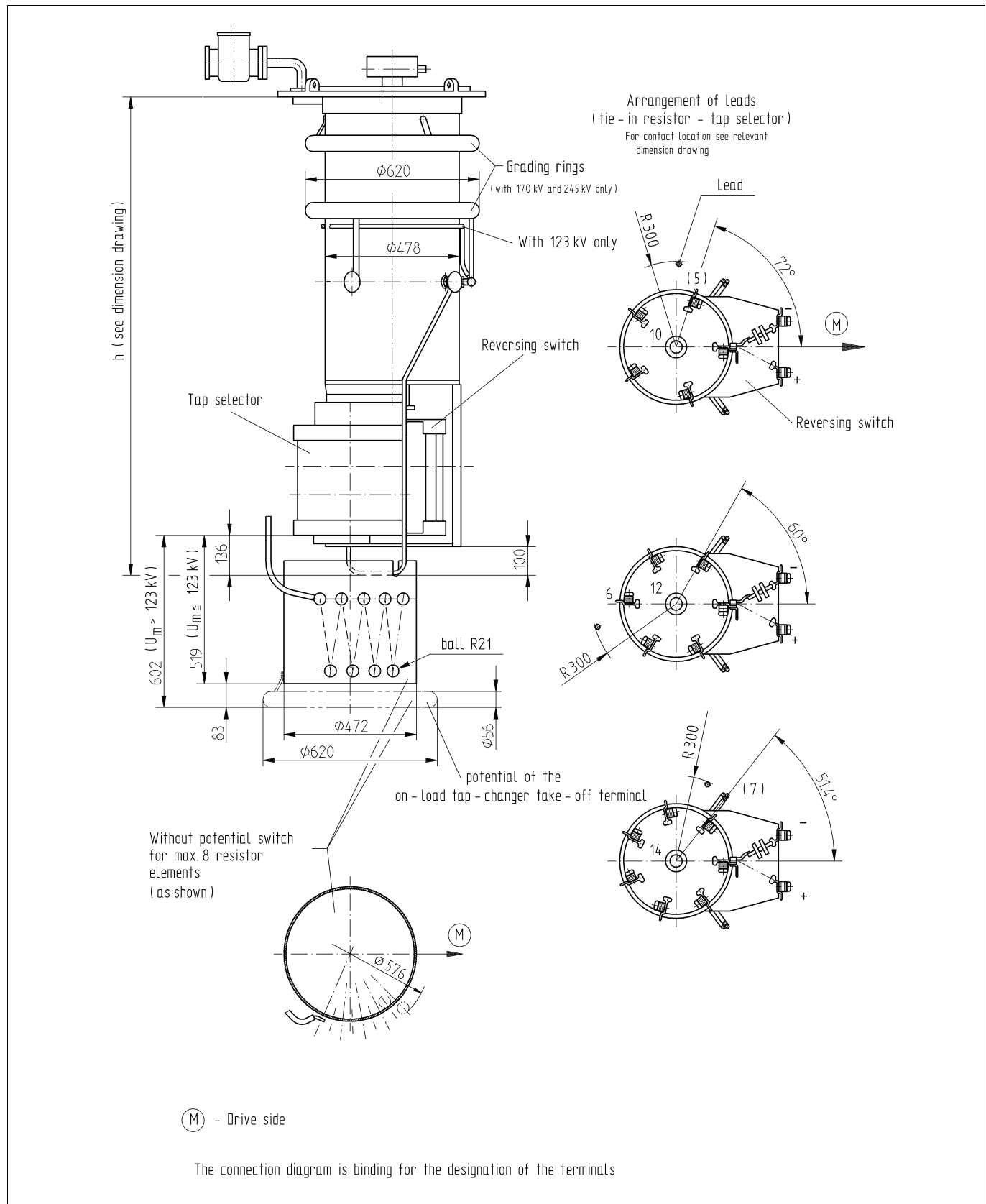
The connection diagram is binding for the designation of the terminals and phases

3.1.16 VACUTAP® VM II 302 - tie-in resistors without potential switch (898694)

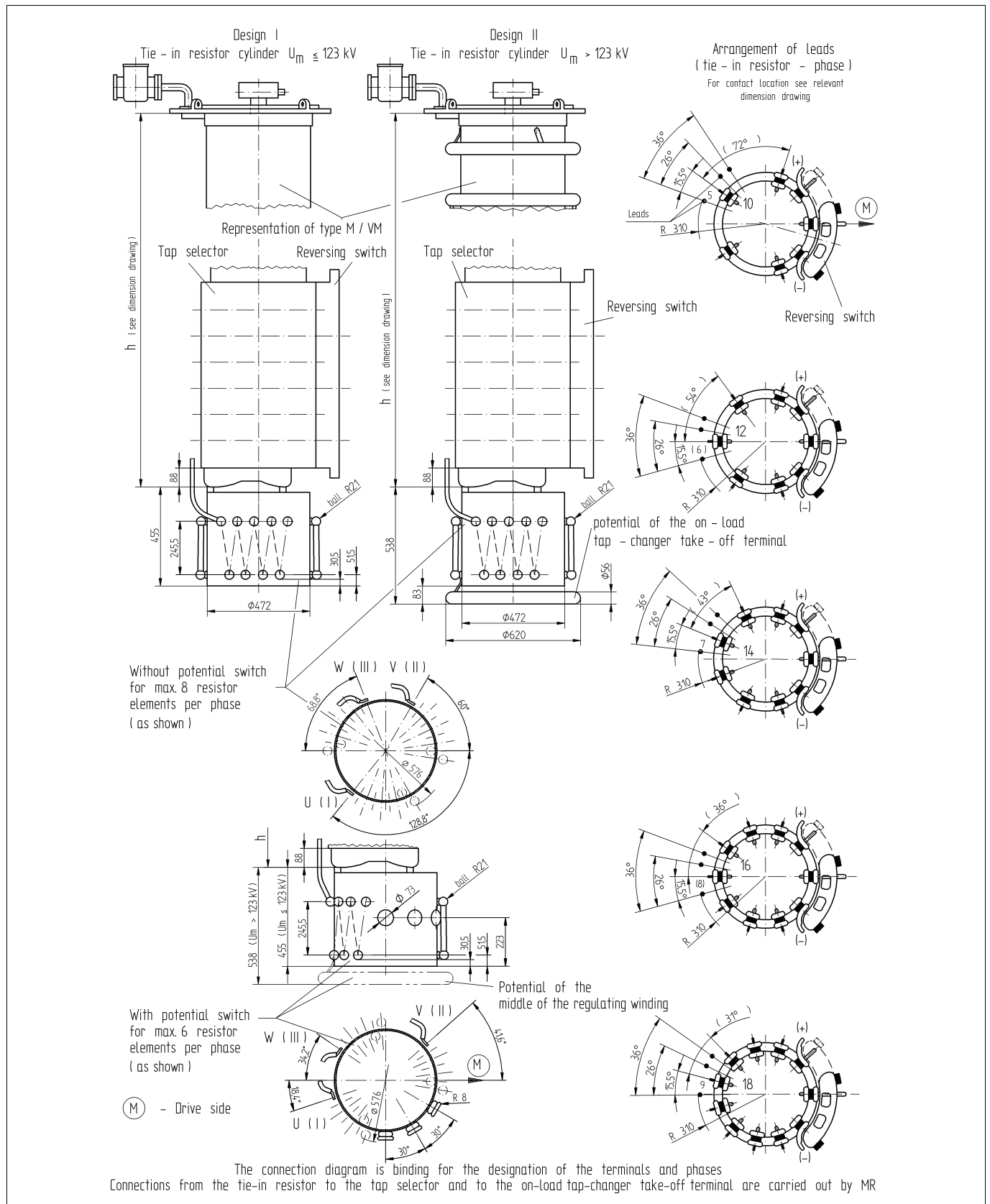


The connection diagram is binding for the designation of the terminals and phases

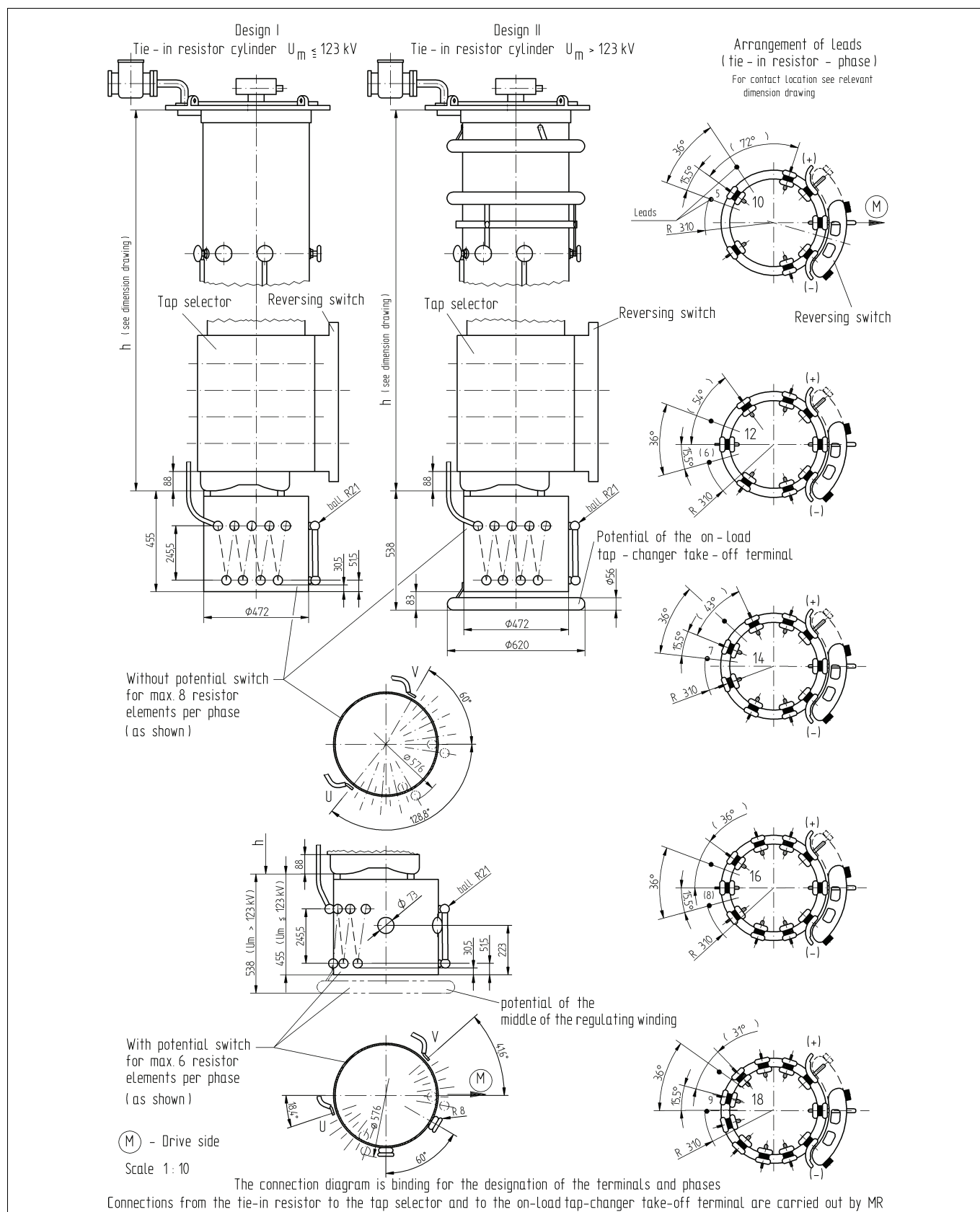
3.1.17 VACUTAP® VM I 301 - tie-in resistors without potential switch (898693)



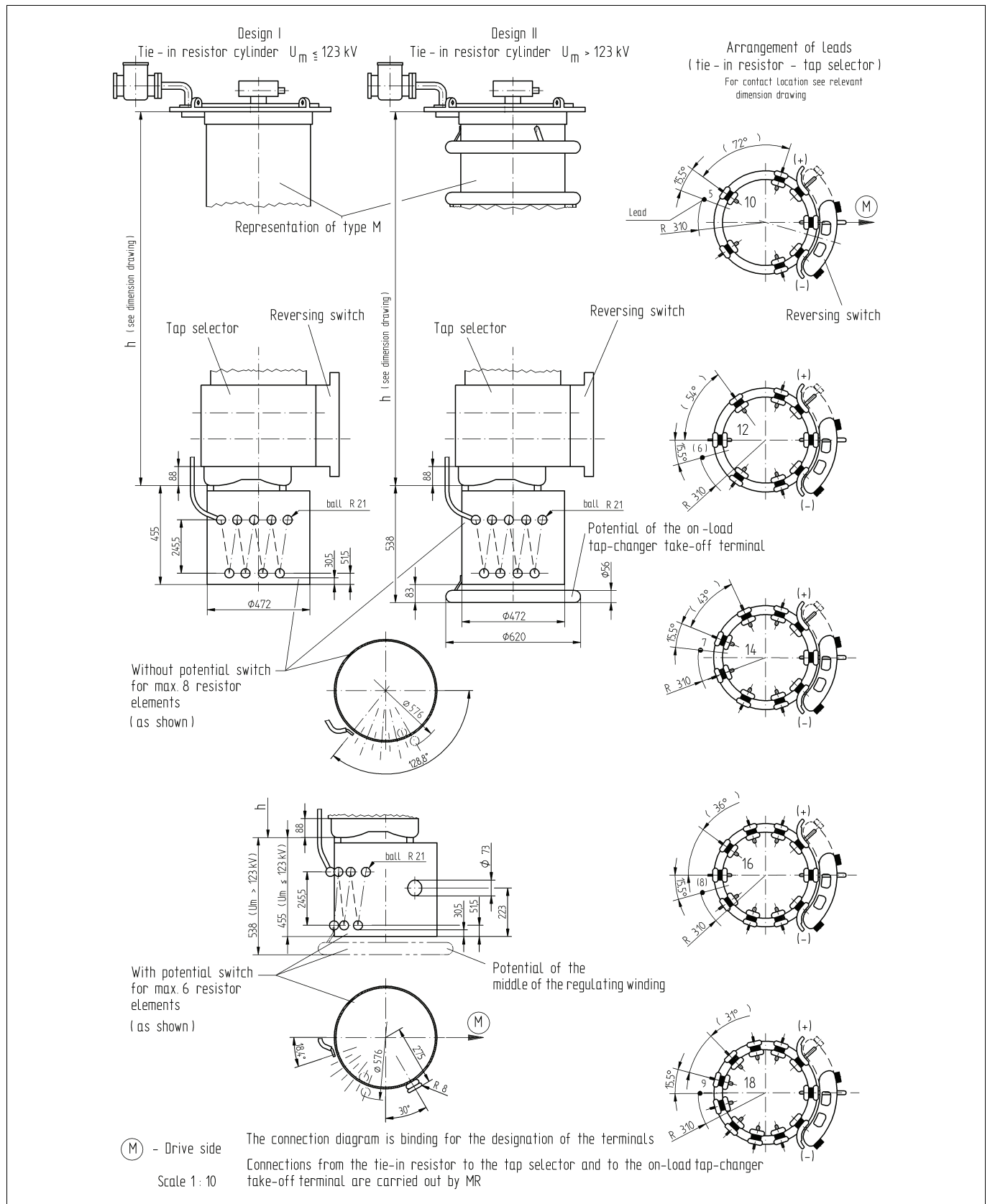
3.1.18 VACUTAP® VM III Y - tie-in resistors with/without potential switch (898692)



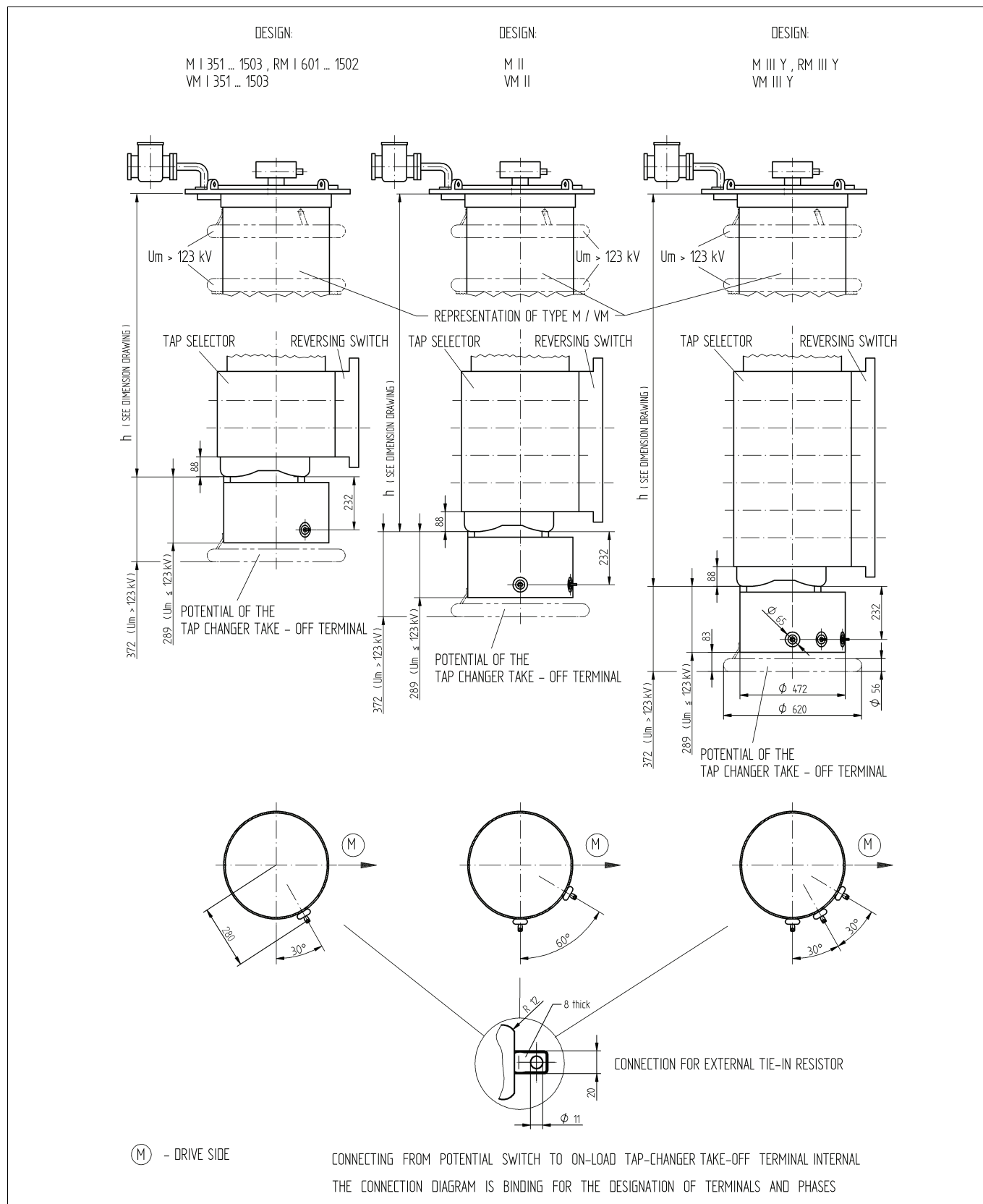
3.1.19 VACUTAP® VM II - tie-in resistors with/without potential switch (898691)



3.1.20 VACUTAP® VM I - tie-in resistors with/without potential switch (898690)



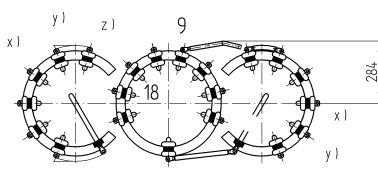
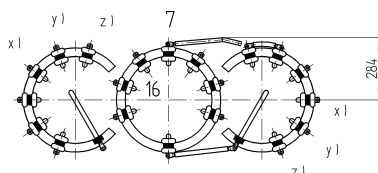
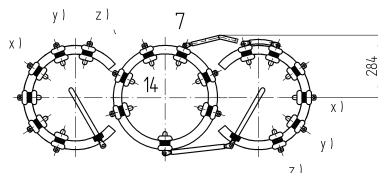
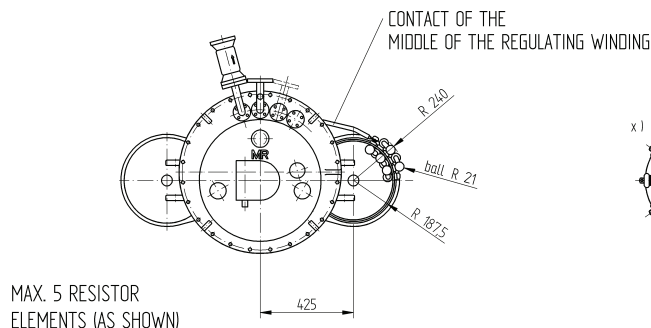
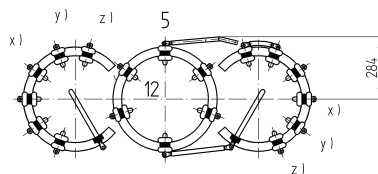
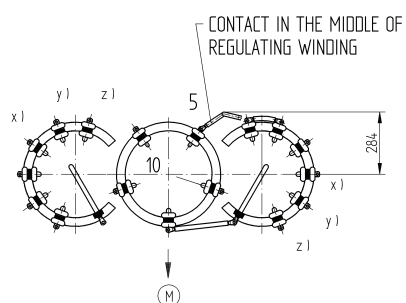
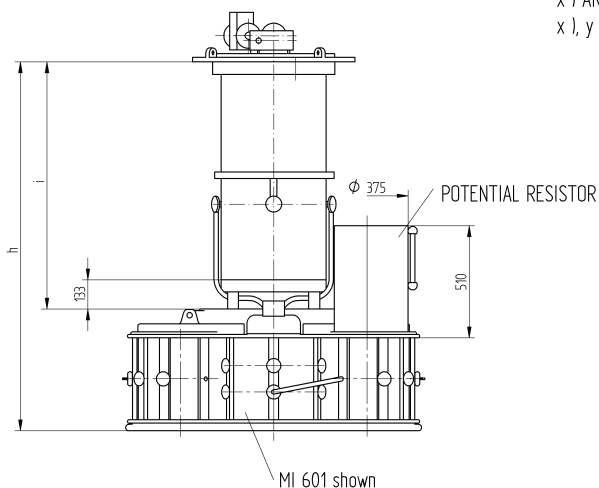
3.1.21 VACUTAP® VM I 351...1503, tie-in resistor cylinder with potential switch without tie-in resistors (898804)



3.1.22 VACUTAP® VM I 651...1503 with multiple coarse change-over selector (719733)

ARRANGEMENT OF TAP SELECTOR CONTACTS, 2-5 COARSE TAPS
(PLAN VIEW)

x) FOR 3 COARSE TAPS
x) AND y) FOR 4 COARSE TAPS
x), y) AND z) FOR 5 COARSE TAPS



THE CONNECTION DIAGRAM IS BINDING FOR THE
DESIGNATION OF THE TERMINALS AND PHASES

(M) DRIVE SIDE



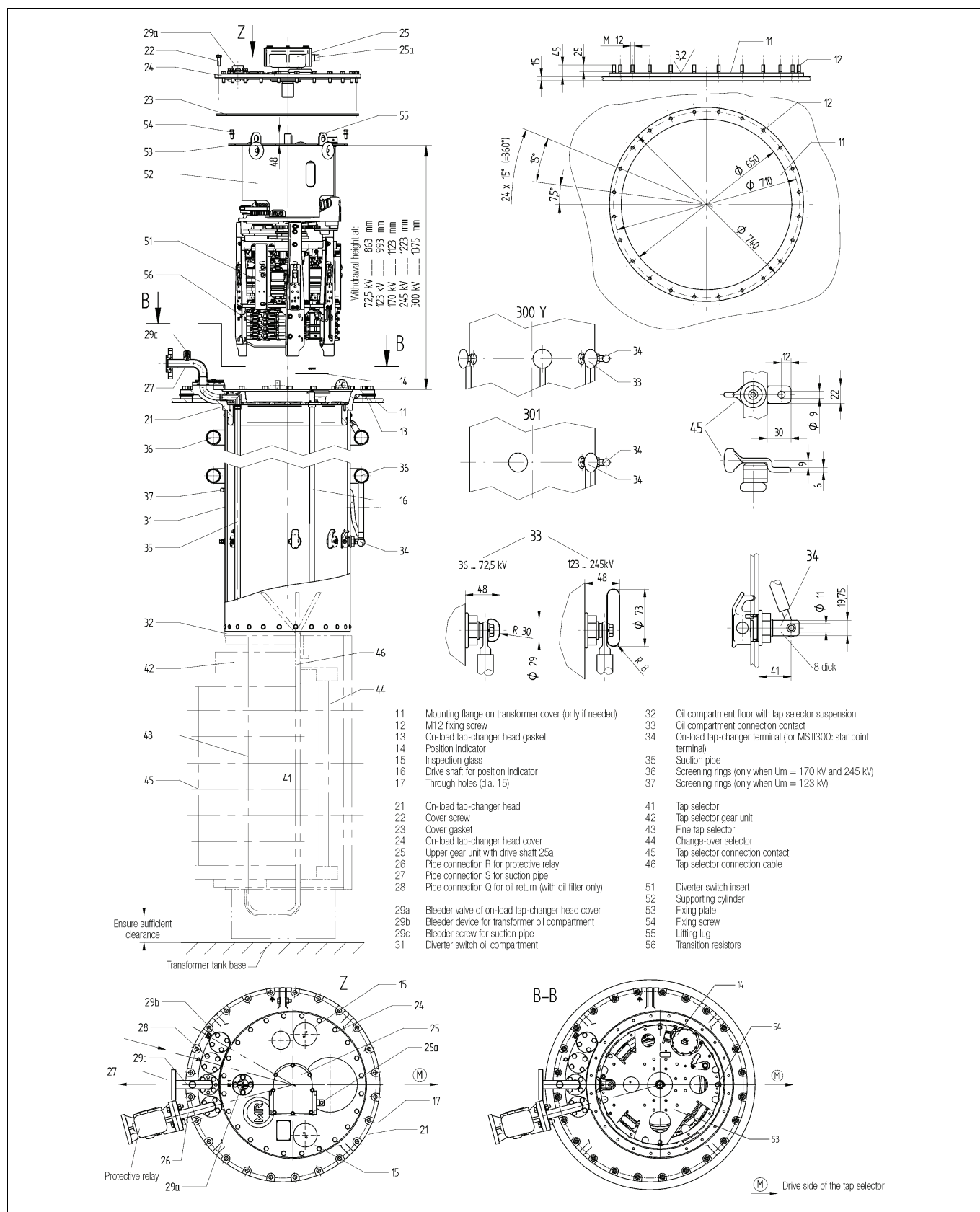
3.1.23 VACUTAP® VM 300, associated drawings (898026)

CONTACT ARRANGEMENT ON TAP SELECTOR	_____898 041
BASIC CONNECTION DIAGRAMS	_____893 819
ARRANGEMENT OF TAP SELECTOR CONTACTS	_____891 114
ON-LOAD TAP-CHANGER HEAD	_____893 899
VARIANTS OF ON-LOAD TAP-CHANGER HEAD	_____720 026
INSTALLATION DRAWING	
MS	_____893 900
VM 300	_____765 192
HORIZONTAL DRIVE SHAFT	_____893 896

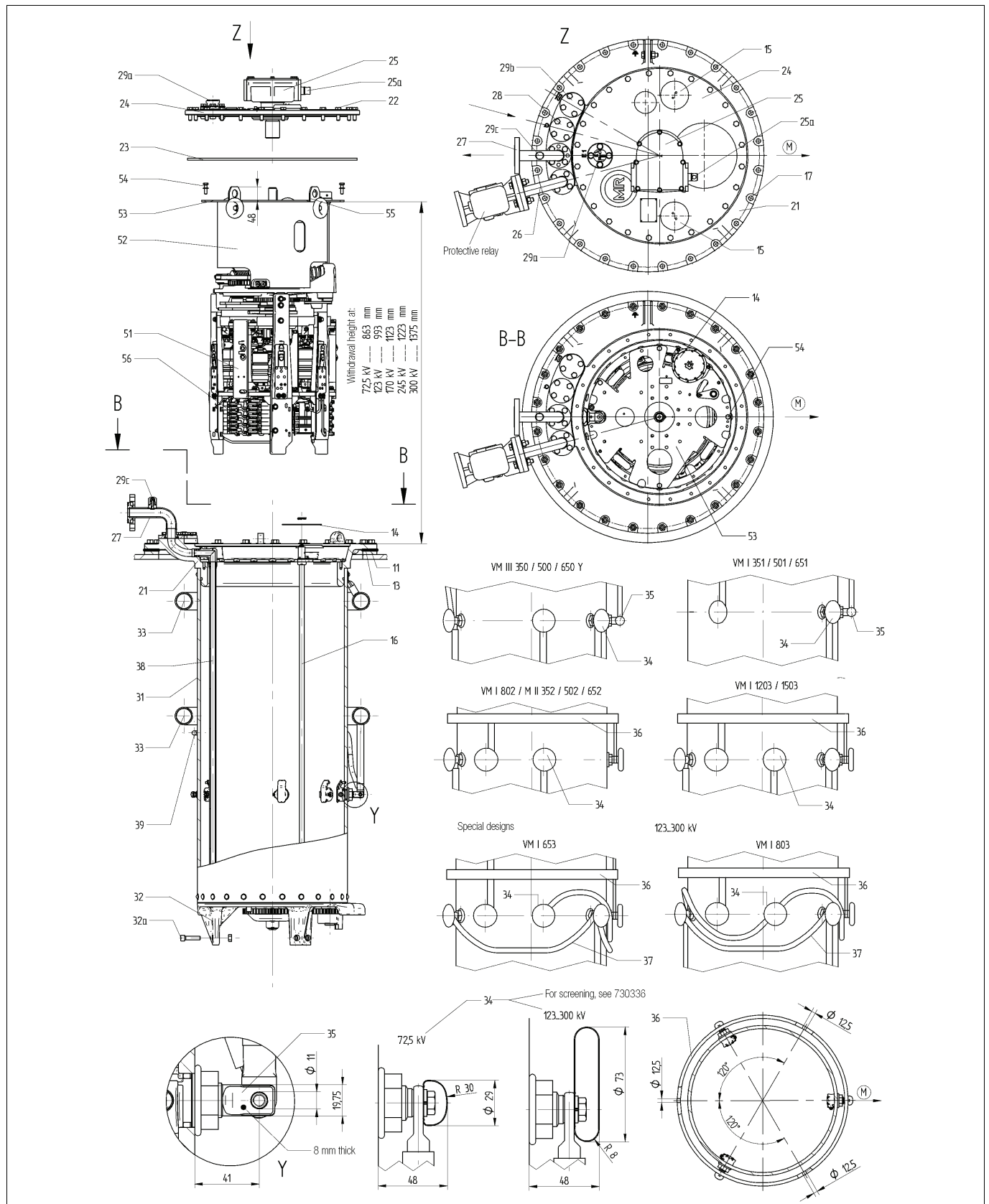
**3.1.24 VACUTAP® VM®, associated drawings (898012)**

BASIC CONNECTION DIAGRAMS	890 616
INSTALLATION DRAWINGS	
M III 350 Y ... M I 1503	893 978
VM III 350 Y ... VM I 1503	746 230
ON-LOAD TAP-CHANGER HEAD M III 350 Y ... M I 1503	
AND VM III 350 Y ... VM I 1503	893 899
VARIANTS OF THE ON-LOAD TAP-CHANGER HEAD	720 026
SWIVELLING RANGE OF THE GEAR UNIT	720 027
SCREENINGS ON OIL COMPARTMENT TERMINALS	730 336
CONTACT ARRANGEMENT ON TAP SELECTOR	
898 013	
MOUNTING POSITION OF THE TAP SELECTOR TERMINALS _ _	890 477
CONNECTING LEAD 3W, 1G, 3G	723 590
SCREENINGS AT TAP SELECTOR	
AND CHANGE- OVER SELECTOR	730 335
PARALLEL BRIDGES	899 598
HORIZONTAL DRIVE SHAFT	893 896
ARRANGEMENT OF TAP SELECTOR CONTACTS	
M III 350 / 500 / 600 Y , M II 352 / 502 / 602	
VM III 350 / 500 / 650 Y , VM II 352 / 502 / 652 _ _ _	891 107
M I 351 / 501 / 601	
VM I 351 / 501 / 651	891 108
M I 1203 / 1503	
VM I 1203 / 1503	891 109
M I 802	
VM I 802 / 1002	891 110

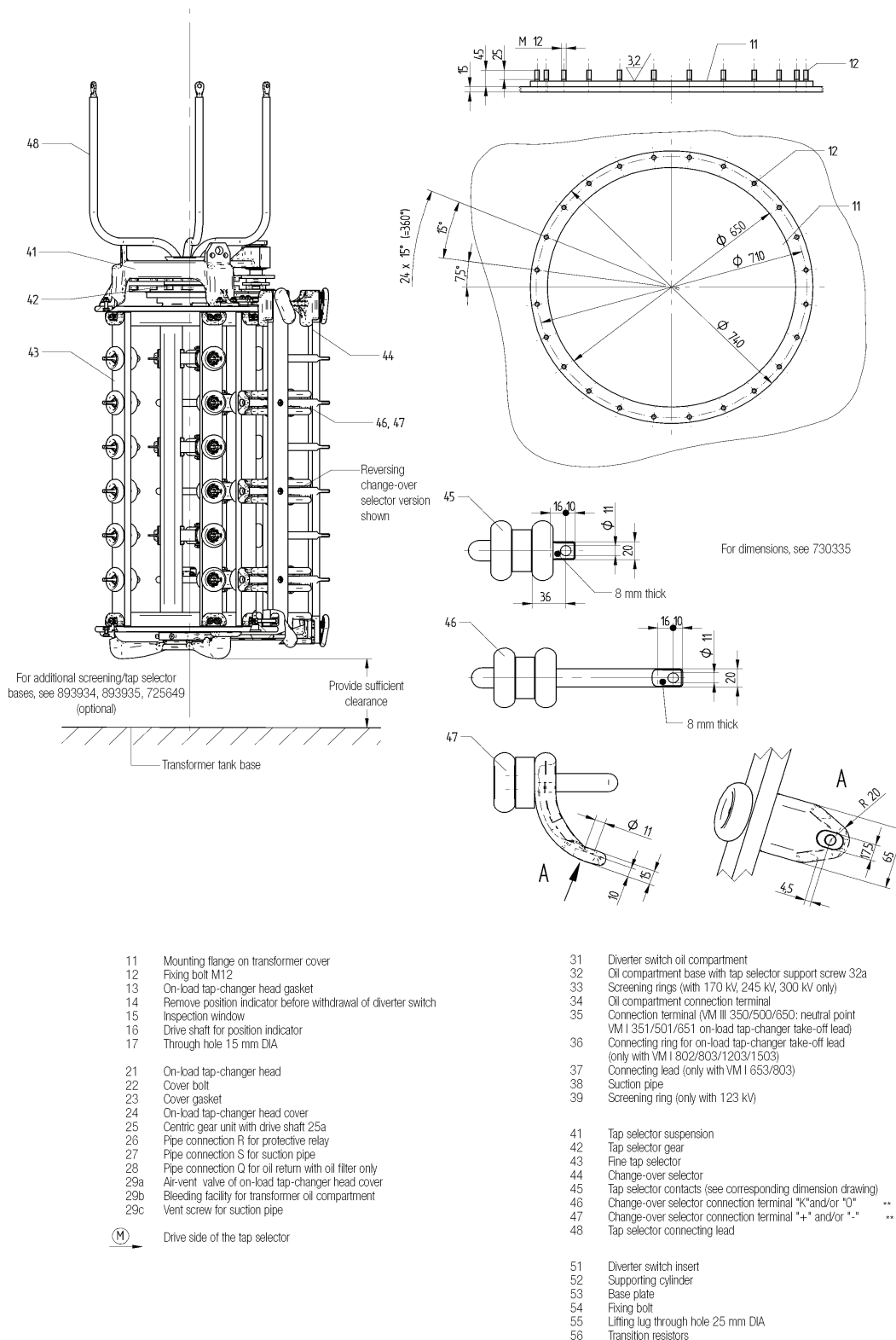
3.1.25 VACUTAP® VM 300 – installation drawing of central drive (765192)



3.1.26 VACUTAP® VM® – installation drawing of central drive (746230)



VACUTAP® VM® – installation drawing of central drive (746230)



** not with multiple coarse change-over selector

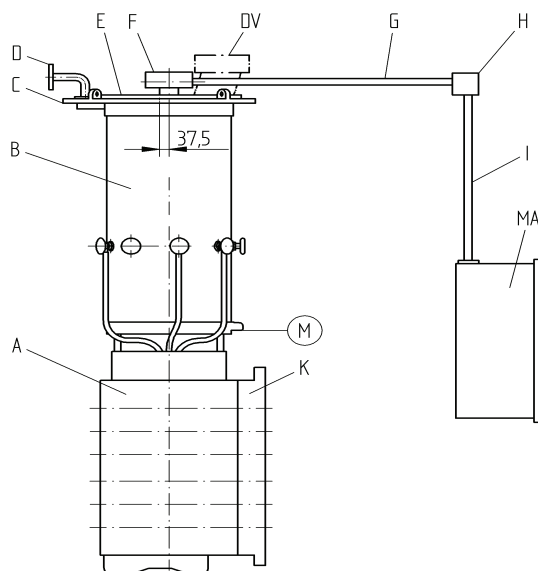
[illegible]

E1 = BLEEDING FACILITY FOR ON-LOAD TAP-CHANGER HEAD
E2 = BLEEDING FACILITY FOR SPACE UNDER THE HEAD OUTSIDE
THE TAP-CHANGER OIL COMPARTMENT (SAME PIPE CONNECTION AS R, S, Q OR BLEEDER SCREW CAN BE USED)
Q = CONNECTION FOR OIL RETURN PIPE (ONLY FOR OIL FILTER) - CONNECTIONS ORIENTABLE THROUGH 360°
S = CONNECTION FOR SUCTION PIPE - CONNECTIONS ORIENTABLE THROUGH 360°
R = CONNECTION FOR PROTECTIVE RELAY (EXCHANGEABLE WITH CONNECTION Q) - CONNECTIONS ORIENTABLE THROUGH 360°
T = THERMOMETER BAG (OPTIONALLY)
SR = INSPECTION WINDOW, RIGHT
SL = INSPECTION WINDOW, LEFT

SCALE 1:2,5 (1:1)

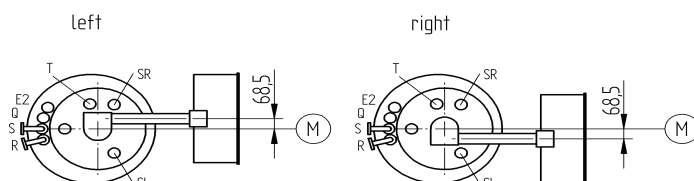
(M) DRIVE SIDE OF THE TAP SELECTOR

3.1.28 VACUTAP® VM® – variants of on-load tap-changer head (720026)

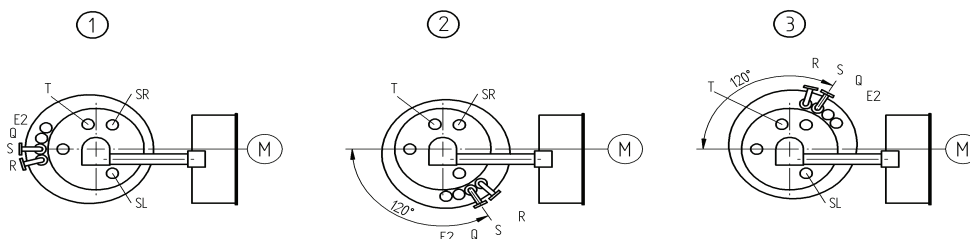


- A = tap selector
 K = change-over selector
 B = diverter switch oil compartment
 C = on-load tap-changer head
 D = pipe connections (R, S, Q, E2)
 DV = pressure relief valve
 E = on-load tap-changer head cover
 F = upper gear unit
 G = drive shaft, horizontal
 H = bevel gear
 I = drive shaft, vertical
 MA = motor drive unit
 (M) = drive side of tap selector
 SR = inspection window on the right
 SL = inspection window on the left
 T = temperature sensor
- the represented version
Typ M

Position of drive shaft of gear unit



Head variants



Swivelling ranges

A considerable number of variants of the on-load tap-changer head are available for adapting the horizontal part of the drive shaft to the transformer tank.

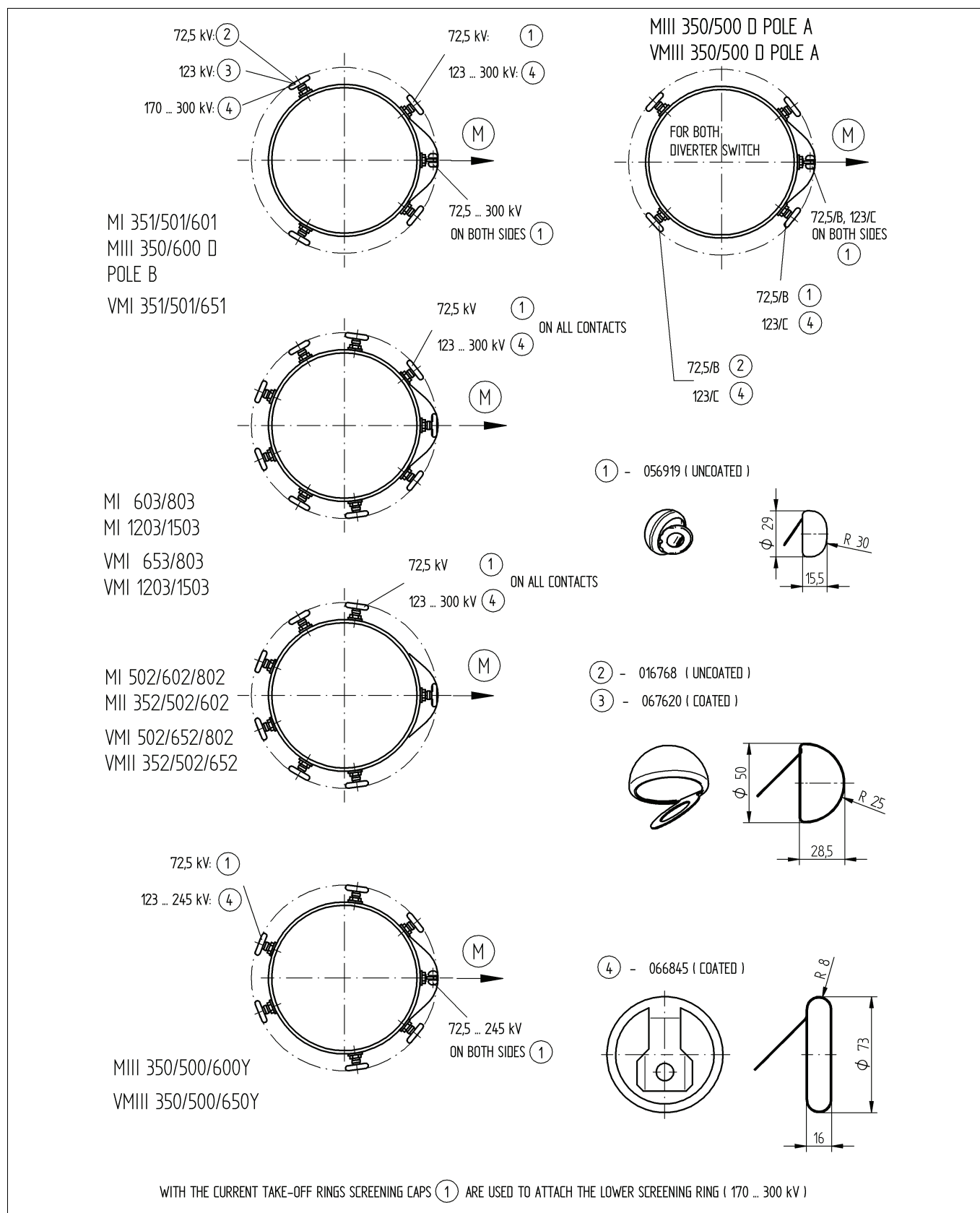
The mounting position of the tap selector A and diverter switch oil compartment B is determined by the drive side of tap selector (M).

The on-load tap-changer head C together with its pipe connections D may be turned through 120 degrees clockwise or anti-clockwise. This results in the variants 1, 2 and 3.

The upper gear unit F can be turned continuously on its own axis. Table 720027: Lists the limitation of the swivelling range for the particular head variant. The angel specifications refer to the center of rotation of the gear unit. Pay particular attention to the offset of the drive shaft.

3.1.29 VACUTAP® VM® – swivel range of gear unit (720027)

SKETCH	HEAD VERSION COMPONENTS USED	LIMITATION OF THE SWIVELLING RANGE
	DRIVE SHAFT RIGHT HEAD VERSION 1	
	PIPE CONNECTION R	-168° -140°
	PIPE CONNECTION S	-155°
	PIPE CONNECTION Q	-171°
	PIPE CONNECTION E2	162° 174°
	PRESSURE RELIEF VALVE DV	-150° -35° 175°
	TEMPERATURE SENSOR T	96° 175°
	DRIVE SHAFT RIGHT HEAD VERSION 2	
	PIPE CONNECTION R	-48° -21°
	PIPE CONNECTION S	-63° -36°
	PIPE CONNECTION Q	-78° -51°
	PIPE CONNECTION E2	-93° -66°
	PRESSURE RELIEF VALVE DV	-150° -35° 175°
	TEMPERATURE SENSOR T	96° 175°
	DRIVE SHAFT RIGHT HEAD VERSION 3	
	PIPE CONNECTION R	72° 99°
	PIPE CONNECTION S	57° 84°
	PIPE CONNECTION Q	42° 69°
	PIPE CONNECTION E2	27° 54°
	PRESSURE RELIEF VALVE DV	-150° -35° 175°
	TEMPERATURE SENSOR T	96° 175°
	DRIVE SHAFT LEFT HEAD VERSION 1	
	PIPE CONNECTION R	-162°
	PIPE CONNECTION S	-177°
	PIPE CONNECTION Q	141° 168°
	PIPE CONNECTION E2	126° 153°
	PRESSURE RELIEF VALVE DV	35° 150°
	TEMPERATURE SENSOR T	34° 114°
	DRIVE SHAFT LEFT HEAD VERSION 2	
	PIPE CONNECTION R	-69° -42°
	PIPE CONNECTION S	-84° -57°
	PIPE CONNECTION Q	-99° -72°
	PIPE CONNECTION E2	-114° -87°
	PRESSURE RELIEF VALVE DV	35° 150°
	TEMPERATURE SENSOR T	34° 114°
	DRIVE SHAFT LEFT HEAD VERSION 3	
	PIPE CONNECTION R	50° 78°
	PIPE CONNECTION S	35° 62°
	PIPE CONNECTION Q	21° 48°
	PIPE CONNECTION E2	6° 33°
	PRESSURE RELIEF VALVE DV	35° 150°
	TEMPERATURE SENSOR T	34° 114°

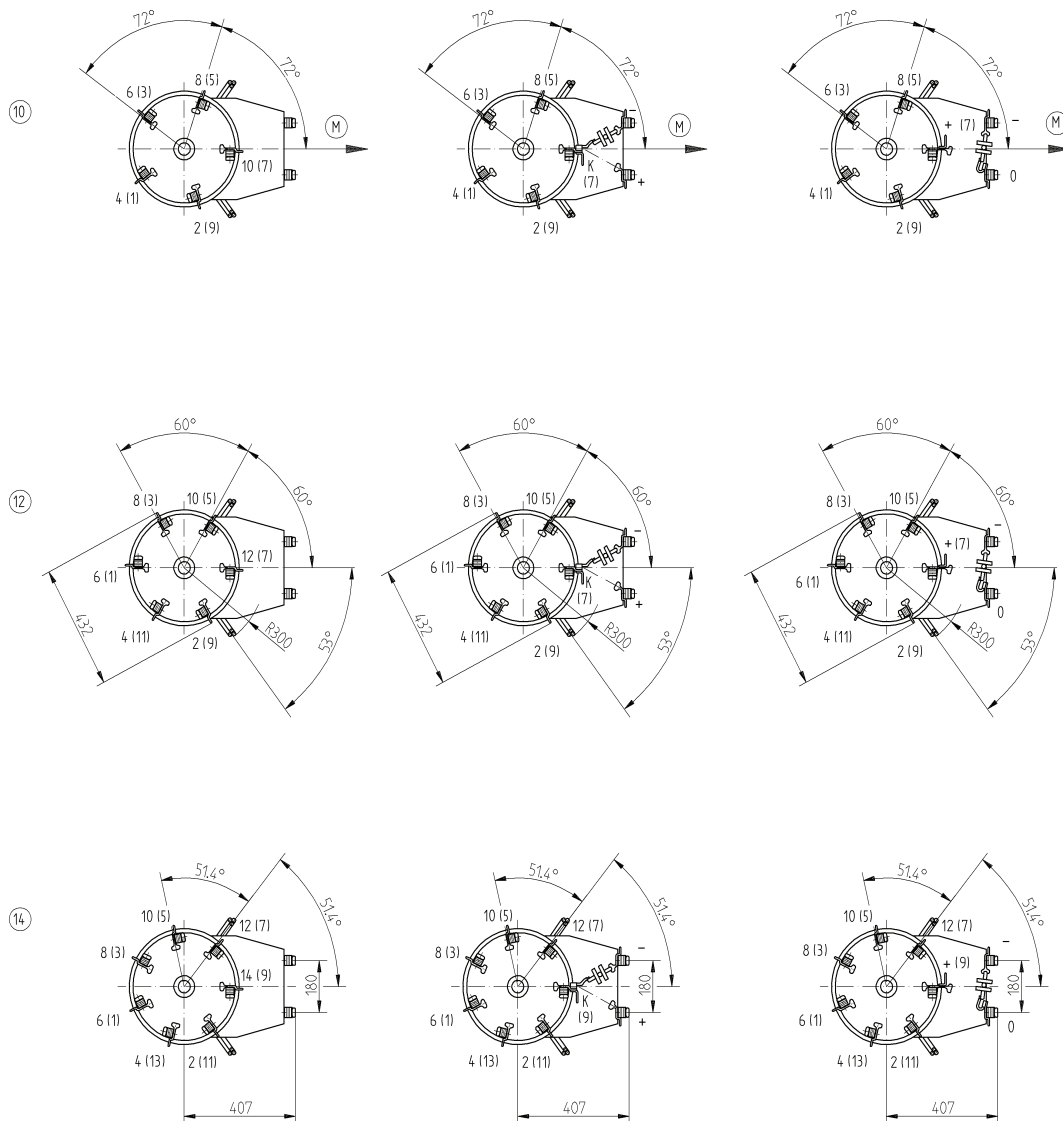
3.1.30 VACUTAP® VM® – screenings on oil compartment contacts (730336)


3.1.31 VACUTAP® VM 300 – tap selector cross-sections (898041)

TAP SELECTOR WITHOUT
CHANGE-OVER SELECTOR:

TAP SELECTOR WITH
REVERSING SWITCH:

TAP SELECTOR WITH
COARSE TAP SELECTOR:



DESIGNATION OF TAP SELECTOR TERMINALS:

(PLAN VIEW)

E. G.: 4 UPPER CONTACT PLANE

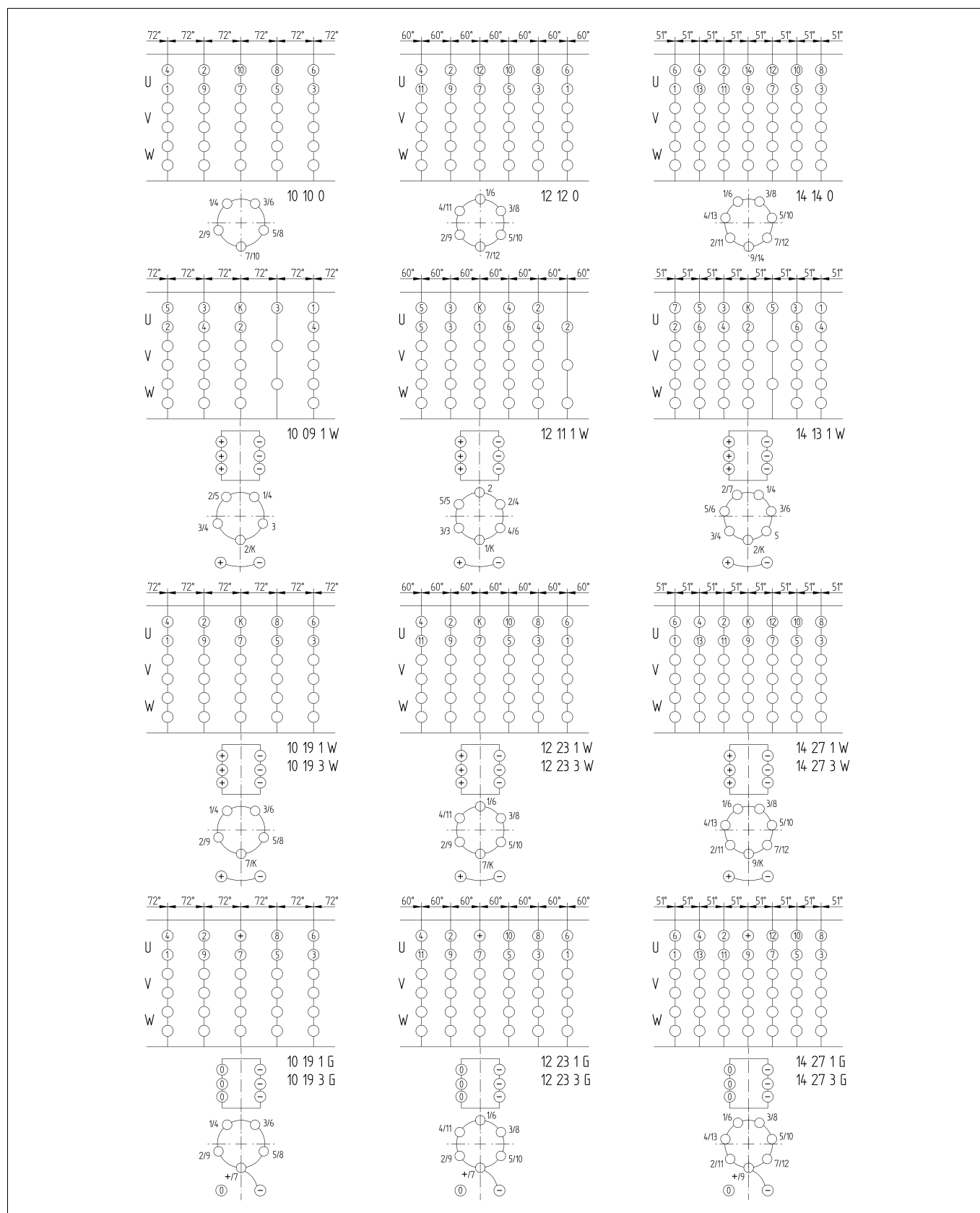
(13) LOWER CONTACT PLANE

(M) DRIVE SIDE OF TAP SELECTOR

(10) (12) (14) TAP SELECTOR PITCH

FOR BINDING DESIGNATIONS OF TERMINALS AND PHASES REFER TO THE RELEVANT CONNECTION DIAGRAM

3.1.32 VACUTAP® VM 300 – arrangement of contacts on tap selector (891114)



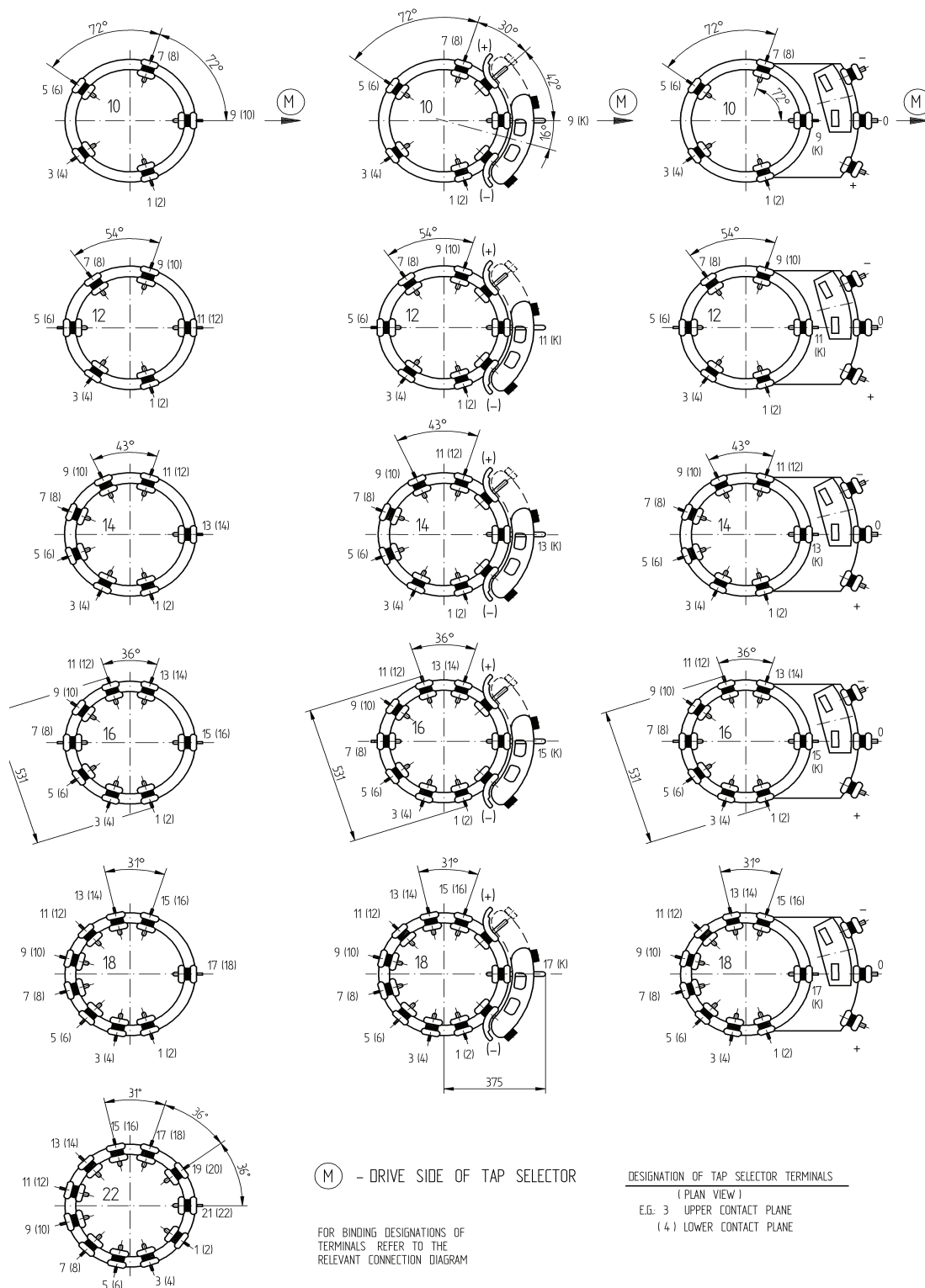
3.1.33 VACUTAP® VM® – arrangement of contacts on tap selector, tap selector division 10...22 (898013)

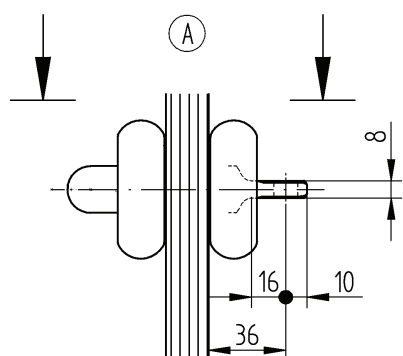
TAP_SELECTOR WITHOUT
CHANGE - OVER_SELECTOR :

TAP_SELECTOR WITH REVERSING_SWITCH :

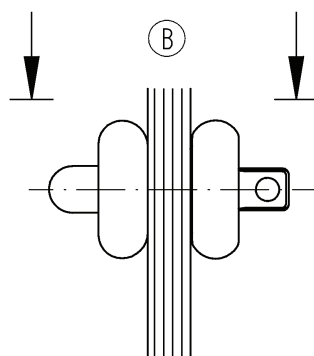
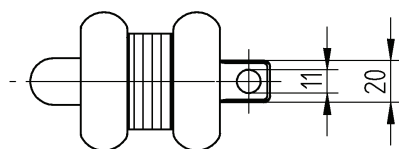
TAP_SELECTOR WITH COARSE
TAP_SELECTOR :

REPRESENTATION APPLIES TO TYPES M III 350/500/600 Y AND M II 352/502/602
THE UPPER AND LOWER TAP_SELECTOR PLANE ARE INTERCHANGED IN TYPES M I 351 - M I 1503
AND TO TYPES VM III 350/500/650 Y AND VM II 352/502/652
THE UPPER AND LOWER TAP_SELECTOR PLANE ARE INTERCHANGED IN TYPES VM I 351 - VM I 1503

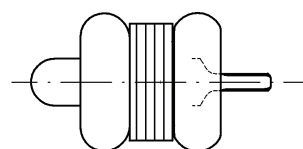


3.1.34 VACUTAP® VM® – installation position of tap selector connection contacts (890477)


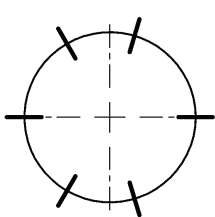
Through hole, vertical



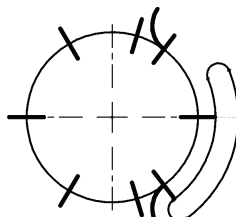
Through hole, horizontal



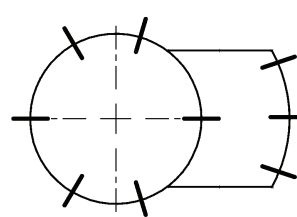
(A)



M III 350 / 500 / 600Y - 0
VM III 350 / 500 / 650Y - 0



M III 350 / 500 / 600Y - W
VM III 350 / 500 / 650Y - W



M III 350 / 500 / 600Y - G
VM III 350 / 500 / 650Y - G

M II 352 / 502 / 602 - 0
VM II 352 / 502 / 652 - 0

M II 352 / 502 / 602 - W
VM II 352 / 502 / 652 - W

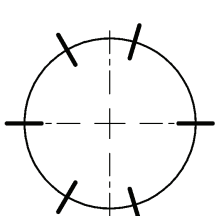
M II 352 / 502 / 602 - G
VM II 352 / 502 / 652 - G

M I 351 / 501 / 601 - 0
VM I 351 / 501 / 651 - 0

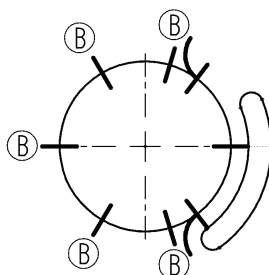
M I 351 / 501 / 601 - W
VM I 351 / 501 / 651 - W

M I 351 / 501 / 601 - G
VM I 351 / 501 / 651 - G

(B)

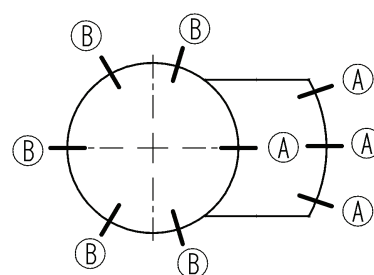


M I 802 - 0
VM I 802 - 0
VM I 1002 - 0
M I 1203 / 1503 - 0
VM I 1203 / 1503 - 0



M I 802 - W
VM I 802 - W
VM I 1002 - W
M I 1203 / 1503 - W
VM I 1203 / 1503 - W

(A)



M I 802 - G
VM I 802 - G
VM I 1002 - G
M I 1203 / 1503 - G
VM I 1203 / 1503 - G

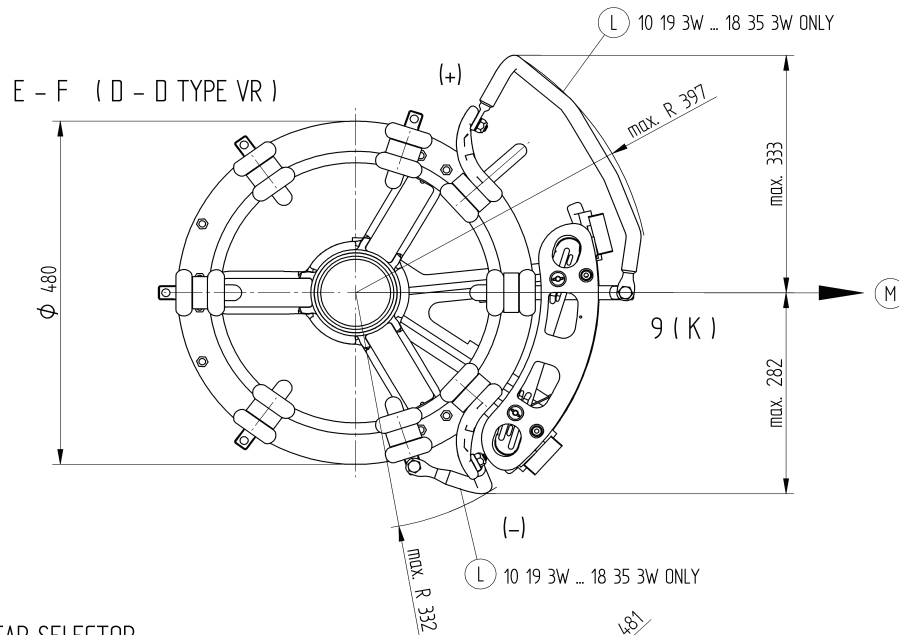
(A) + (B)

3.1.35 VACUTAP® VM® – connecting leads 3W, 1G, 3G (723590)

REVERSING SWITCH:

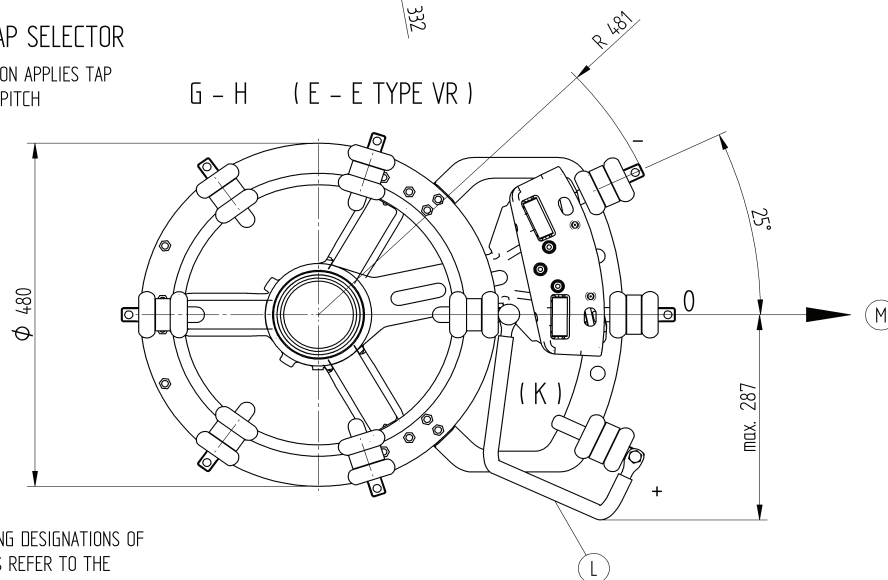
REPRESENTATION APPLIES TO TYPES M III 350/500/600Y AND M II 352/502/602, 12-PITCH
THE UPPER AND LOWER TAP SELECTOR PLANE ARE INTERCHANGED IN TYPES M I 351 ... M I 1503

AND REPRESENTATION APPLIES TO TYPES VM III 350/500/650Y AND VM II 352/502/652, 12-PITCH
THE UPPER AND LOWER TAP SELECTOR PLANE ARE INTERCHANGED IN TYPES VM I 351 ... VM I 1503



COARSE TAP SELECTOR

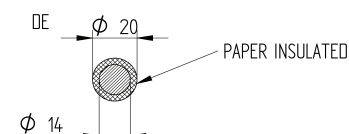
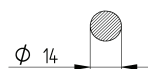
REPRESENTATION APPLIES TAP SELECTOR, 12-PITCH



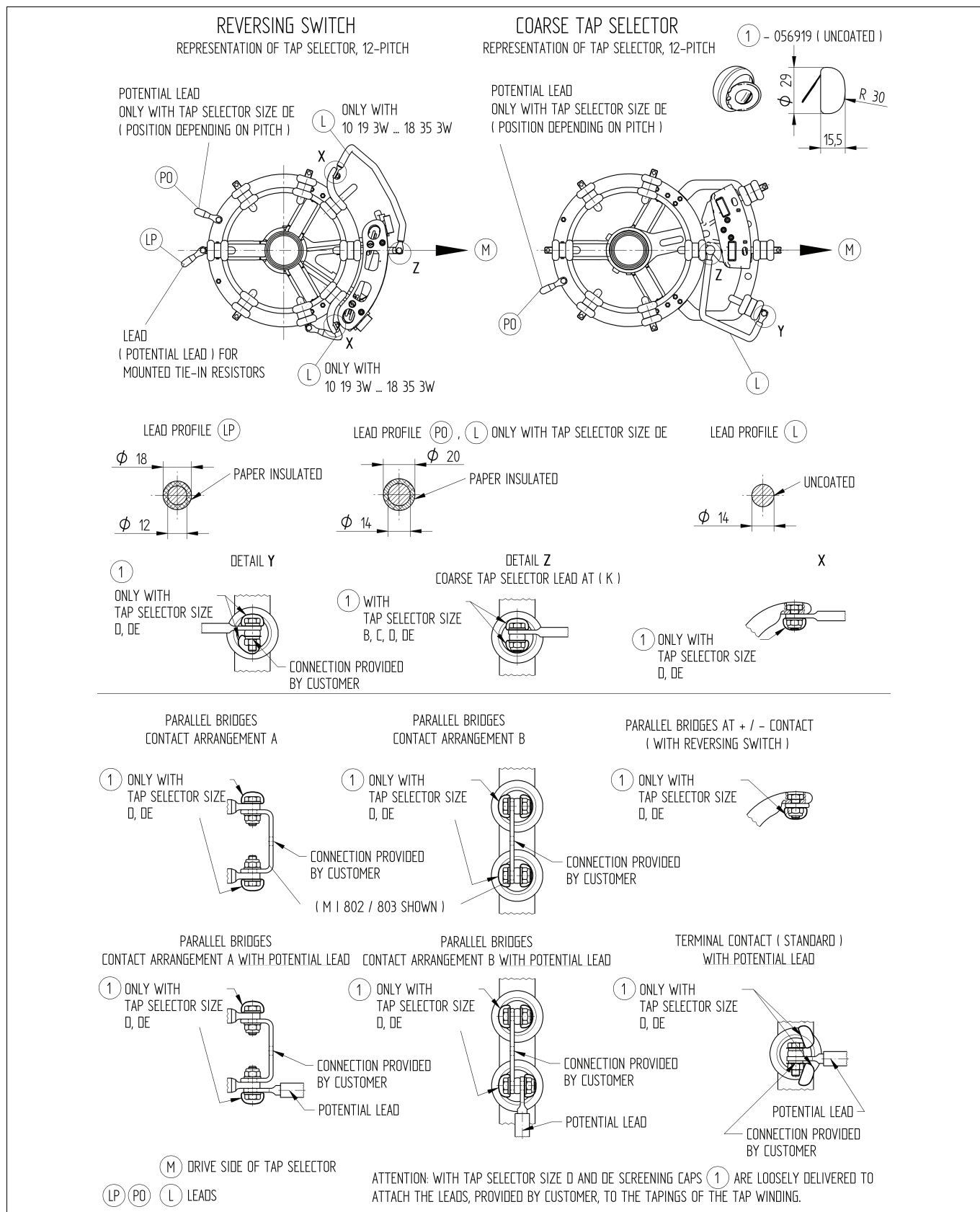
FOR BINDING DESIGNATIONS OF
TERMINALS REFER TO THE
RELEVANT CONNECTION DIAGRAM

- (M) DRIVE SIDE OF TAP SELECTOR
- (L) CONNECTING LEAD

TAP SELECTOR SIZE
B, C, D



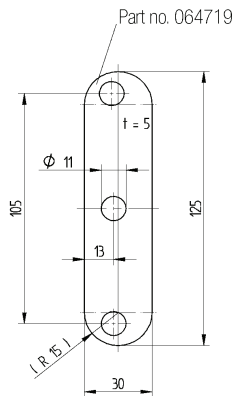
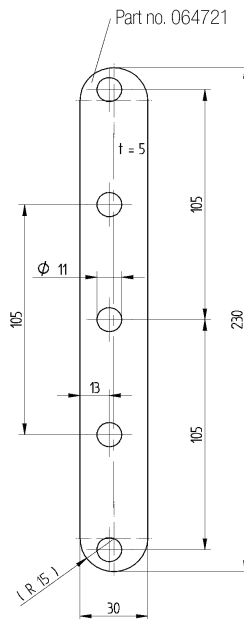
3.1.36



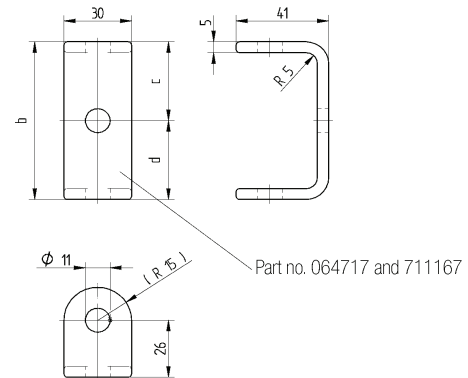
3.1.37

VACUTAP® VM 802/803/1203/1503 – bridges to parallel connection of tap selector connection contacts (899598)

Arrangement of contacts B

M | 802 / 803
VM | 802 / 803 / 1002

M | 1203 / 1503
VM | 1203 / 1503


Arrangement of contacts A

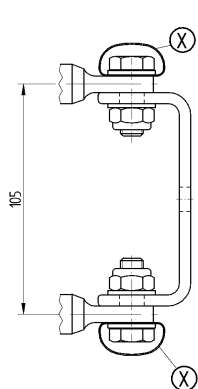
M | 802 / 803 / 1203 / 1503
VM | 802 / 803 / 1203 / 1503


Arrangement of contacts (see 890477:)	Part no.	Dimension b	Dimension c	Dimension d
A without lead	064717	97	48,5	48,5
A with lead	711167	91	48,5	42,5

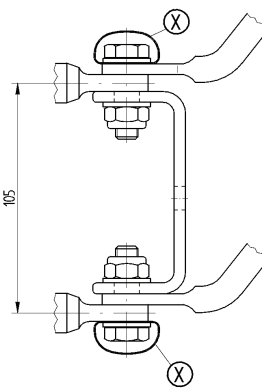
Installation of parallel bridges for contact arrangement A with
and without connecting lead for 3W tap change operation

M | 802 / 803
VM | 802 / 803 / 1002

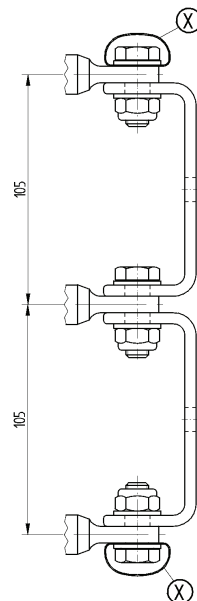
Without lead



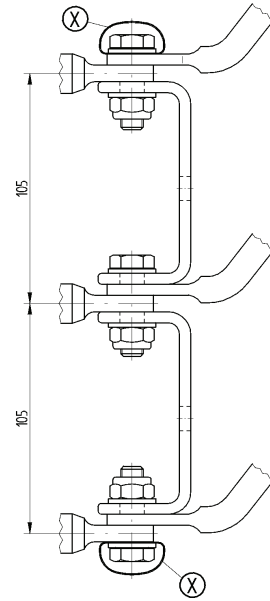
With lead


M | 1203 / 1503
VM | 1203 / 1503

Without lead



With lead

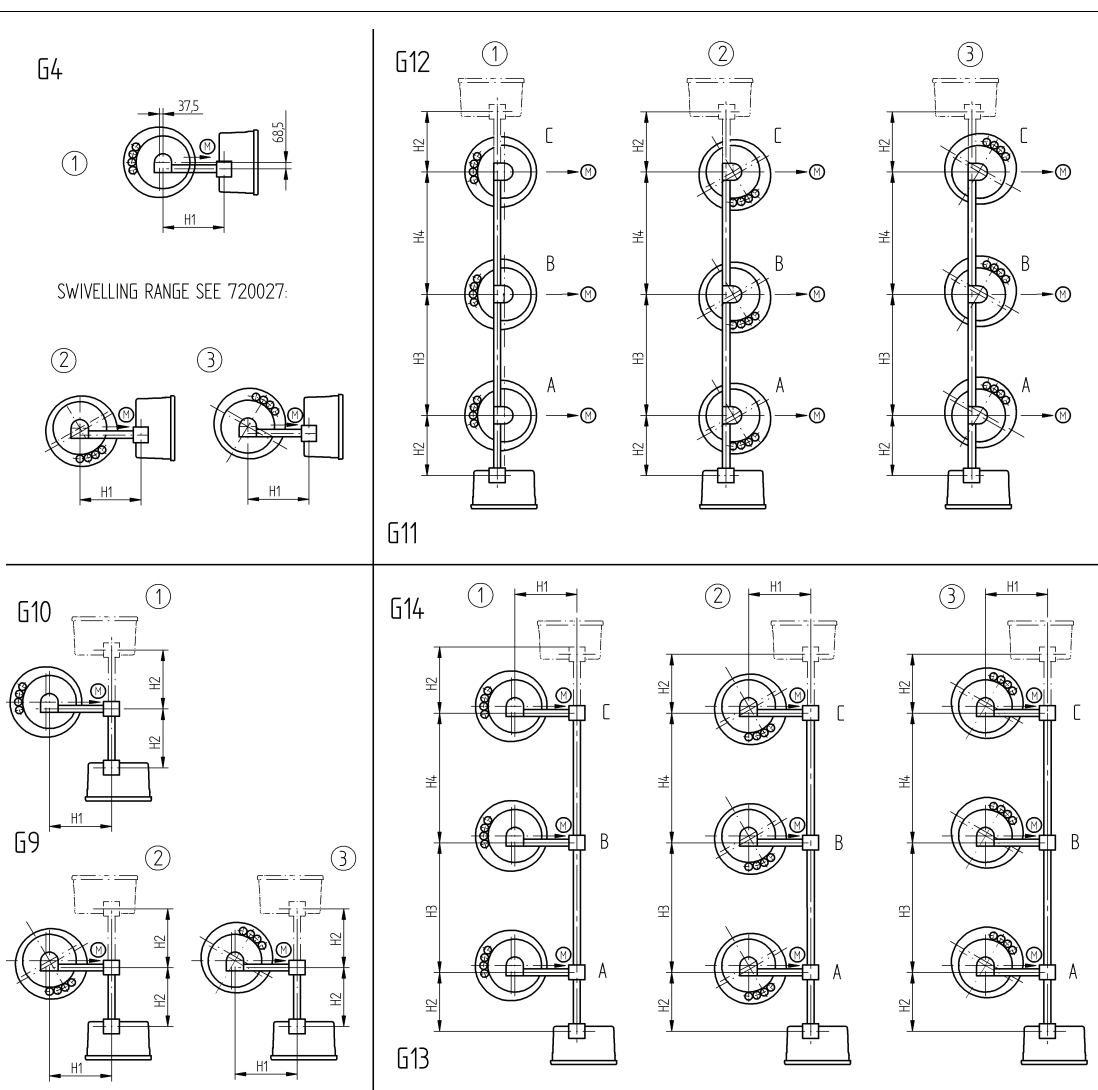


(X) only with tap selector size 0 and OE

Caution: Parallel bridges are not included in the standard scope of supply

Scale 1 : 1

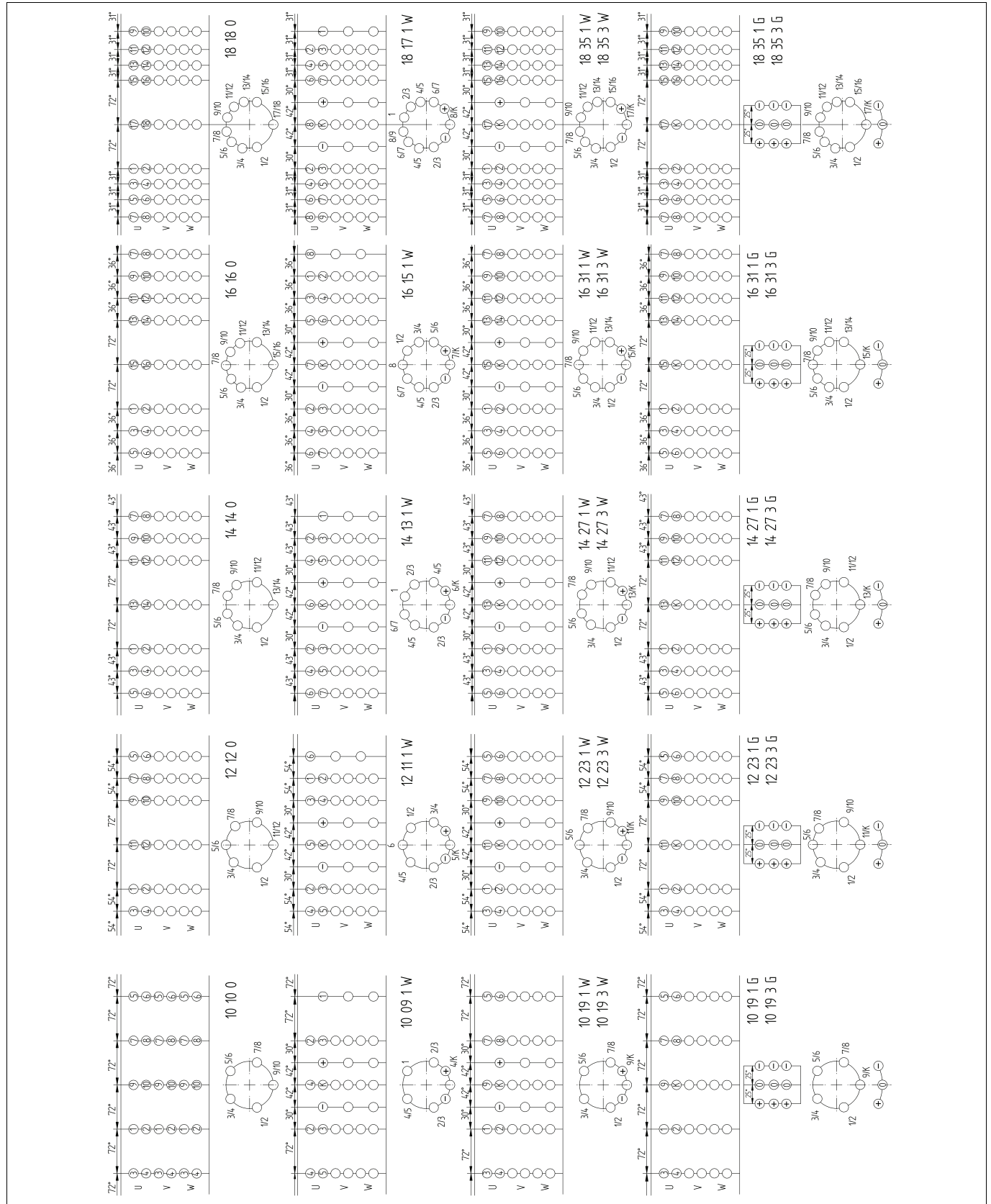
3.1.38 VACUTAP® VM® – horizontal drive shaft, central drive (893896)



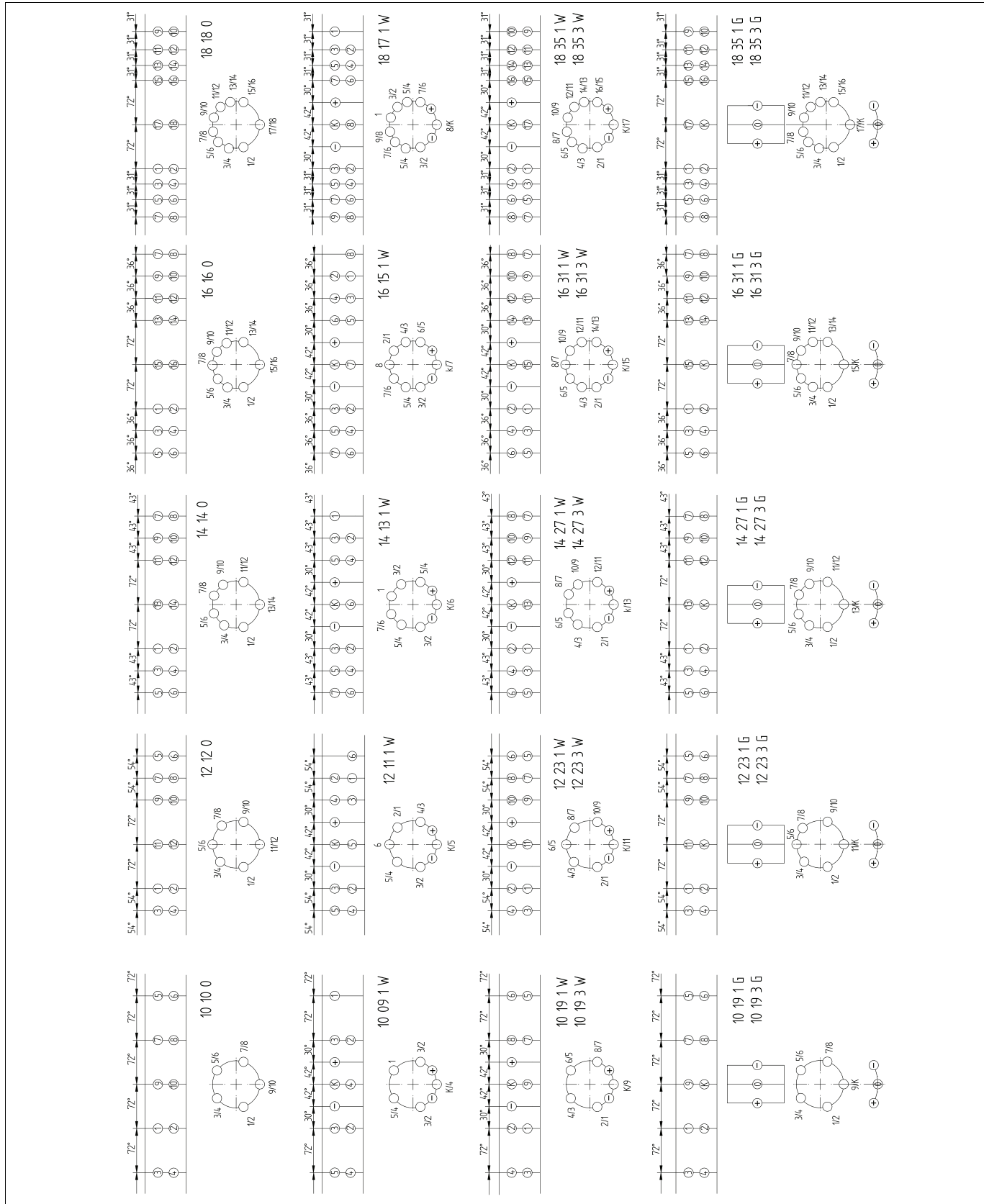
ARRANGEMENT		G4	G9, G10	G11, G12	G13, G14
STANDARD DESIGN		■		■	
SPECIAL DESIGN			■		■
MINIMUM DIMENSIONS ¹⁾ (DETERMINED FOR MECHANICAL REASONS; NECESSARY INSULATING DISTANCES NOT CONSIDERED !)	H1	535	545	-	545
	H2	-	318	515	318
	H3 ²⁾	-	-	840	840
	H4 ²⁾	-	-	840	840
NOTE: 1) FOR OLTCs WITH THE CHANGE-OVER SELECTOR ATTACHED Laterally, THE DIMENSIONS OF THE CHANGE-OVER SELECTOR AFTER INSTALLED IN POSITION HAVE TO BE TAKEN INTO ACCOUNT (SEE THE CORRESPONDING OLTC DIMENSION DRAWING) 2) IN GENERAL DETERMINED BY THE INSULATION DISTANCE BETWEEN POLES A, B, C.					
INTERMEDIATE BEARING FOR	H1 >	2254	2309	-	2309
	H2 >	-	2259	2254	2259
	H3 >	-	-	2249	2259
	H4 >	-	-	2249	2259

① ② ③ – HEAD VERSION
 → (M) – DRIVE SIDE OF TAP SELECTOR/ SIDE OF CHANGE-OVER SELECTOR

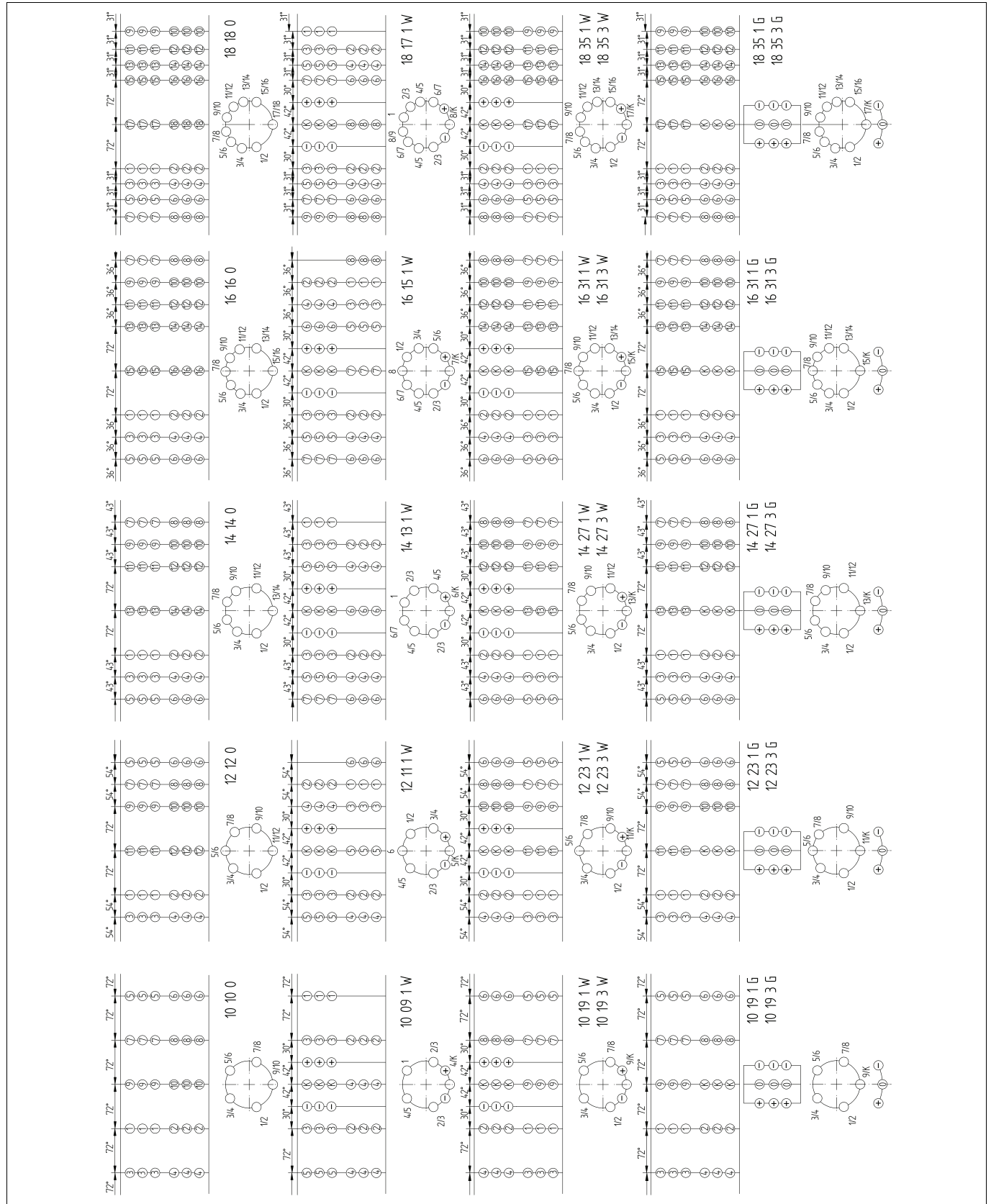
3.1.39 VACUTAP® VM III 350/500/650 Y – arrangement of tap selectors (891107)



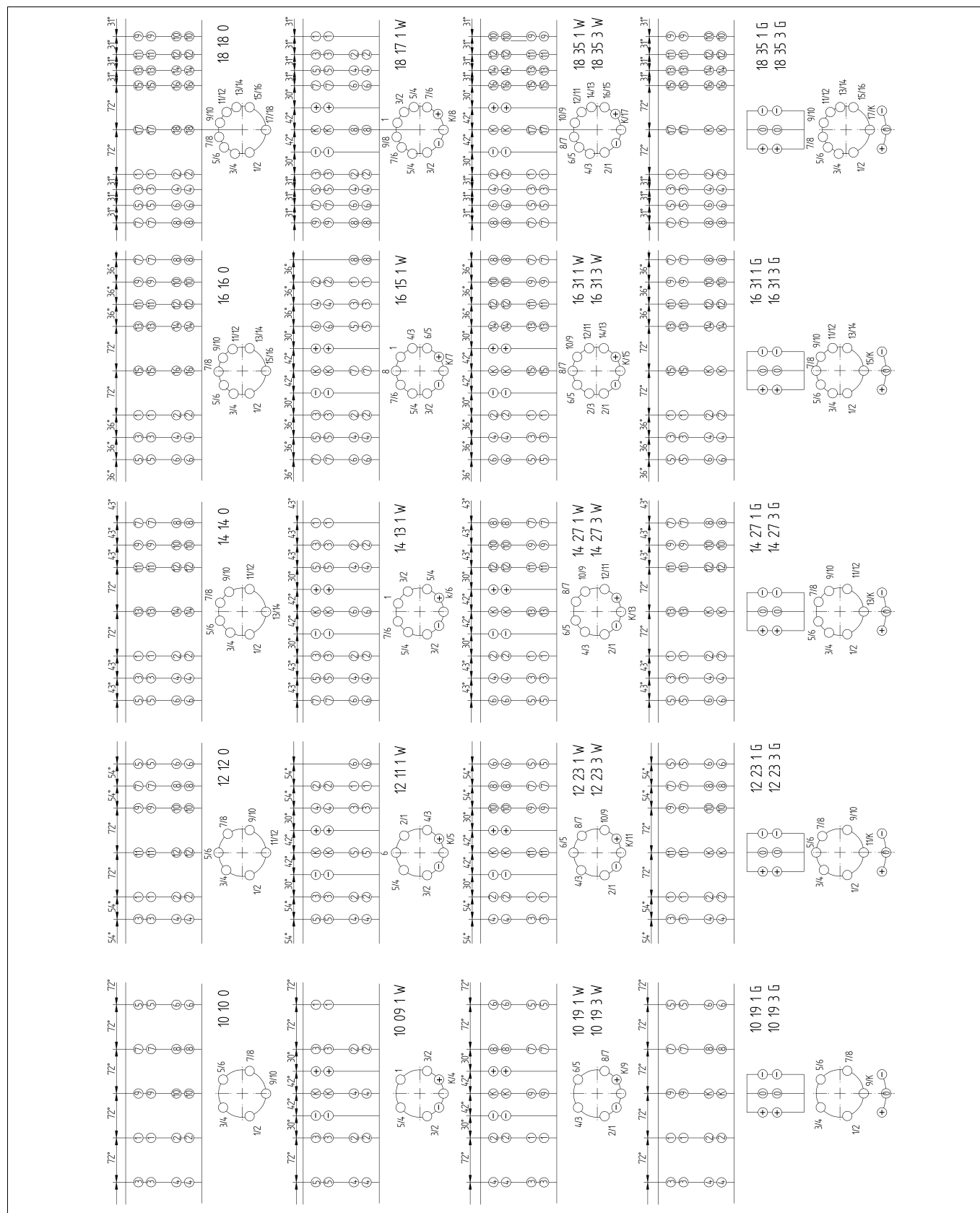
3.1.40 VACUTAP® VM I 351/501/651 – arrangement of tap selectors (891108)



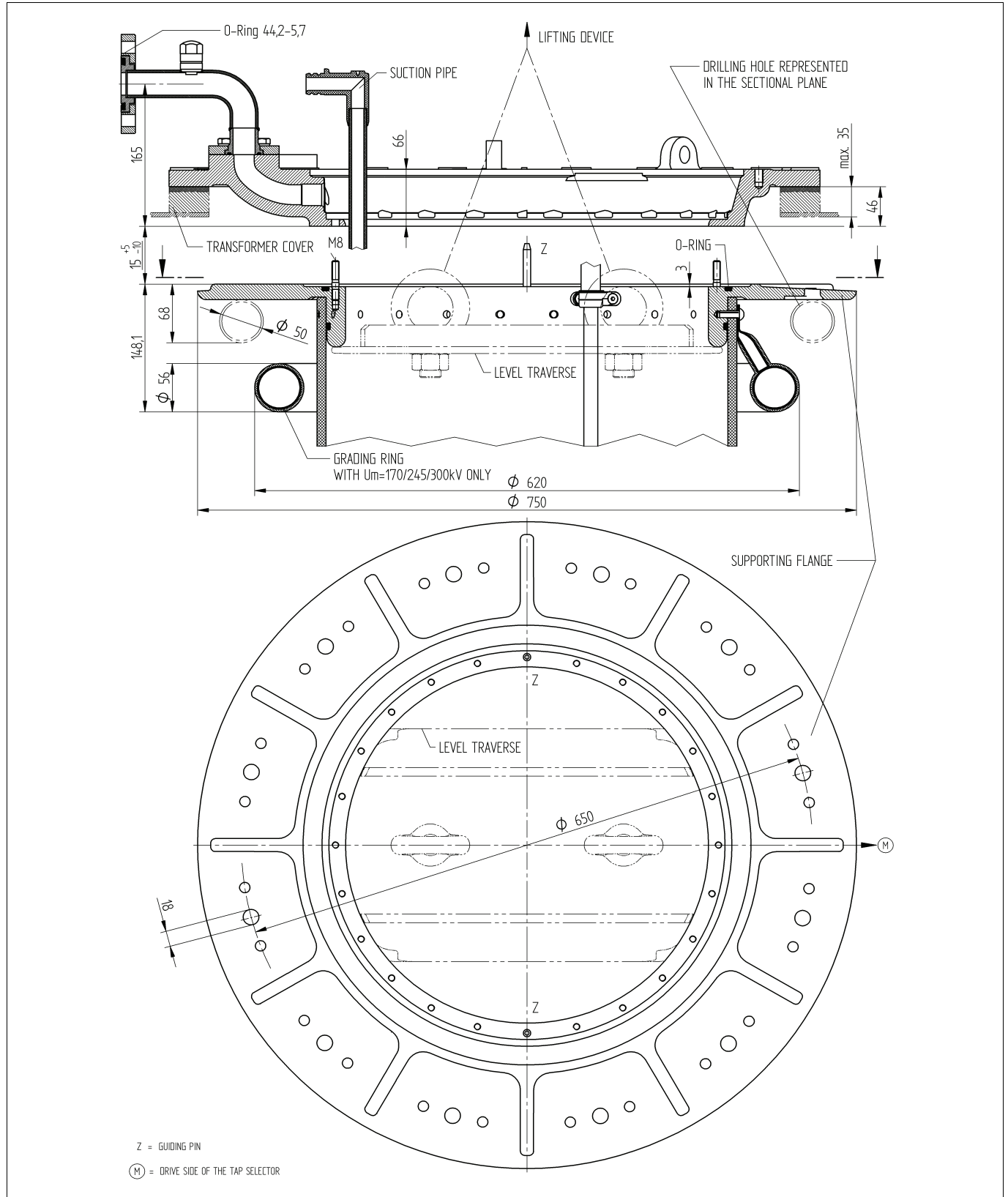
3.1.41 VACUTAP® VM I 1503 – arrangement of tap selectors (891109)



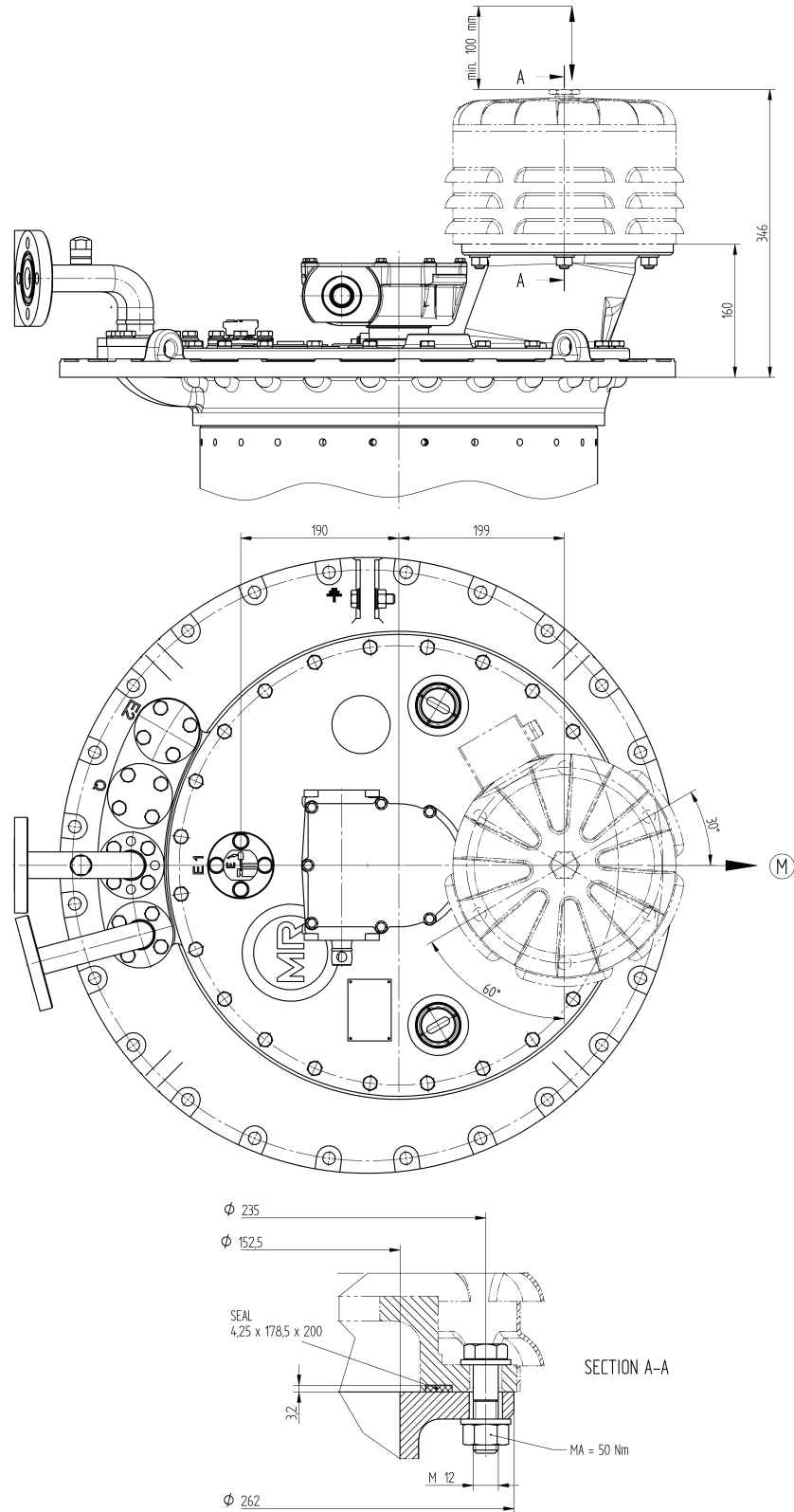
3.1.42 VACUTAP® VM I 802/1002 – arrangement of tap selectors (891110)



3.1.43

VACUTAP® VM®, supporting flange, special design for bell-tank installation for U_m up to 300 kV (896762)


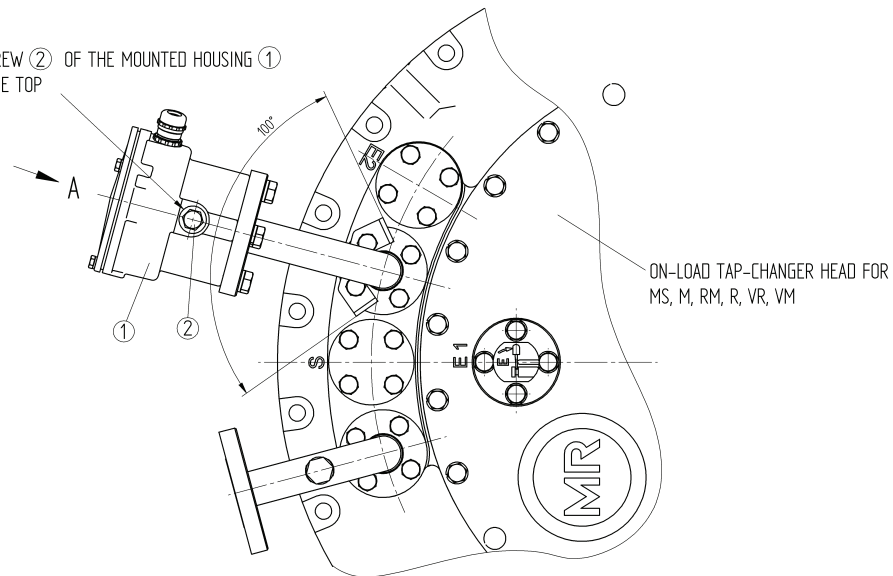
3.1.44 VACUTAP® VM® – flange for pressure relief valve (895168)



3.1.45 VACUTAP® VM® - pipe connection with tap-change supervisory control (766161)

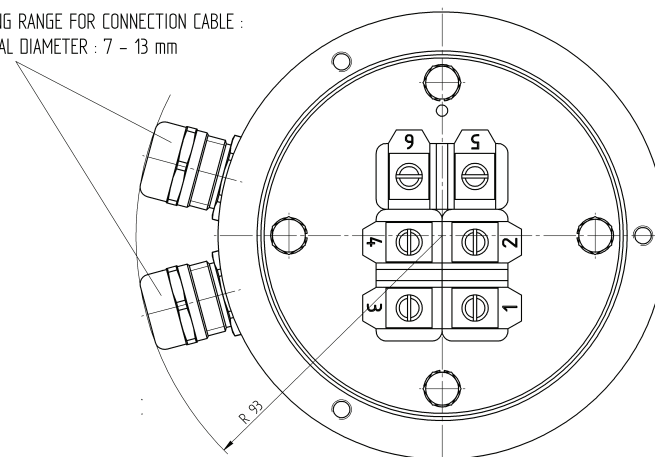
PIPE CONNECTION WITH SUPERVISORY CONTROL BUSHING WITHOUT OIL FILTER UNIT

CAUTION !
THE BLEEDER SCREW ② OF THE MOUNTED HOUSING ①
HAS TO BE ON THE TOP



ANSICHT A (1,5 : 1)
SHOWN WITHOUT COVER

M20x1,5
CLAMPING RANGE FOR CONNECTION CABLE :
EXTERNAL DIAMETER : 7 - 13 mm



CONNECTING TERMINALS FOR MONITORING DEVICE

WIRING SEE CONNECTION DIAGRAM OF THE
MOTOR DRIVE UNIT

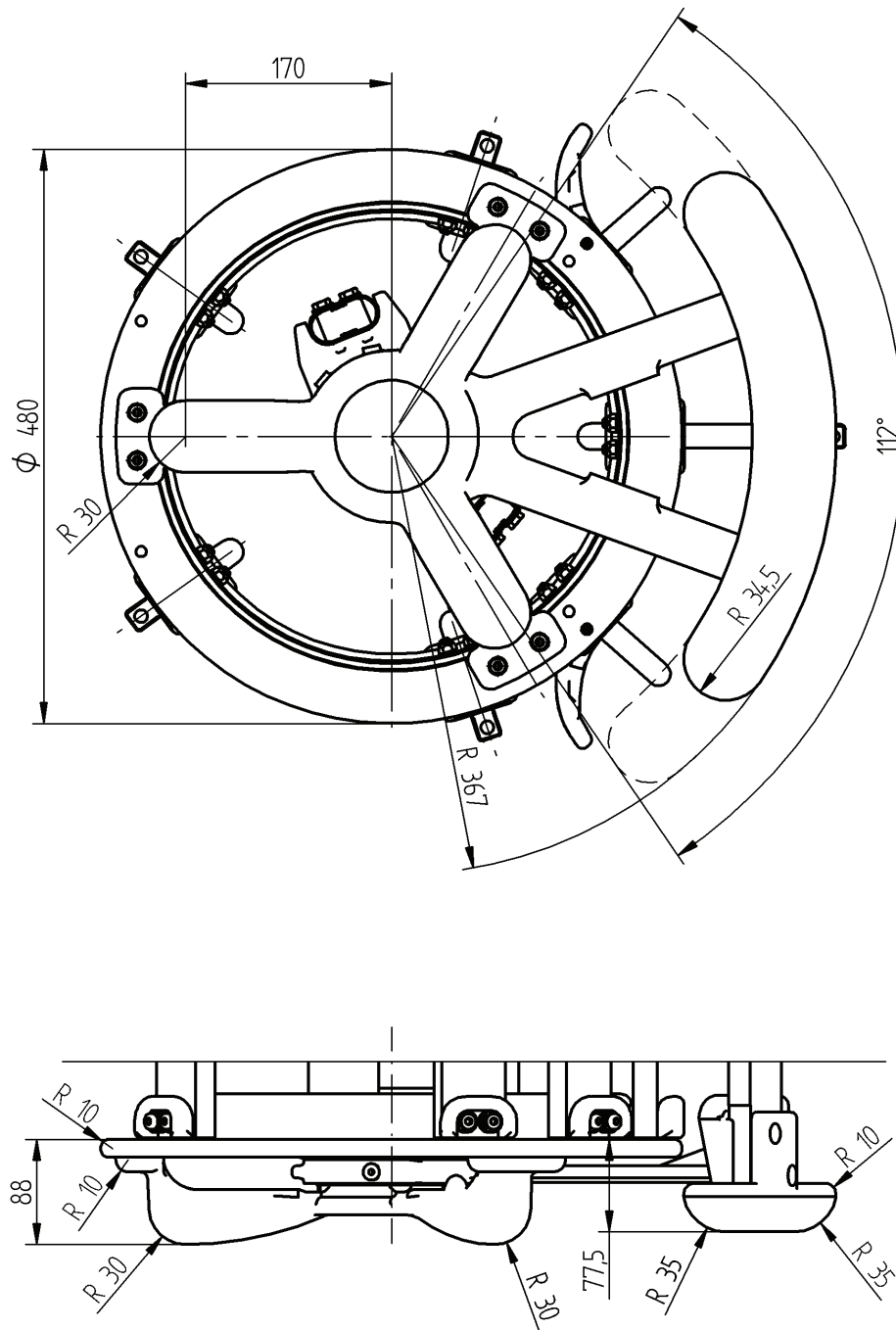
FUNCTION DIAGRAM FOR MONITORING DEVICE
SEE MOTOR DRIVE CIRCUIT DIAGRAM

CONTINUOUS RATED CURRENT: 2A
RATED VOLTAGE DC/AC (50HZ): 24V ... 250V
DIELECTRIC STRENGTH: 1150V / 50HZ / 1 MIN.

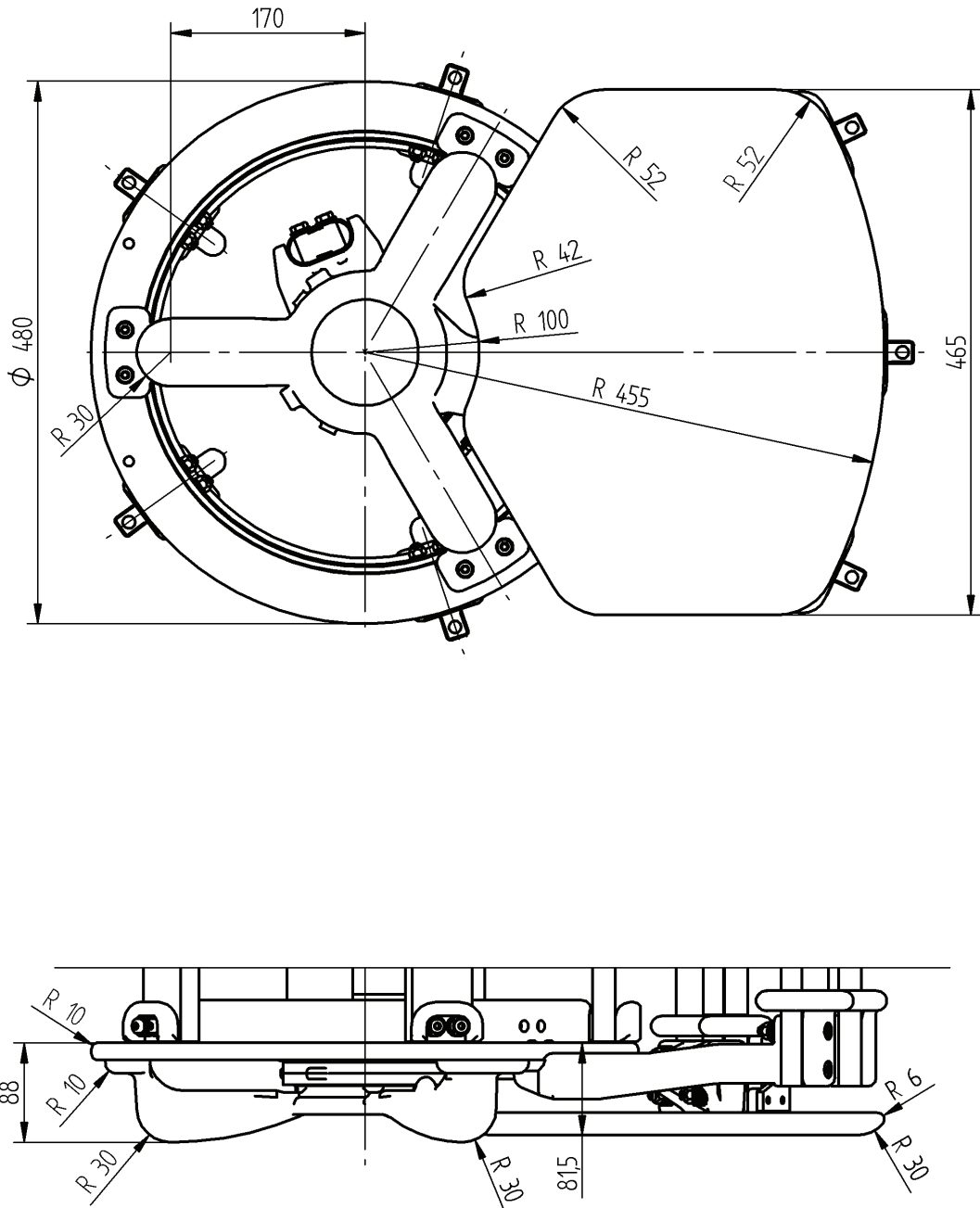
DIELECTRIC TEST OF ALL VOLTAGE-CARRYING
CONNECTIONS AGAINST GROUND PARTS:
2000V AC , 50HZ , TEST-DURATION 1 MIN.

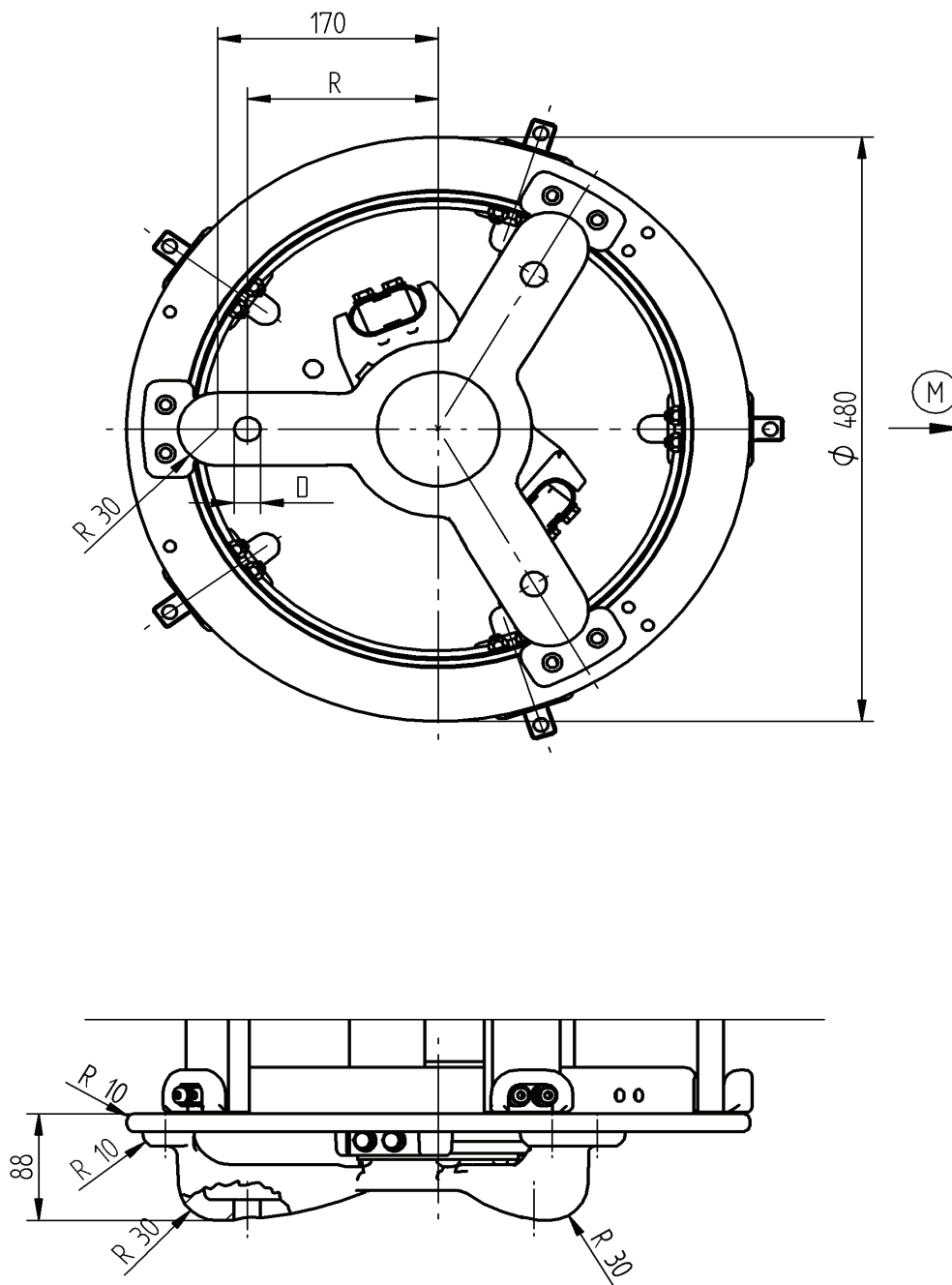
SCALE 1 : 2 (1,5 : 1)

3.1.46



3.1.47

VACUTAP® VM® – tap selector base with additional screening, coarse tap connection design (893935)


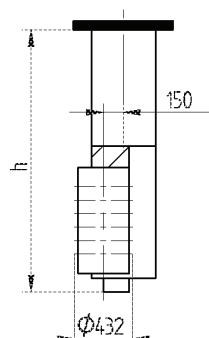
3.1.48 VACUTAP® VM® – tap selector base with through hole D20 and D13 (725649)


(M) DRIVE SIDE OF TAP SELECTOR

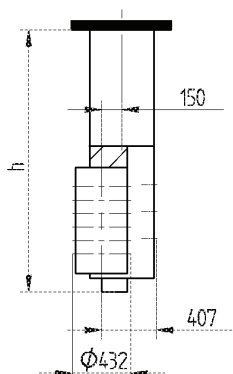
R	D	BOTTOM OF TAP SELECTOR
147	20	097251
160	13	097252

3.2 VACUTAP® VM 300 – overview of on-load tap-changer designs (765835)

without
change-over selector



with
change-over selector



Installation length h in mm

VM III 300 Y

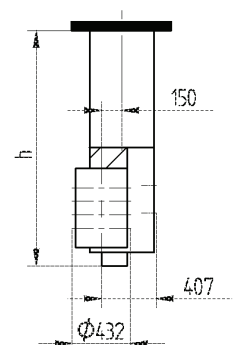
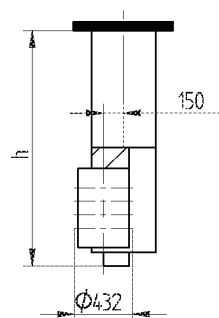
U _m	Tap selector size B
72,5	1942
123	2072
170	2202
245	2302

VM I 301

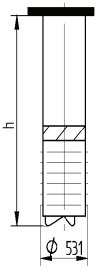
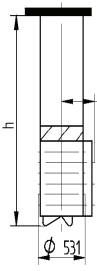
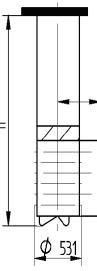
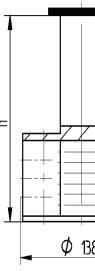
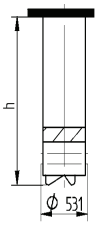
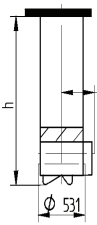
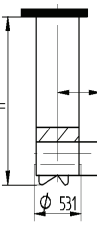
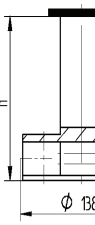
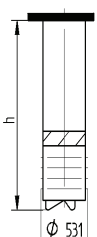
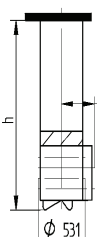
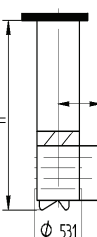
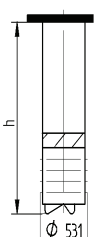
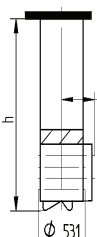
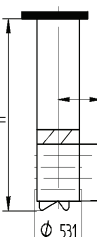
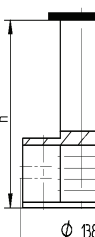
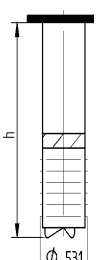
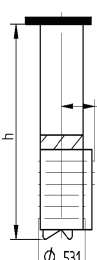
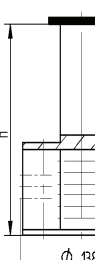
U _m	Tap selector size B
72,5	1542
123	1672
170	1802
245	1902

VM II 302

U _m	Tap selector size B
72,5	1742
123	1872
170	2002
245	2102

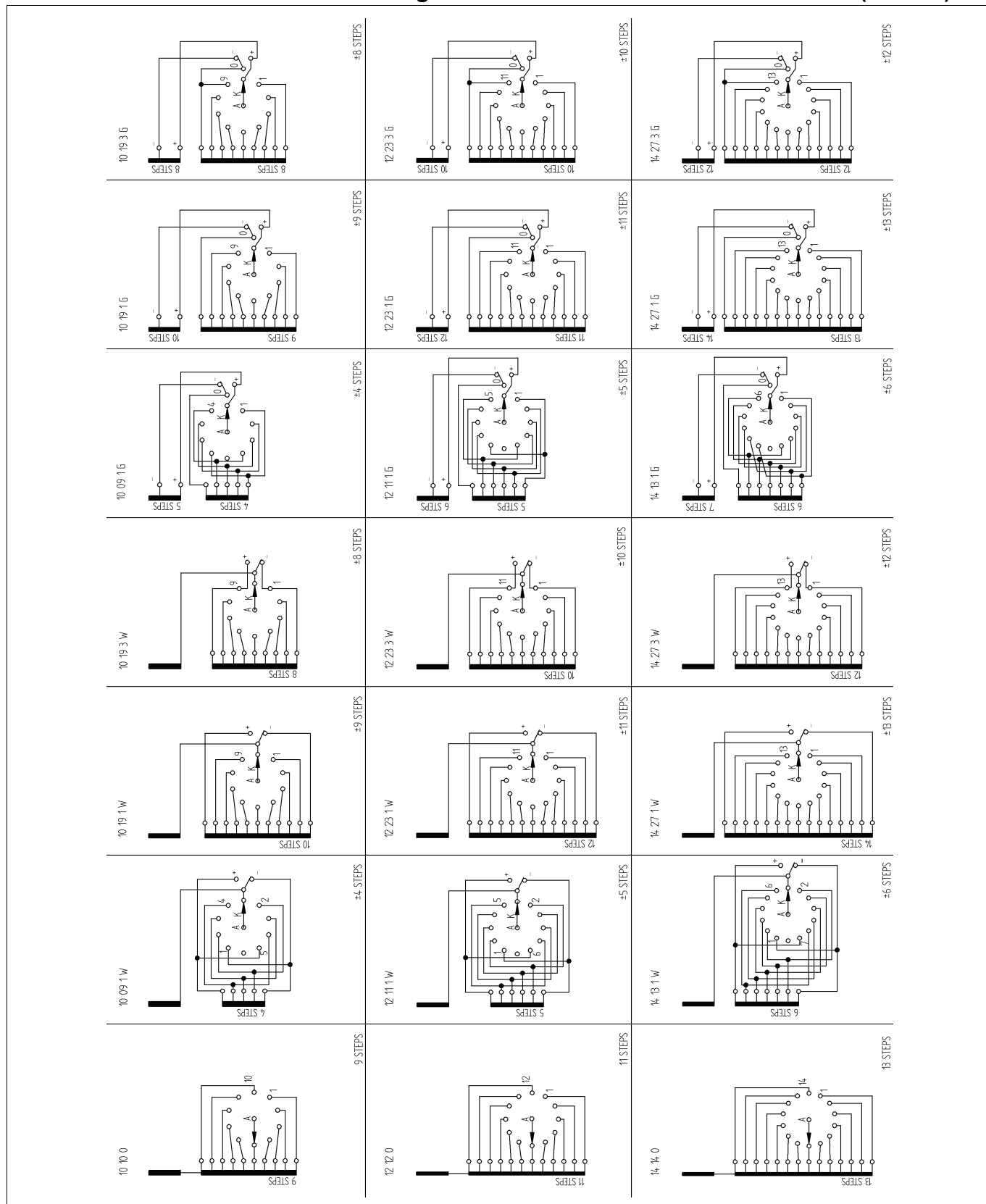


3.3 VACUTAP® VM® – overview of on-load tap-changer designs (899740)

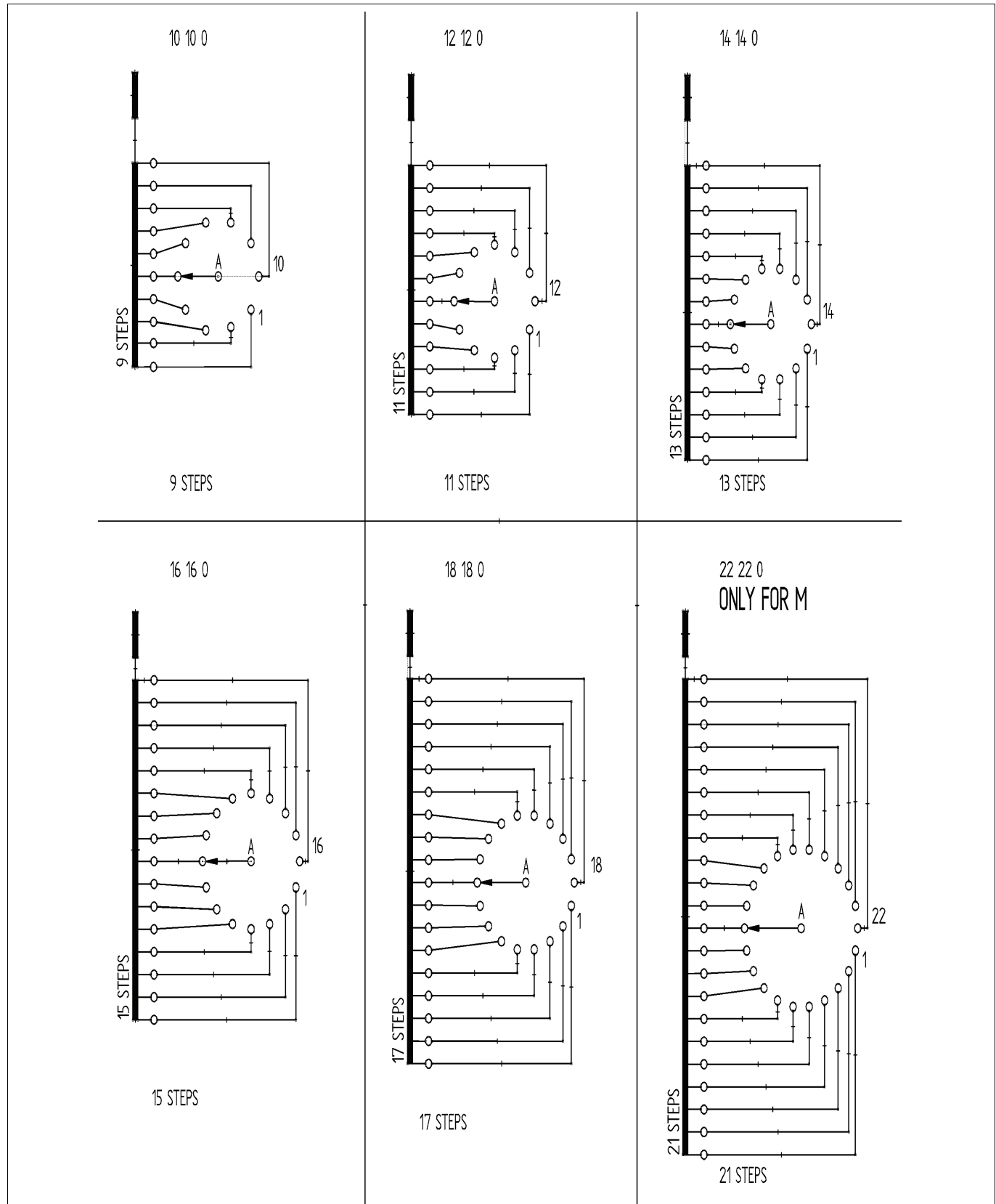
	Without change-over selector	With reversing change-over selector	With coarse change-over selector	with multiple coarse change-over selector (MG) only size B, C and O	Installation length h in mm																																																							
M III 350 Y *) M III 500 Y *) M III 600 Y (*) not as MG VM III 350 Y *) VM III 500 Y *) VM III 650 Y (*) not as MG					<table><tr><th colspan="6">Tap selector size</th></tr><tr><th></th><th colspan="2">B</th><th colspan="2">C</th><th>D/DE</th><th>D</th></tr><tr><th>U_m</th><th>0/W/G</th><th>MG</th><th>0/W/G</th><th>MG</th><th>0/W/G</th><th>MG</th></tr><tr><td>72,5</td><td>1894</td><td>1856</td><td>2069</td><td>2031</td><td>2524</td><td>2486</td></tr><tr><td>123</td><td>2024</td><td>1986</td><td>2199</td><td>2161</td><td>2654</td><td>2616</td></tr><tr><td>170</td><td>2154</td><td>2116</td><td>2329</td><td>2291</td><td>2784</td><td>2746</td></tr><tr><td>245</td><td>2254</td><td>2216</td><td>2429</td><td>2391</td><td>2884</td><td>2846</td></tr></table>	Tap selector size							B		C		D/DE	D	U _m	0/W/G	MG	0/W/G	MG	0/W/G	MG	72,5	1894	1856	2069	2031	2524	2486	123	2024	1986	2199	2161	2654	2616	170	2154	2116	2329	2291	2784	2746	245	2254	2216	2429	2391	2884	2846							
Tap selector size																																																												
	B		C		D/DE	D																																																						
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123	2024	1986	2199	2161	2654	2616																																																						
170	2154	2116	2329	2291	2784	2746																																																						
245	2254	2216	2429	2391	2884	2846																																																						
M I 351 *) M I 501 *) M I 601 (*) not as MG VM I 351 *) VM I 501 *) VM I 651 (*) not as MG					<table><tr><th colspan="6">Tap selector size</th></tr><tr><th></th><th colspan="2">B</th><th colspan="2">C</th><th>D/DE</th><th>D</th></tr><tr><th>U_m</th><th>0/W/G</th><th>MG</th><th>0/W/G</th><th>MG</th><th>0/W/G</th><th>MG</th></tr><tr><td>72,5</td><td>1514</td><td>1476</td><td>1589</td><td>1551</td><td>1784</td><td>1746</td></tr><tr><td>123</td><td>1644</td><td>1606</td><td>1719</td><td>1681</td><td>1914</td><td>1876</td></tr><tr><td>170</td><td>1774</td><td>1736</td><td>1849</td><td>1811</td><td>2044</td><td>2006</td></tr><tr><td>245</td><td>1874</td><td>1836</td><td>1949</td><td>1911</td><td>2144</td><td>2106</td></tr><tr><td>300</td><td>2026</td><td>1988</td><td>2101</td><td>2063</td><td>2296</td><td>2258</td></tr></table>	Tap selector size							B		C		D/DE	D	U _m	0/W/G	MG	0/W/G	MG	0/W/G	MG	72,5	1514	1476	1589	1551	1784	1746	123	1644	1606	1719	1681	1914	1876	170	1774	1736	1849	1811	2044	2006	245	1874	1836	1949	1911	2144	2106	300	2026	1988	2101	2063	2296	2258
Tap selector size																																																												
	B		C		D/DE	D																																																						
U _m	0/W/G	MG	0/W/G	MG	0/W/G	MG																																																						
72,5	1514	1476	1589	1551	1784	1746																																																						
123	1644	1606	1719	1681	1914	1876																																																						
170	1774	1736	1849	1811	2044	2006																																																						
245	1874	1836	1949	1911	2144	2106																																																						
300	2026	1988	2101	2063	2296	2258																																																						
M II 352 M II 502 M II 602 VM II 352 VM II 502 VM II 652					<table><tr><th colspan="4">Tap selector size</th></tr><tr><th></th><th>B</th><th>C</th><th>D/DE</th></tr><tr><th>U_m</th><th>0/W/G</th><th>0/W/G</th><th>0/W/G</th></tr><tr><td>72,5</td><td>1704</td><td>1829</td><td>2154</td></tr><tr><td>123</td><td>1834</td><td>1959</td><td>2284</td></tr><tr><td>170</td><td>1964</td><td>2089</td><td>2414</td></tr><tr><td>245</td><td>2064</td><td>2189</td><td>2514</td></tr><tr><td>300</td><td>2216</td><td>2341</td><td>2666</td></tr></table>	Tap selector size					B	C	D/DE	U _m	0/W/G	0/W/G	0/W/G	72,5	1704	1829	2154	123	1834	1959	2284	170	1964	2089	2414	245	2064	2189	2514	300	2216	2341	2666																							
Tap selector size																																																												
	B	C	D/DE																																																									
U _m	0/W/G	0/W/G	0/W/G																																																									
72,5	1704	1829	2154																																																									
123	1834	1959	2284																																																									
170	1964	2089	2414																																																									
245	2064	2189	2514																																																									
300	2216	2341	2666																																																									
M I 802 VM I 802 VM I 1002					<table><tr><th colspan="6">Tap selector size</th></tr><tr><th></th><th colspan="2">B</th><th colspan="2">C</th><th>D/DE</th><th>D</th></tr><tr><th>U_m</th><th>0/W/G</th><th>MG</th><th>0/W/G</th><th>MG</th><th>0/W/G</th><th>MG</th></tr><tr><td>72,5</td><td>1724</td><td>1686</td><td>1799</td><td>1761</td><td>1994</td><td>1956</td></tr><tr><td>123</td><td>1854</td><td>1816</td><td>1929</td><td>1891</td><td>2124</td><td>2086</td></tr><tr><td>170</td><td>1984</td><td>1956</td><td>2059</td><td>2021</td><td>2254</td><td>2216</td></tr><tr><td>245</td><td>2084</td><td>2046</td><td>2159</td><td>2121</td><td>2354</td><td>2316</td></tr><tr><td>300</td><td>2236</td><td>2198</td><td>2311</td><td>2273</td><td>2506</td><td>2468</td></tr></table>	Tap selector size							B		C		D/DE	D	U _m	0/W/G	MG	0/W/G	MG	0/W/G	MG	72,5	1724	1686	1799	1761	1994	1956	123	1854	1816	1929	1891	2124	2086	170	1984	1956	2059	2021	2254	2216	245	2084	2046	2159	2121	2354	2316	300	2236	2198	2311	2273	2506	2468
Tap selector size																																																												
	B		C		D/DE	D																																																						
U _m	0/W/G	MG	0/W/G	MG	0/W/G	MG																																																						
72,5	1724	1686	1799	1761	1994	1956																																																						
123	1854	1816	1929	1891	2124	2086																																																						
170	1984	1956	2059	2021	2254	2216																																																						
245	2084	2046	2159	2121	2354	2316																																																						
300	2236	2198	2311	2273	2506	2468																																																						
M I 1203 M I 1503 VM I 1203 VM I 1503					<table><tr><th colspan="6">Tap selector size</th></tr><tr><th></th><th colspan="2">B</th><th colspan="2">C</th><th>D/DE</th><th>D</th></tr><tr><th>U_m</th><th>0/W/G</th><th>MG</th><th>0/W/G</th><th>MG</th><th>0/W/G</th><th>MG</th></tr><tr><td>72,5</td><td>1934</td><td>1896</td><td>2009</td><td>1971</td><td>2204</td><td>2166</td></tr><tr><td>123</td><td>2064</td><td>2026</td><td>2139</td><td>2101</td><td>2334</td><td>2296</td></tr><tr><td>170</td><td>2194</td><td>2156</td><td>2269</td><td>2231</td><td>2464</td><td>2426</td></tr><tr><td>245</td><td>2294</td><td>2256</td><td>2369</td><td>2331</td><td>2564</td><td>2526</td></tr><tr><td>300</td><td>2446</td><td>2408</td><td>2521</td><td>2483</td><td>2716</td><td>2678</td></tr></table>	Tap selector size							B		C		D/DE	D	U _m	0/W/G	MG	0/W/G	MG	0/W/G	MG	72,5	1934	1896	2009	1971	2204	2166	123	2064	2026	2139	2101	2334	2296	170	2194	2156	2269	2231	2464	2426	245	2294	2256	2369	2331	2564	2526	300	2446	2408	2521	2483	2716	2678
Tap selector size																																																												
	B		C		D/DE	D																																																						
U _m	0/W/G	MG	0/W/G	MG	0/W/G	MG																																																						
72,5	1934	1896	2009	1971	2204	2166																																																						
123	2064	2026	2139	2101	2334	2296																																																						
170	2194	2156	2269	2231	2464	2426																																																						
245	2294	2256	2369	2331	2564	2526																																																						
300	2446	2408	2521	2483	2716	2678																																																						

3.4

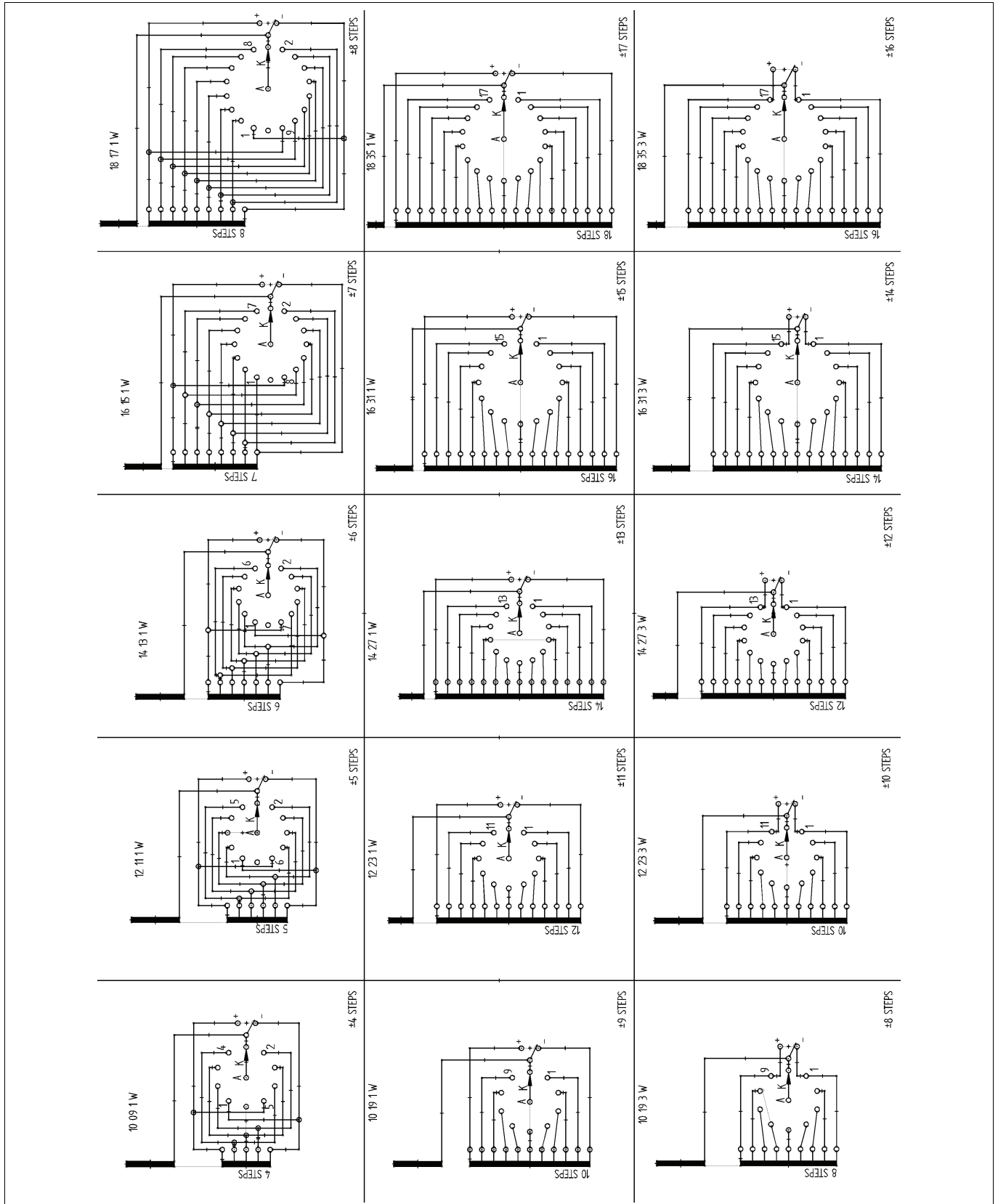
VACUTAP® VM 300 - Overview of basic connection diagrams with tap selector connection contacts designated in accordance with MR standard (893819)



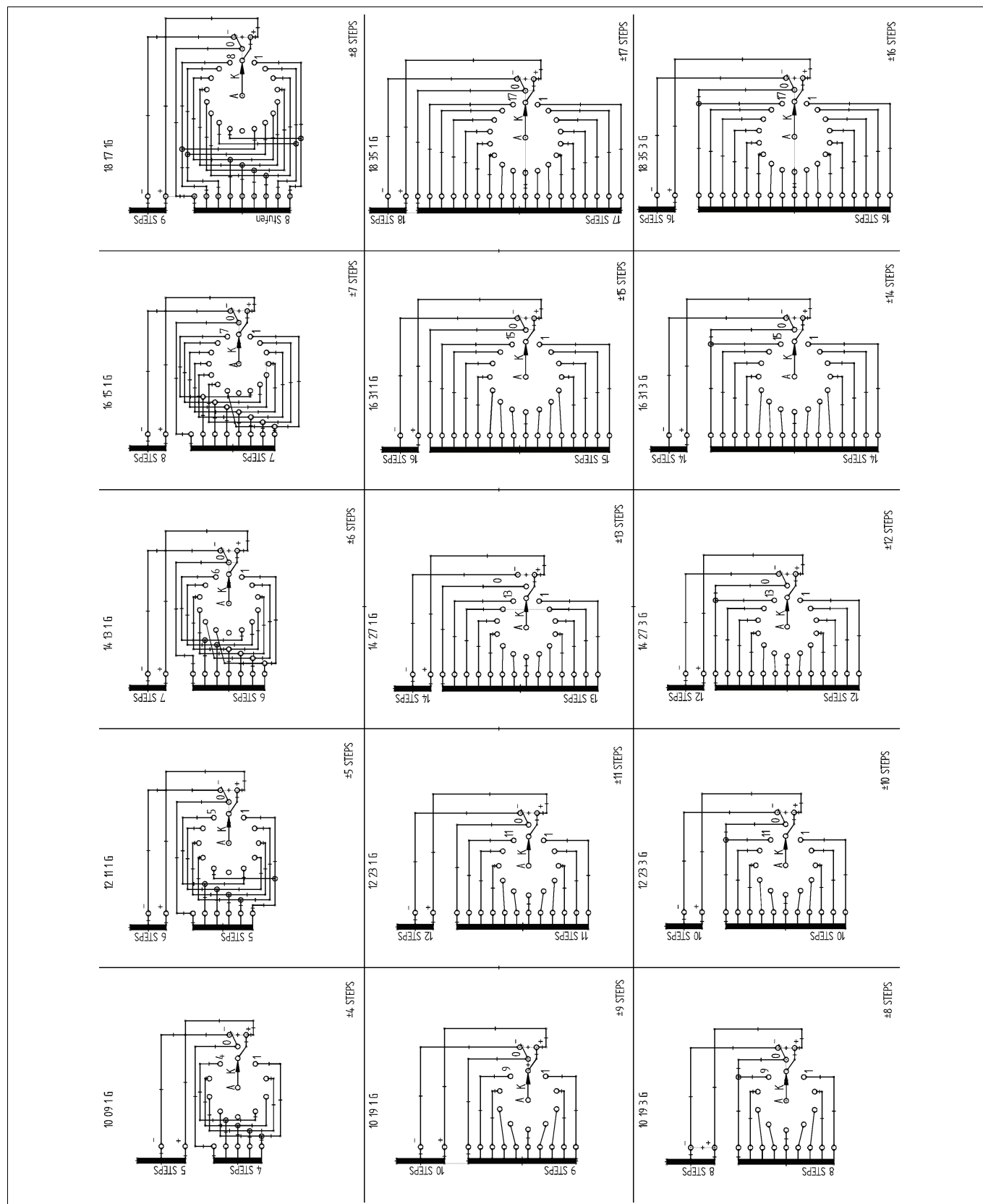
3.5 Overview of basic connection diagrams with tap selector connection contacts designated in accordance with MR standard (890616)



Overview of basic connection diagrams with tap selector connection contacts designated in accordance with MR standard (890616)

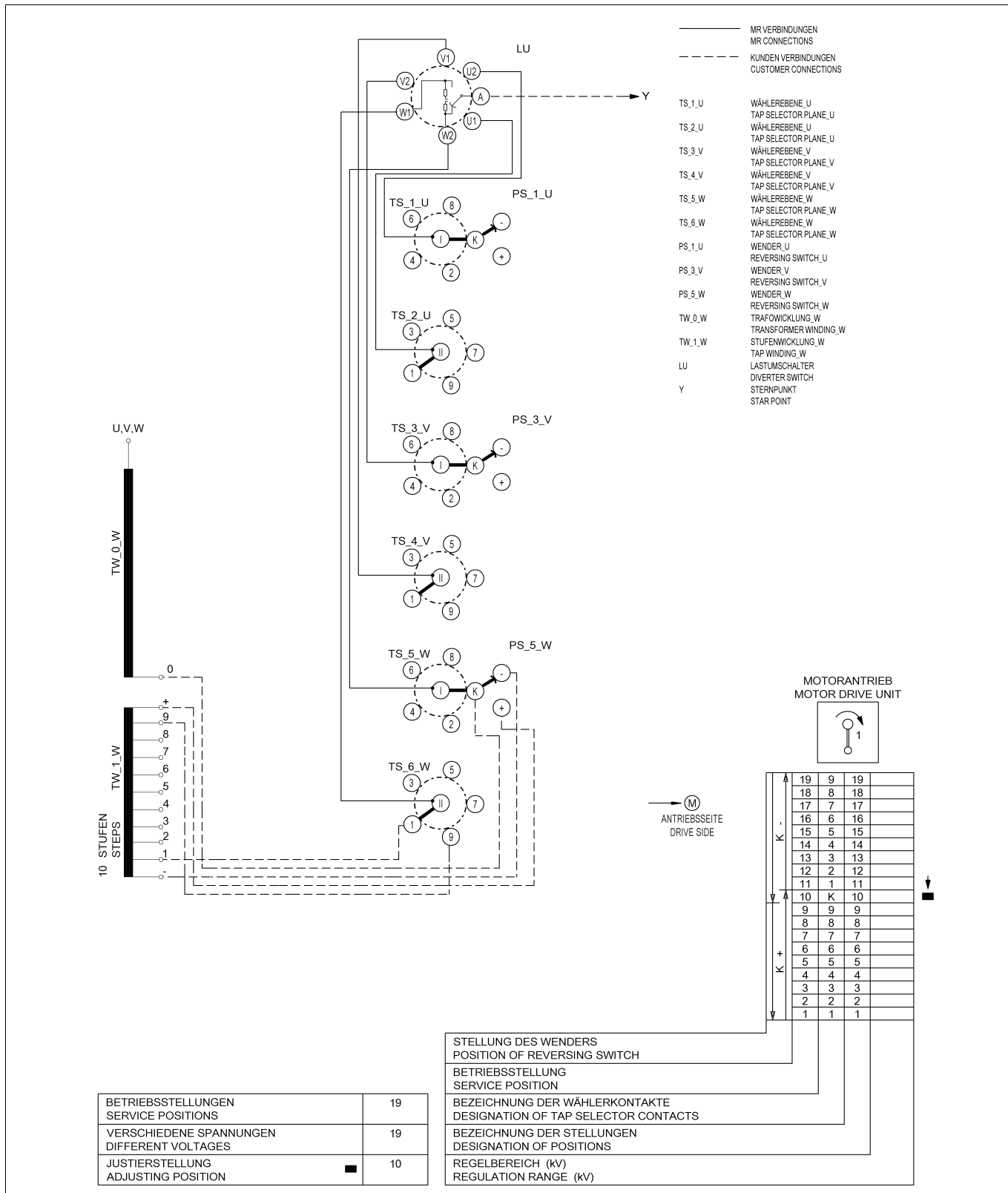


Overview of basic connection diagrams with tap selector connection contacts designated in accordance with MR standard (890616)



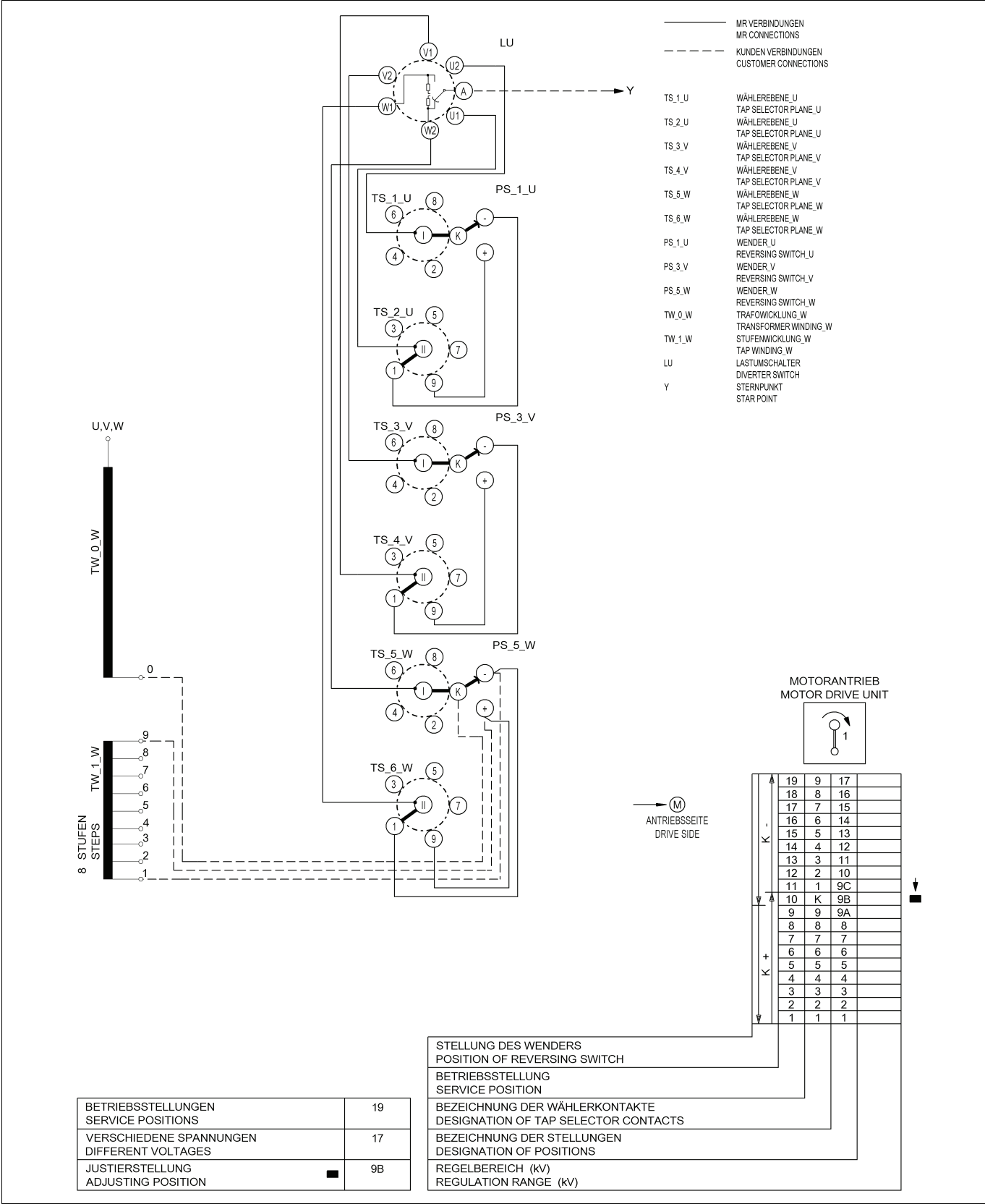
3.6 Connection diagram (contacts designated in accordance with MR standard)

3.6.1 VACUTAP® VM III 300 Y, basic connection diagram 10 19 1 W (2414642)

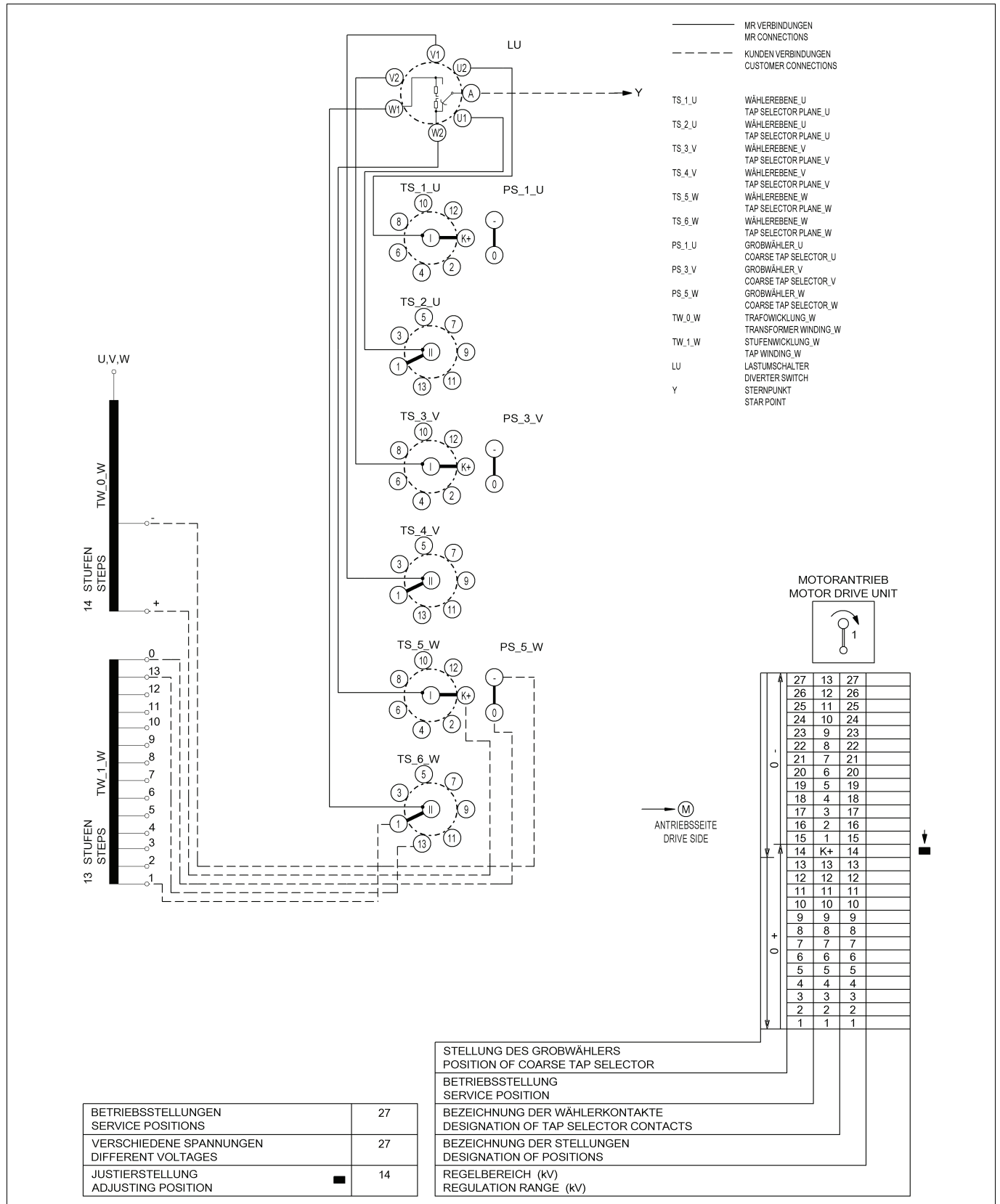




3.6.2 VACUTAP® VM III 300 Y, basic connection diagram 10 19 3 W (2414644)

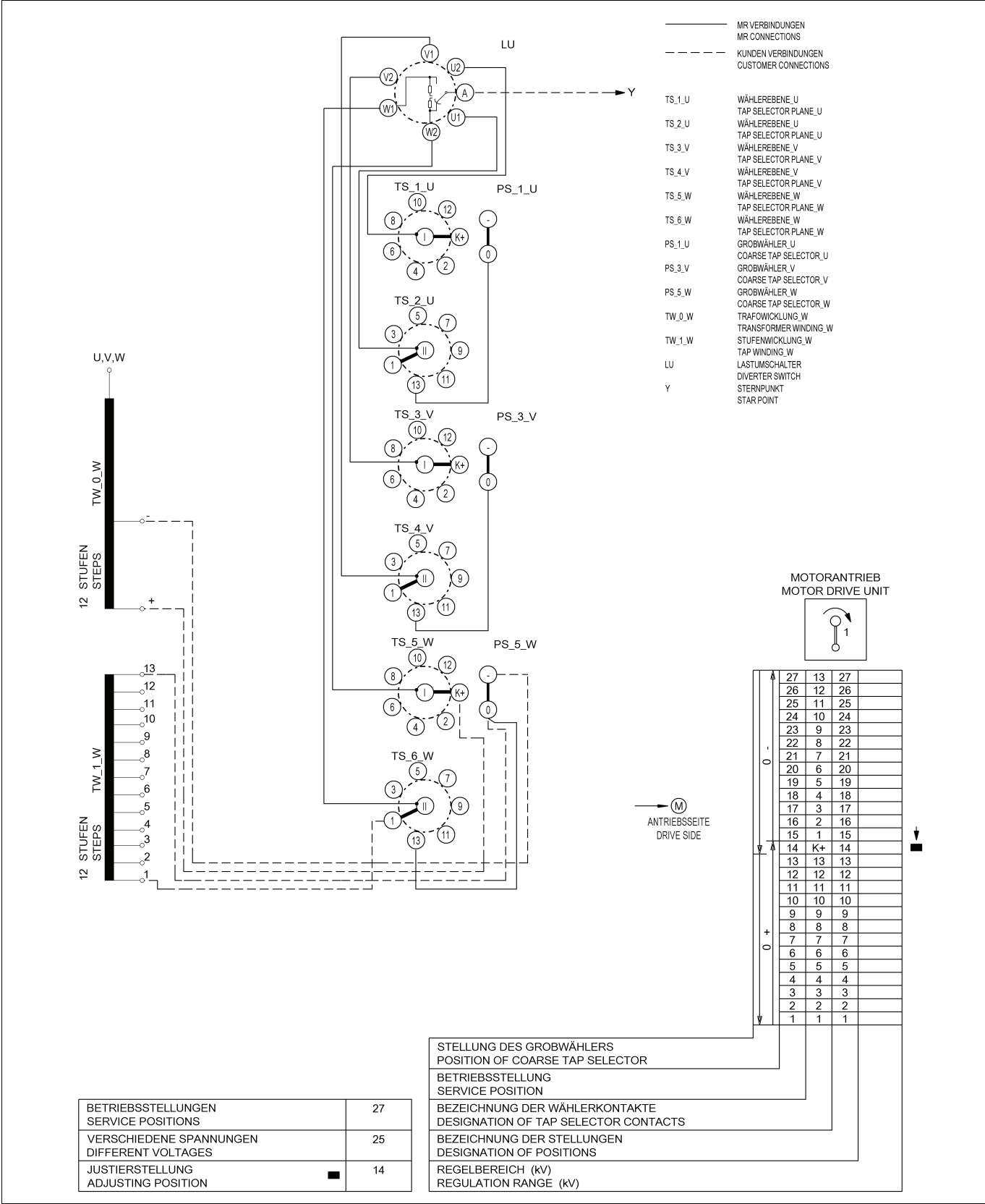


3.6.3 VACUTAP® VM III 300 Y, basic connection diagram 14 27 1 G (2414648)

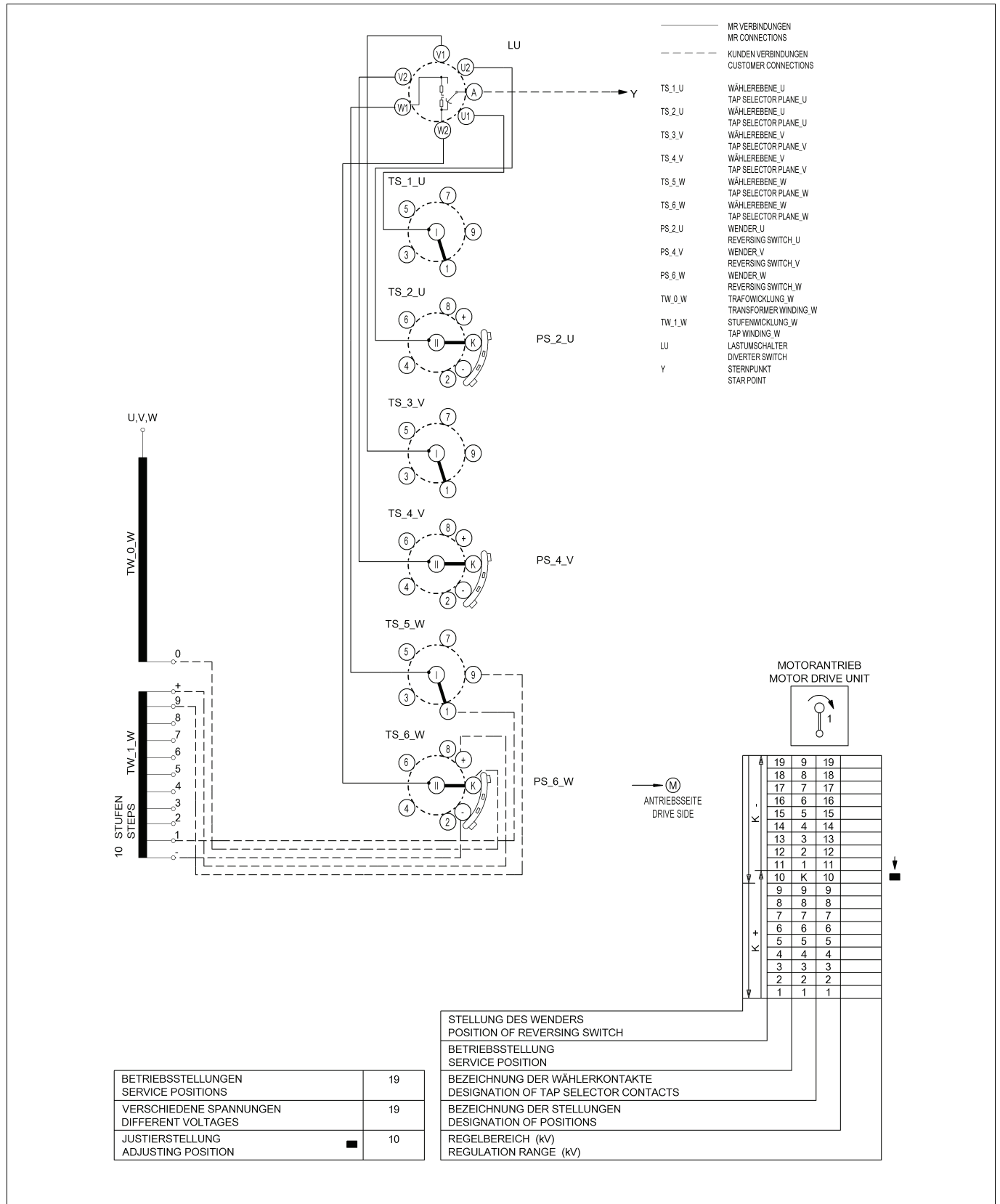




3.6.4 VACUTAP® VM III 300 Y, basic connection diagram 14 27 3 G (2414649)

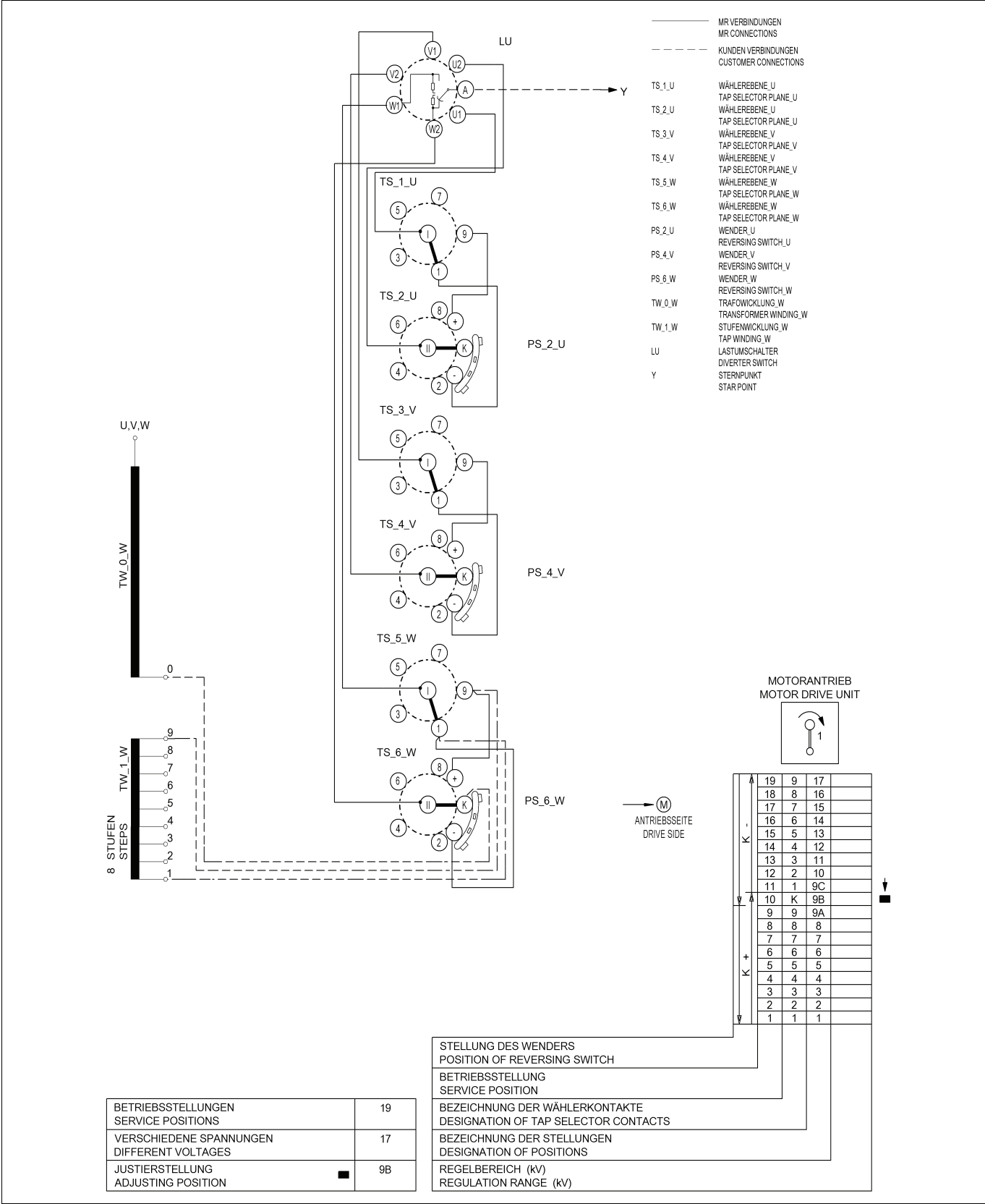


3.6.5 VACUTAP® VM III 350/500/650 Y, basic connection diagram 10 19 1 W (2414658)

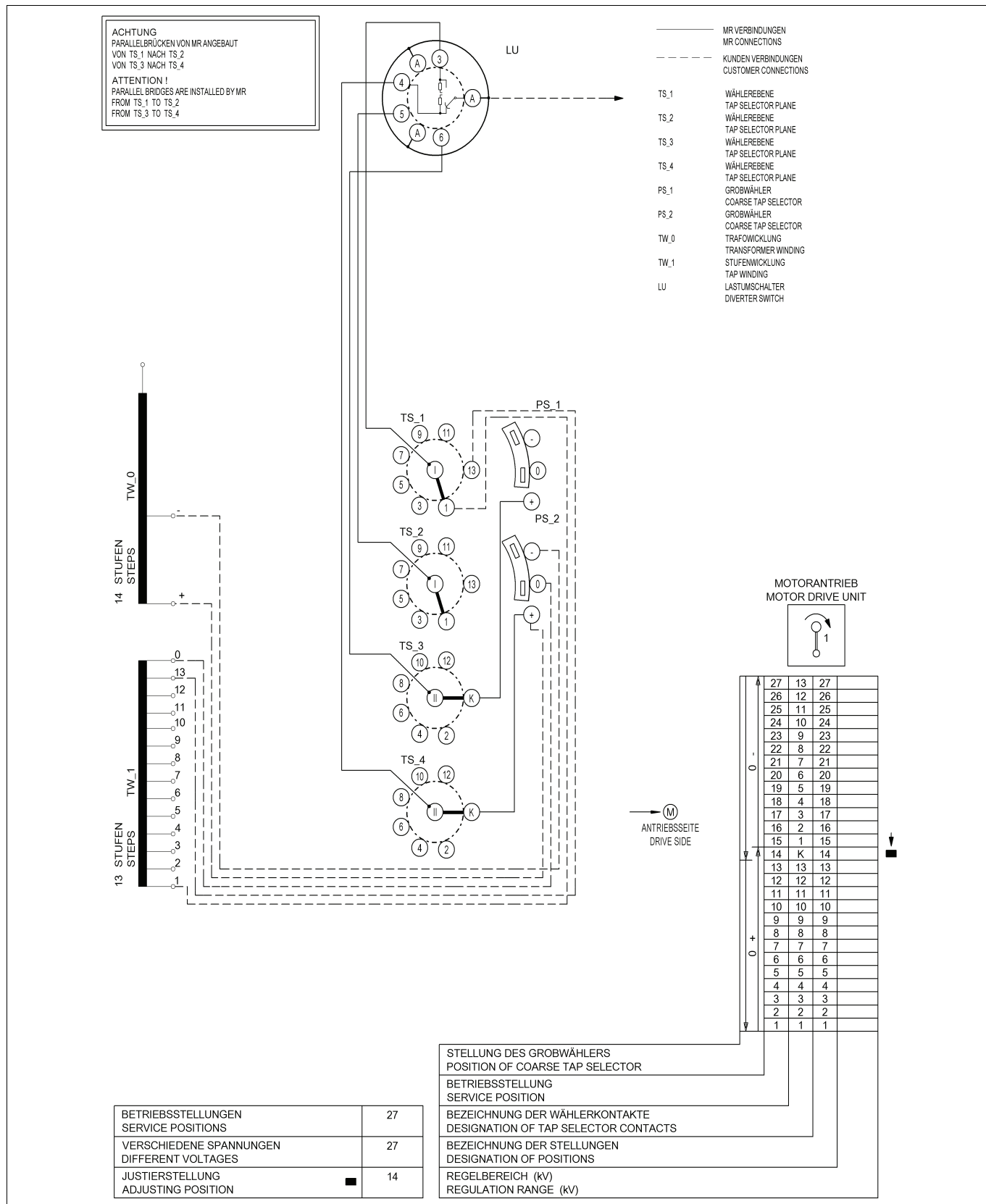




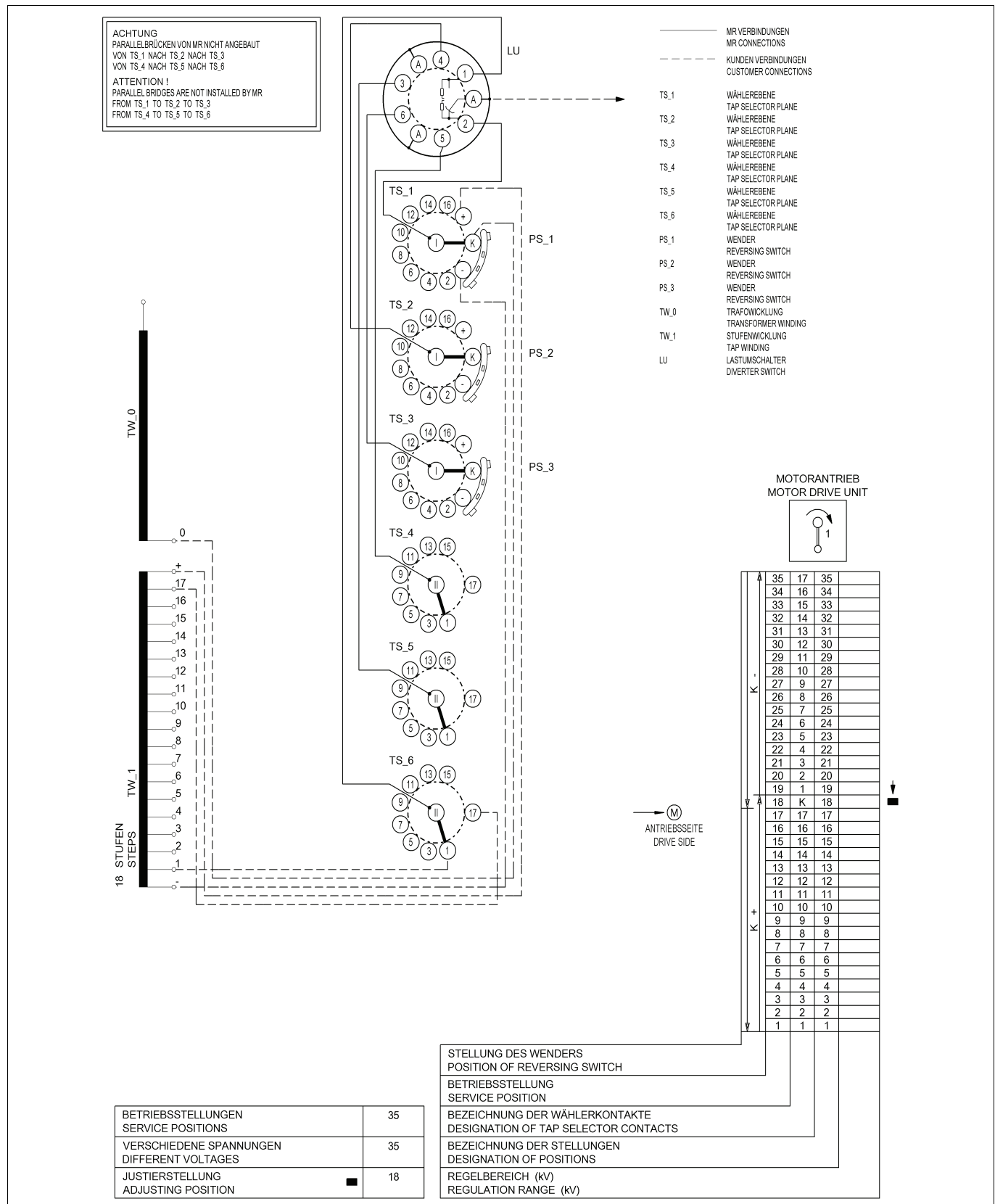
3.6.6 VACUTAP® VM III 350/500/650 Y, basic connection diagram 10 19 3 W (2414670)



3.6.7 VACUTAP® VM I 802/1002, basic connection diagram 14 27 1 G (2414631)



3.6.8 VACUTAP® VM I 1203/1503, basic connection diagram 18 35 1 W (2414636)



VACUTAP® VM I 802/1002, basic connection diagram 16 79 1 G, multiple coarse change-over selector design (2407535)





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